Rev. 8 - October 7th, 2026

DT5560SE / R5560 / R5560SE

32 and 128 Channel 14 bit 125 MS/s Open FPGA Digitizers



DT5560SE

32 ch, desktop, single-ended on LEMO



R5560SE

128 ch, 3U rack, single-ended on MCX



R5560

128 ch, 2U rack, differential on 32x RJ45

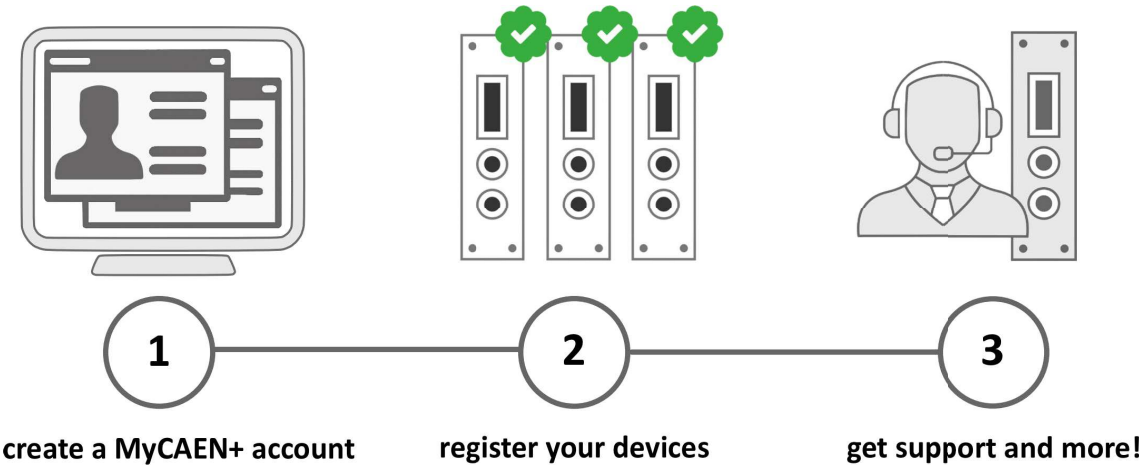
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User Manual
UM6952

Register your device

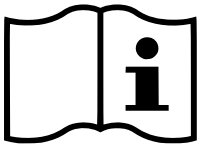
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Purpose of this Manual



This User Manual contains the full description of the DT5560SE / R5560 / R5560SE 32 and 128 Channel 14 bit 125 MS/s Open FPGA Digitizers, of their web interface and a brief guide to the Open Hardware Readout Software. The three models share the same DAQ architecture and the same firmware and software tools; wherever their behaviour differs, the text states which model it refers to.

Change Document Record

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Symbols, Abbreviated Terms and Notation

ADC	Analog to Digital Converter
AFE	Analog Frontend
ASIC	Application Specific Integrated Circuit
DAQ	Data Acquisition
DMA	Direct Memory Access
FIFO	First In First Out
FPGA	Field Programmable Gate Array
FTP	File Transfer Protocol
H-LINK	Horizontal Link (R5560/R5560SE inter-section link)
LVDS	Low Voltage Differential Signaling
OS	Operating system
PID	Product Identifier
RNDIS	Remote Network Driver Interface Specification
SCSP	Slow Control and Sync Processor (R5560/R5560SE only)
SDK	Software Development Kit
SoC	System on Chip
VHDCI	Very High Density Cable Interconnect

Reference Documents

[RD1] GD6520 - SCI-Compiler Quick Start Guide

[RD2] UM6519 - SCI-Compiler User Manual

All CAEN documents can be downloaded at:
`www.caen.it/support-services/documentation-area`

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Limitation of Responsibility

If the warnings contained in this manual are not followed, CAEN will not be responsible for damage caused by improper use of the device. The manufacturer declines all responsibility for damage resulting from failure to comply with the instructions for use of the product. The equipment must be used as described in the user manual, with particular regard to the intended use, using only accessories as specified by the manufacturer. No modification or repair can be performed.

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We remark that all our boards have been designed and assembled in Italy. In a challenging environment where a competitive edge is often obtained at the cost of lower wages and declining working conditions, we proudly acknowledge that all those who participated in the production and distribution process of our devices were reasonably paid and worked in a safe environment (while this is true for the boards marked “MADE IN ITALY”, we cannot guarantee for third-party manufactures).



Index

Purpose of this Manual . . . 3

Change document record . . . 3

Symbols, abbreviated terms and notation . . . 3

Reference Documents . . . 4

Manufacturer Contacts . . . 5

Limitation of Responsibility . . . 5

Disclaimer . . . 5

Made in Italy . . . 5

List of Figures . . . 9

List of Tables . . . 12

1 Introduction . . . 13

2 Block diagram . . . 15
2.1 DT5560SE . . . 15
2.2 R5560/R5560SE . . . 16

3 Technical specifications . . . 18
3.1 Operating limits . . . 20

4 Packaging and compliancy . . . 21
4.1 SCI-Compiler license . . . 23

5 PID (Product Identifier) . . . 24

6 Power requirements . . . 25

7 Cooling management . . . 26
7.1 Cleaning the air vents . . . 26

8 Installing the device . . . 27
8.1 Desktop unit (DT5560SE) . . . 27
8.2 Rack units (R5560/R5560SE) . . . 27
8.3 Safety precautions . . . 28

9 Panels description . . . 29
9.1 Dimensions and panel views . . . 29
9.1.1 DT5560SE . . . 29
9.1.2 R5560 . . . 31
9.1.3 R5560SE . . . 31
9.2 Front panel . . . 32
9.3 Rear panel . . . 36

10 Functional description . . . 37
10.1 System architecture . . . 37
10.1.1 DT5560SE . . . 37
10.1.2 R5560/R5560SE . . . 37

10.2 DAQ section . . . 39
10.2.1 R5560 differential inputs . . . 40
10.2.2 Analog frontend (DT5560SE/R5560SE only) . . . 42
10.3 Digital stage . . . 42
10.3.1 VHDCI connectors (R5560/R5560SE only) . . . 43
10.4 Data readout . . . 44
10.4.1 Readout using DAQ section Ethernet port . . . 44
10.4.2 Readout using slow control Ethernet port (R5560/R5560SE only) . . . 44
10.4.3 Readout using the USB 3.0 (R5560/R5560SE only) . . . 45
10.4.4 Readout using optical link . . . 45
10.4.5 Optical daisy chain (R5560/R5560SE only) . . . 46
10.4.6 Parallel readout (R5560/R5560SE only) . . . 46
10.5 Clock distribution . . . 47
10.6 Synchronization among multiple boards . . . 48
10.7 DAQ datapath (R5560/R5560SE only) . . . 50

11 Touchscreen display guide . . . 52
11.1 Ethernet settings . . . 52
11.1.1 DT5560SE . . . 52
11.1.2 R5560/R5560SE . . . 53
11.2 Signal routing settings . . . 54
11.2.1 DT5560SE . . . 54
11.2.2 R5560/R5560SE . . . 56
11.3 Info and status . . . 58
11.3.1 DT5560SE . . . 58
11.3.2 R5560/R5560SE . . . 59
11.4 Cleaning the Touchscreen . . . 59

12 Web interface . . . 60
12.1 Baseboard web interface . . . 60
12.1.1 Connection . . . 60
12.1.2 Device status and firmware upgrade . . . 60
12.1.3 Analog settings . . . 63
12.1.4 I/O settings . . . 64
12.1.5 Ethernet settings . . . 66
12.1.6 Firmware settings (R5560/R5560SE only) . . . 66
12.2 DAQ section web interface (R5560/R5560SE only) . . . 67
12.2.1 Connection . . . 67
12.2.2 Device status and firmware upgrade . . . 67
12.2.3 Analog settings (R5560SE only) . . . 69
12.2.4 Ethernet settings . . . 69
12.3 Resource Explorer (beta) . . . 70
12.3.1 Starting the Resource Explorer . . . 70
12.3.2 Panels . . . 71
12.3.3 Panel toolbar . . . 72

13 Firmware developing . . . 75
13.1 Firmware design: the processing core . . . 75
13.2 Default firmware . . . 76
13.3 Firmware upgrade (Legacy) . . . 78
13.4 Operating system upgrade (Legacy) . . . 80

14 Drivers & libraries . . . 82
14.1 Drivers installation . . . 82
14.1.1 DT5560SE . . . 82
14.1.2 R5560/R5560SE . . . 82
14.2 Software development kit . . . 82
14.3 Libraries for the default firmware (Legacy) . . . 83

14.3.1 Start connection . . . 84
14.3.2 Close connection . . . 84
14.3.3 Write register . . . 85
14.3.4 Read register . . . 85
14.3.5 Write data . . . 85
14.3.6 Read data . . . 86
14.3.7 Read FIFO . . . 86

15 Open Hardware Readout Software . . . 88
15.1 Software installation . . . 88
15.2 Board connection . . . 92
15.3 Software GUI description . . . 93
15.3.1 Control bar . . . 94
15.3.2 MCA Settings tab . . . 95
15.3.3 Oscilloscope tab . . . 98
15.3.4 Spectrum tab . . . 100
15.3.5 Statistics tab . . . 102
15.3.6 Imaging module . . . 103
15.4 How to perform a fit . . . 104
15.5 How to save data . . . 105

16 Appendix . . . 108
16.1 Ethernet protocol . . . 108
16.1.1 TCP web server . . . 108
16.1.2 FIFO server . . . 109
16.1.3 JSON server . . . 110
16.2 Optical link protocol . . . 110
16.2.1 Zynq 7030 (DT5560SE, R5560A, R5560SEA) . . . 111
16.2.2 Zynq 7035 (R5560B, R5560SEB) . . . 112
16.3 Libraries usage examples (Legacy) . . . 112

Safety Notices . . . 115

17 Instructions for Cleaning . . . 117
17.1 Cleaning the Air Vents . . . 117
17.2 General Cleaning Safety Precautions . . . 117

18 Device Decommissioning . . . 118

19 Disposal . . . 119

20 Technical Support . . . 120

List of Figures

Fig. 2.1 DT5560SE block diagram. . . . 16
Fig. 2.2 Block diagram of the R5560/R5560SE. AFE sections are implemented in the R5560SE version only. . . 17

Fig. 5.1 PID location (the PID in the pictures and the device models are purely indicative). On the desktop unit the PID can be found on the top side of the module, together with the plate data of the module, and/or on the rear panel. . . . 24

Fig. 8.1 Installing the DT5560SE. . . . 27
Fig. 8.2 Installing the device into a 19” rack. . . . 28

Fig. 9.1 DT5560SE front panel view and dimensions. . . . 29
Fig. 9.2 DT5560SE rear panel view. . . . 30
Fig. 9.3 DT5560SE side view and dimensions. . . . 30
Fig. 9.4 R5560 front panel view and dimensions. . . . 31
Fig. 9.5 R5560 rear panel view and dimensions. . . . 31
Fig. 9.6 R5560 side view and dimensions. . . . 31
Fig. 9.7 R5560SE front panel view and dimensions. . . . 32
Fig. 9.8 R5560SE rear panel view and dimensions. . . . 32
Fig. 9.9 R5560SE side view and dimensions. . . . 32

Fig. 10.1 The complete block diagram of the R5560SE. The R5560 works the same without the AFE boards. . . 38
Fig. 10.2 Block diagram of a single DAQ section. Signals interconnecting the DAQ with the base board and AFE boards are shown. This diagram is identical on all models of the family. . . . 40
Fig. 10.3 RJ45 analog connector pinout: the pinout for the first group of channels [1:4] is shown. The same ordering can be applied to the subsequent groups. . . . 41
Fig. 10.4 Block diagram of an AFE section. . . . 42
Fig. 10.5 Pinout of the LVDS digital connectors of the R5560/R5560SE. . . . 43
Fig. 10.6 Readout using the DAQ section Ethernet port, one port per section, through an external Ethernet switch. . . . 44
Fig. 10.7 Readout using the slow control Ethernet port on the SCSP board. . . . 45
Fig. 10.8 Readout using the USB 3.0 connector. . . . 45
Fig. 10.9 Readout using the optical links, one link pair per DAQ section, connected to the facility readout system. 46
Fig. 10.10 Optical daisy chain readout topology. . . . 46
Fig. 10.11 Parallel readout using the LVDS interface of each DAQ section. . . . 47
Fig. 10.12 Scheme of the clock distribution. The R5560/R5560SE feeds four DAQ sections and the SCSP from the same PLL; the DT5560SE feeds its single DAQ section. . . . 48
Fig. 10.13 Pinout of the SYNC RJ45 connector. . . . 48
Fig. 10.14 Synchronization connection between two boards using a single Ethernet cable. The slave board receives the synchronization signals on its SYNC IN connector from the master board SYNC OUT. . . . 49
Fig. 10.15 Example of the router operation for data transfer. . . . 50
Fig. 10.16 Data transfer between two DAQ sections, using the router or the H-LINK. . . . 51

Fig. 11.1 The main page of the Touchscreen display. . . . 52
Fig. 11.2 The sequence on the Touchscreen display to set the Ethernet configuration of the DT5560SE. . . . 53
Fig. 11.3 The Ethernet tabs on the Touchscreen display of the R5560/R5560SE. . . . 54
Fig. 11.4 The CLOCK tab of the DT5560SE. . . . 54
Fig. 11.5 The DAQ SYNC IN tab of the DT5560SE. . . . 55
Fig. 11.6 The SYNC OUT tab of the DT5560SE. . . . 55
Fig. 11.7 The LEMO OUT tab of the DT5560SE. . . . 56

Fig. 11.8 The CLOCK INPUT tab of the R5560/R5560SE. . . . 56
Fig. 11.9 The DAQ SYNC tab of the R5560/R5560SE. . . . 57
Fig. 11.10 The SYNC OUT tab of the R5560/R5560SE. . . . 57
Fig. 11.11 The LEMO OUT tab of the R5560/R5560SE. . . . 57
Fig. 11.12 The PULSER tab of the R5560/R5560SE. . . . 58
Fig. 11.13 The sequence on the Touchscreen display of the DT5560SE to retrieve a pre-loaded firmware. . . . 58
Fig. 11.14 The temperature tab of the DT5560SE. . . . 59
Fig. 11.15 The TEMPERATURE, DAQ LAN STATUS and DAQ INFO tabs of the R5560/R5560SE. . . . 59

Fig. 12.1 Connection scheme to access the baseboard Web interface, shown on an R5560SE; the scheme is the same on the R5560. The baseboard port can be connected to the network or directly to a PC Ethernet port. . . . 60
Fig. 12.2 The Home page of the Web interface. . . . 61
Fig. 12.3 Upgrade of the OS of the board from the Web interface. . . . 62
Fig. 12.4 Upgrade of the DAQ firmware from the Web interface of the DT5560SE. . . . 63
Fig. 12.5 The Analog Settings page of the Web interface. . . . 64
Fig. 12.6 The I/O Settings page of the Web interface of the DT5560SE. . . . 65
Fig. 12.7 The I/O Settings page of the baseboard Web interface of the R5560/R5560SE. . . . 66
Fig. 12.8 The Ethernet Settings page of the Web interface. . . . 66
Fig. 12.9 The Firmware Settings page of the baseboard Web interface. . . . 67
Fig. 12.10 Connection scheme to access the DAQ section Web interface. Each port can be connected independently to the network or directly to a PC Ethernet port. . . . 67
Fig. 12.11 The Home page of the DAQ section Web interface. . . . 68
Fig. 12.12 Upgrade of the OS of the board from the DAQ Web interface. . . . 68
Fig. 12.13 Upgrade of the firmware from the DAQ section Web interface. . . . 69
Fig. 12.14 The Analog Settings page of the DAQ section Web interface. . . . 69
Fig. 12.15 The Ethernet Settings page of the DAQ section Web interface. . . . 70
Fig. 12.16 The Resource Explorer panel of the web interface. . . . 70
Fig. 12.17 Status of the Resource Explorer once started. . . . 70
Fig. 12.18 The Resource Explorer at the first opening: the tree is loaded with Get Tree, then an element is dragged from the tree to the panel. . . . 71
Fig. 12.19 Creation of a new panel. . . . 71
Fig. 12.20 A panel with the oscilloscope of a channel, acquisition not started yet. . . . 72
Fig. 12.21 The Auto Save menu and the version information of the top bar. . . . 72
Fig. 12.22 The panel toolbar and the zoom submenu. . . . 72
Fig. 12.23 The windows opened from the panel toolbar. . . . 73
Fig. 12.24 The oscilloscope of a channel during the acquisition. . . . 74

Fig. 13.1 SCI-Compiler scheme for single channel data processing in the default firmware. . . . 77
Fig. 13.2 The trapezoid method used in the board default firmware. . . . 77
Fig. 13.3 Hardware selection in the OpenHardware Firmware Upgrader. . . . 78
Fig. 13.4 IP address of the board to be upgraded. . . . 79
Fig. 13.5 Loading the firmware file. . . . 79
Fig. 13.6 Firmware upgrade completed. . . . 79
Fig. 13.7 Loading the Operating System file. . . . 80
Fig. 13.8 Operating System upgrade completed. . . . 81

Fig. 15.1 Choice of the destination folder. . . . 89
Fig. 15.2 Choice of the Start Menu folder. . . . 89
Fig. 15.3 Summary of the settings before the installation. . . . 90
Fig. 15.4 Installation in progress. . . . 90
Fig. 15.5 Completion of the software installation, with the selection of the drivers to install. . . . 91
Fig. 15.6 Extraction of the FTDI USB Drivers. . . . 91
Fig. 15.7 Installation of the FTDI USB Drivers. . . . 92
Fig. 15.8 The “Connection” window at start-up of Open Hardware Readout Software. . . . 93
Fig. 15.9 The main window of Open Hardware Readout Software. The main areas are highlighted. . . . 94
Fig. 15.10 The Control Bar of Open Hardware Readout Software. . . . 94
Fig. 15.11 The MCA Settings Tab of Open Hardware Readout Software. . . . 96

Fig. 15.12 Scheme of the trapezoidal energy filter parameters to be set in Open Hardware Readout Software. . 97
Fig. 15.13 The AFE Settings tab in Open Hardware Readout Software. 98
Fig. 15.14 The Oscilloscope Tab of Open Hardware Readout Software. Here only signals for channel 0 are displayed. 99
Fig. 15.15 Printing (left) and saving to file (right) the current view of the Oscilloscope Tab. 100
Fig. 15.16 The Spectrum Tab of Open Hardware Readout Software. Here only the spectrum for channel 0 is displayed. 101
Fig. 15.17 Printing (left) and saving to file (right) the current view of the Spectrum Tab. 102
Fig. 15.18 The Statistics Tab of Open Hardware Readout Software, with the instantaneous rates. 103
Fig. 15.19 The Imaging Module of Open Hardware Readout Software, where it is possible to see 5 channels acquiring energy data. 104
Fig. 15.20 The fitting tool of Open Hardware Readout Software. 105
Fig. 15.21 The Data Recording Configuration window. 106

Fig. 16.1 Format of the packet sent from the client to the server. 108
Fig. 16.2 Format of the packet sent from the server to the client. 109
Fig. 16.3 Scheme for optical links operation with the Zynq 7030 SoC. 111
Fig. 16.4 Scheme for optical links operation with the Zynq 7035 SoC. 112

List of Tables

Tab. 1.1 Table of available board models and accessories. 14

Tab. 3.1 Technical specifications for the DT5560SE / R5560 / R5560SE. 20
Tab. 3.2 Operating limits of the connectors. 20

Tab. 4.1 Delivered kit, DT5560SE. 21
Tab. 4.2 Delivered kit, R5560/R5560SE. 22

Tab. 10.1 Analog RJ45 connector cable pinout. 40
Tab. 10.2 Differential parameters for the R5560 analog input. 41

Tab. 14.1 Arguments of the Start Connection function. 84
Tab. 14.2 Arguments of the Close Connection function. 84
Tab. 14.3 Arguments of the Write Register function. 85
Tab. 14.4 Arguments of the Read Register function. 85
Tab. 14.5 Arguments of the Write Data function. 86
Tab. 14.6 Arguments of the Read Data function. 86
Tab. 14.7 Arguments of the Read FIFO function. 87

1 Introduction

The DT5560SE/R5560/R5560SE are Open FPGA Digitizers sampling at 14 bit and 125 MS/s, designed to attain **programmable data processing capabilities**. They share the same DAQ architecture, the same default firmware and the same development and readout tools, and differ in channel count, form factor and analog input stage:

- the DT5560SE is a **32 channel** unit in a handy **desktop** form factor, with **single-ended analog inputs on LEMO** connectors;
- the R5560 is a **128 channel 19" rack-mount** unit with **differential analog inputs on RJ45** connectors;
- the R5560SE is a **128 channel 19" rack-mount** unit with **single-ended analog inputs on MCX** connectors.

The device is designed to operate in laboratory environment under the supervision of skilled technicians.

The desktop form factor and the convenient connectors make the DT5560SE an optimal solution for **R&D and lab development** with commonly used HEP and nuclear physics photodetectors. It offers the possibility to develop **custom firmware** (using SCI-Compiler tool) and easily readout data through convenient 1 Gb Ethernet and USB 2.0 connectivity. It is also possible to interface the device to a larger experiment readout system developing an **Optical Link** protocol in the board firmware.

The R5560/R5560SE is an optimal solution for large experiments, usually requiring fast digitization of analog signals and usage of several digital lines to interface with external systems. The board supports multi-board synchronization through a single CAT5e cable, with the possibility to scale up to thousands of channels. Moreover, the rack-mount form factor simplifies the experimental setup in case of multi-board systems, where an effective space management is often a constraint.

Thanks to its programmability, the board can fit several applications. In fact, it is possible to take advantage of the powerful SoC mounted onboard (Xilinx Zynq®-7030 or 7035 models, depending on the model and ordering option) to write a custom pulse processing algorithm on the open FPGA as well as build a middleware/software that fits the needs of the application of interest.

The DT5560SE/R5560/R5560SE is capable of managing simultaneously **digital** and **analog** signals to implement many functionalities required by physics experiments: signal digitization, complex trigger logic, Pulse Height Analysis with MCA capabilities, Time Tagging, Pulse Shape Discrimination, Scaler, Counters and so on.

The DT5560SE/R5560SE features **single-ended analog inputs** with an **advanced and versatile analog frontend**: it is possible to configure the gain, input impedance, offset on each single channel and also integrate a programmable shaper, making the device an optimum choice for gamma and X-ray applications. The R5560 features **differential analog inputs** that feed the ADCs directly, with no analog frontend stage.

Thanks to the Open FPGA and programmable analog frontend, the DT5560SE/R5560/R5560SE can be used for a wide range of applications:

- Charge integration for the readout of SiPM, Silicon Detectors, PMTs
- Neutron detectors readout
- Trapezoidal filter PHA for HPGe readout
- Pulse shape discrimination
- Continuous reset or transistor reset preamplifier readout

The DT5560SE/R5560/R5560SE is fully supported by **SCI-Compiler**, a Windows-based graphical development system for **easy FPGA programming**. This tool allows the user to develop and compile the firmware

code using graphical blocks which represent the functionalities needed for firmware implementation (for example oscilloscope, TDC, MCA, charge integration, etc.). SCI-Compiler automatically generates the VHDL firmware code starting only from logic blocks and virtual instruments that can be connected together in the GUI and, moreover, it allows usage of the compatible SciSDK in Windows and Linux (refer to **[RD1]** for more details).

The SCI-Compiler Runtime License, needed to run the firmware generated by SCI-Compiler, is included with every model of the family; the SCI-Compiler license is purchased separately (see Sec. 4.1).

The DT5560SE includes a Board-level SDK (BSDK) that allows the user to configure and modify the Linux OS running onboard SoC in order to perform user custom operations inside the readout application before data are transferred to the user (only via Ethernet and USB). This makes it possible to implement complex custom applications running on the on-board SoC and customize the readout protocol to the facility.

A complete, ready-to-use default firmware is provided for free and open source. The default firmware manages the basic waveform digitization and pulse height analysis, and it is preloaded on the board. The user can open the default firmware project in SCI-Compiler and modify it in order to customize, for example, the trigger logic, the data online processing or integrate it in a larger system.

The **Open Hardware Readout Software** is the free and open source Windows-based software developed to perform acquisitions with the DT5560SE/R5560/R5560SE and other Open FPGA boards. It works in conjunction with the default firmware and it is distributed as open source, so that the user can modify it according to the functions implemented in a custom firmware and for any other need.

Available board models and accessories are listed below.

Ordering options		
DT5560SE	DT5560SE – 32 Ch. 14 bit 125 MS/s Digitizer single-ended	WDT5560SEXAA
R5560A	R5560A – 128 Ch. 14 bit 125 MS/s Digitizer-7030	WR5560AXAAAA
R5560B	R5560B – 128 Ch. 14 bit 125 MS/s Digitizer-7035	WR5560BXAAAA
R5560SE	R5560SE – 128 Ch. 14 bit 125 MS/s Digitizer single-ended – 7030	WR5560SEXAAA
R5560SEB	R5560SEB – 128 Ch. 14 bit 125 MS/s Digitizer single-ended – 7035	WR5560SEBXAA
Accessories		
SW555	SW555 – SCI-Compiler PRO License	WSW555PROXAA
RCS1Y	1 year remote customization service + upgrade for SCI-Compiler	WSW55RCSXAAA
RCS5Y	5 years remote customization service + upgrade for SCI-Compiler	WSW55RCSX5YA

Tab. 1.1: Table of available board models and accessories.

2 Block diagram

The DT5560SE/R5560/R5560SE is based on **Xilinx Zynq-7000**, a powerful System-on-Chip (SoC) that uses a large programmable logic for massive real-time signal processing and a Dual Core 1 GHz ARM processor for Ethernet communication and data post-processing.

All models of the family are built around the same DAQ section, reading out 32 analog channels: the DT5560SE hosts a single DAQ section in a desktop enclosure, while the R5560/R5560SE hosts four of them in a 19" rack unit, for a total of 128 channels, each section also driving 32 configurable digital LVDS I/Os on a VHDCI connector.

2.1 DT5560SE

The DT5560SE is organized in the main sections shown in **Fig. 2.1**:

- the **DAQ** section is the core of the system and it implements the analog-to-digital conversion, running the custom firmware generated by the SCI-Compiler. The DAQ section hosts readout interface (Optical Link, 1 Gb Ethernet) and a JTAG mini-USB connector, that allows for direct access to the FPGA for on-the-fly firmware upload and debug. The JTAG also gives access to the integrated USB-to-UART converter, which is useful to interface with a Linux console.
- the DAQ is connected to two **analog frontend boards (AFE)**. The first AFE manages the analog input channels from 0 to 15 while the second the channels from 16 to 31. The AFE's function is to amplify the signal, regulate the offset, select the impedance, the input division and the shaping of the signal.
- the **baseboard** is used to interconnect all the remaining sections. It features a small FPGA that is used to route the SYNC connectors, LEMO connectors and the DAQ sync signals that make it possible to configure the interconnection between the DAQ and digital I/Os. The baseboard hosts the mini-USB connector to connect the DAQ to the computer with a USB cable.

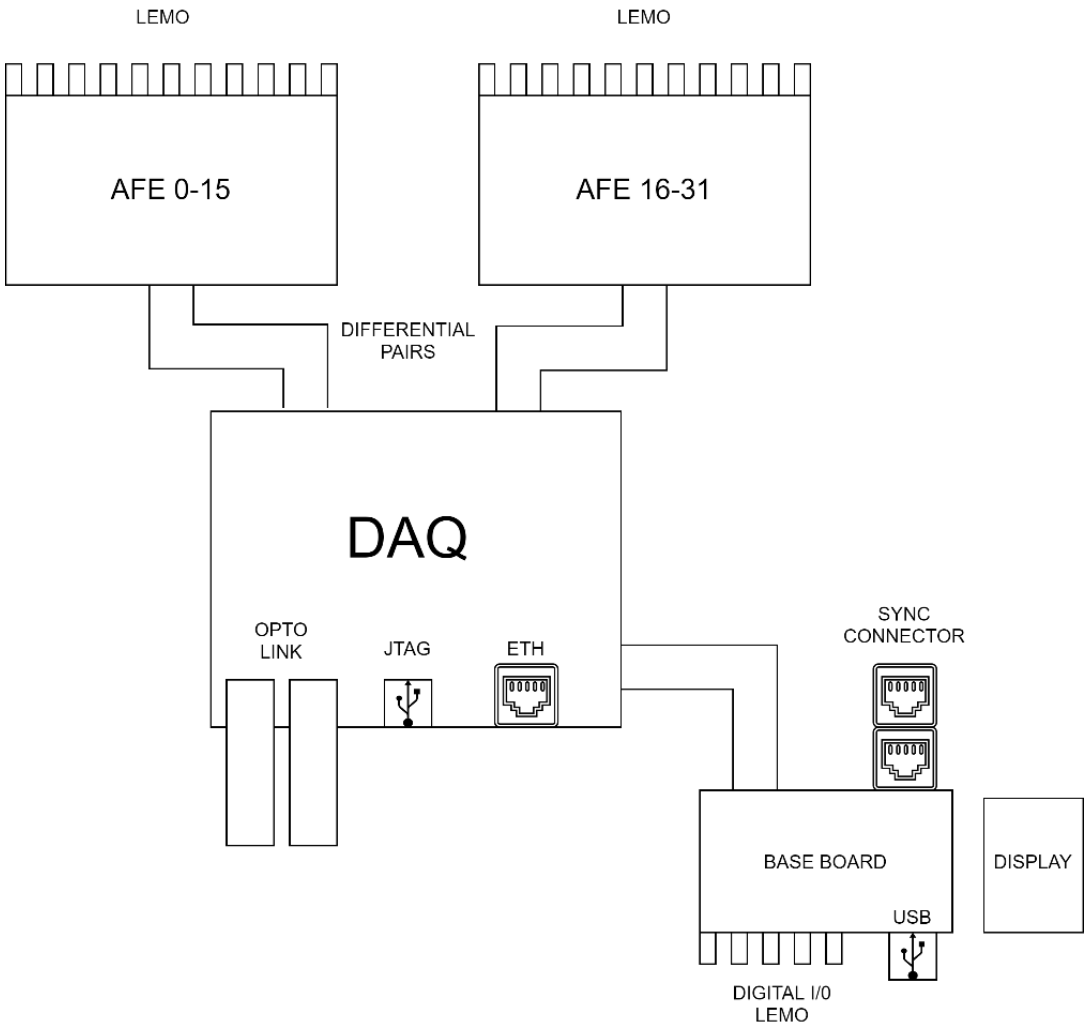


Fig. 2.1: DT5560SE block diagram.

2.2 R5560/R5560SE

The R5560/R5560SE is organized as shown in **Fig. 2.2**:

- **4 DAQ sections**: the DAQ section is the core of the system and it implements the analog-to-digital conversion, running the custom firmware generated by the SCI-Compiler. The DAQ section hosts readout interface (Optical Link, 1 Gb Ethernet) and a JTAG USB connector, that allows for direct access to the FPGA for on-the-fly firmware upload and debug. The JTAG also gives access to the integrated USB-to-UART converter, which is useful to interface with a Linux console.
- **8 AFE sections (R5560SE only)**: each DAQ is connected to two analog frontend boards (AFE). The first AFE manages the analog input channels from 0 to 15 while the second the channels from 16 to 31. The AFE's function is to amplify the signal, regulate the offset, select the impedance, the input division and the shaping of the signal. The R5560 has no AFE: its differential inputs feed the ADCs directly.
- the **baseboard** is used to interconnect all the board sections. It features a system manager to route digital signals from the DAQ to the SYNC connectors and USER IOs (and vice versa). It hosts the USB 3.0 and 1 Gbps Ethernet ports for slow control and cumulative readout of the 4 DAQ sections.

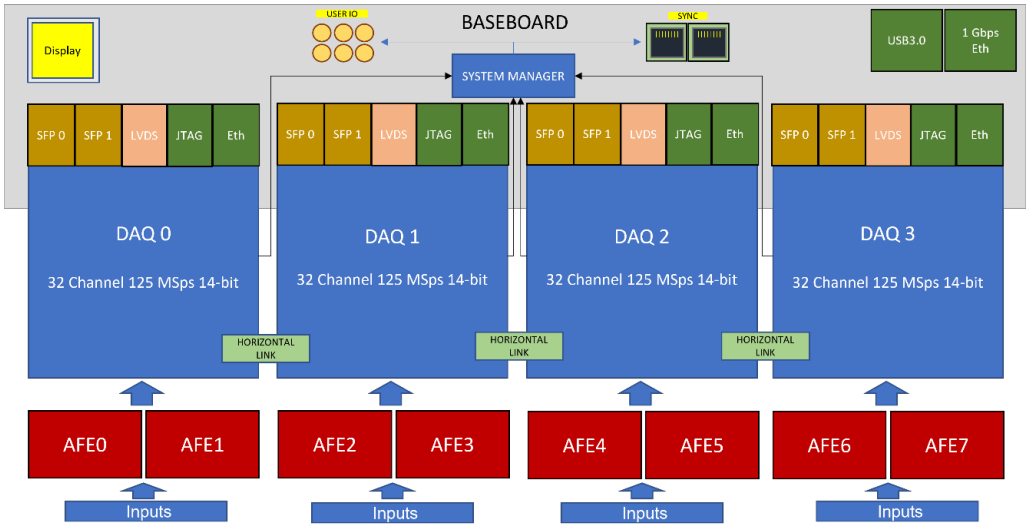


Fig. 2.2: Block diagram of the R5560/R5560SE. AFE sections are implemented in the R5560SE version only.

3 Technical specifications

TECHNICAL		
POWER REQUIREMENTS	**Voltage** 100-240 Vac ± 10% **Frequency** 50/60 Hz **Typ. power consumption** 0.3 A @ 220 Vac (DT5560SE), 0.5 A @ 220 Vac (R5560/R5560SE) **Fuse** 2 A, 250 V – 5 mm x 20 mm	
ANALOG INPUT DT5560SE	**Channels** 32 single-ended inputs on LEMO **Impedance** 50 Ω / 1 kΩ programmable **Full scale range** [0.015 Vpp: 1.5 Vpp]	**Bandwidth** 60 MHz **Analog coarse gain** [x1:x100] Programmable DC offset adjustment on each input in the full scale range
ANALOG INPUT R5560	**Channels** 128 differential inputs **Bandwidth** 125 MHz **Full scale range** 2 Vpp	**Connector** 32x CAT5e RJ45 **Impedance** Z_{diff} = 100 Ω
ANALOG INPUT R5560SE	**Channels** 128 single-ended inputs on MCX **Impedance** 50 Ω / 1 kΩ programmable **Full scale range** [0.015 Vpp: 1.5 Vpp]	**Bandwidth** 60 MHz **Analog coarse gain** [x1:x100] Programmable DC offset adjustment on each input in the full scale range
DIGITAL I/Os R5560/R5560SE	**Channels** 4 x 32 differential **Signal type** LVCMOS 2.5 V / LVDS / BLVDS **Coupling** DC	**Connector** 4x VHDCI **Impedance** Z_{diff} = 100 Ω
USER I/Os	**USER IO (LEMO)** Programmable Digital I/Os, function stated at firmware level. Can be used as Trigger, Start, Busy. Single-ended, Zin / Rt = 50 Ω, DC coupling **Count** 3 IN/OUT pairs (IN0...2, OUT0...2) on every model	
DIGITAL CONVERSION	**Resolution** 14 bits	**Sampling rate** 125 MS/s simultaneously on each channel
CLOCK GENERATION	**125 MHz ADC clock** Clock sources: internal/external Internal 25 MHz oscillator External 25 MHz clock: through programmable LEMO or SYNC connector	
TRIGGER	**Trigger source** Internal/External: managed by the default firmware Complex trigger logic: implementable by the user on the open FPGA **Trigger timestamp** Default FW: 32-bit counter, 8 ns resolution, 26-day range Custom FW: defined by the firmware design	**Trigger propagation** Through USER I/Os and Sync Connector

SYNCHRONIZATION	**Clock propagation** USER I/Os connectors SYNC Connector	**Acquisition synchronization** Through programmable LEMO Through dedicated SYNC Connector
	The SYNC connector makes it possible to cascade multiple units and synchronize them with a single standard CAT5e cable.	
FPGA	**Open FPGA** 1x Xilinx Zynq-7000 SoC Z-7030 (DT5560SE) 4x Xilinx Zynq-7000 SoC: Z-7030 (R5560A, R5560SEA) Z-7035 (R5560B, R5560SEB)	
MEMORY	1 GByte of memory for list readout on each SoC Up to 8 kS/ch for simultaneous waveform readout	
COMMUNICATION INTERFACE DT5560SE	**Ethernet** 1 Gbps **Optical Link** Slots for 2 x 10 Gbps SFP+ transceivers (communication protocol not implemented by default)	**USB 2.0** 1x mini-USB
	The different readout interfaces make it possible to integrate the DT5560SE in existing experimental environments.	
COMMUNICATION INTERFACE R5560/R5560SE	**Ethernet (readout)** 4 x 1 Gbps (4 Gbps cumulative speed) **Ethernet (slow control)** 1 Gbps (additional port, not required if the fast readout Ethernet is used) **Optical Link** Slots for 8 x 10 Gbps SFP+ transceivers (communication protocol not implemented by default)	**USB 3.0** 1x USB 3.0 readout **Parallel TTL/LVDS readout** Readout through the VHDCI digital I/O on custom protocol
	All readout interfaces can perform the same task at different speeds and using different media. They can be used independently or simultaneously. The different readout interfaces make it possible to integrate the R5560/R5560SE in existing experimental environments.	
FIRMWARE	**Default** - Waveform recording and Pulse Height Analysis - Ethernet communication (also USB on the DT5560SE)	**Custom** Use SCI-Compiler to develop your own firmware! Firmware can be upgraded via Ethernet or USB 2.0 debugger (on-the-fly) SCI-COMPILER Open FPGA
SOFTWARE	**Open Hardware Readout Software** to manage the default firmware **SCI-Compiler** for custom firmware development	
MECHANICAL		
FORM FACTOR	DT5560SE – Desktop module R5560 – 19”, 2U Rack-mount R5560SE – 19”, 3U Rack-mount	
DIMENSIONS (W x H x D)	DT5560SE – 257 x 102 x 331 mm, excluding the handle bar R5560 – 482.0 x 88 x 367.0 mm (396 mm deep with handles) R5560SE – 482.0 x 132.5 x 366.0 mm (399.8 mm deep with handles)	
ENVIRONMENTAL		
ENVIRONMENTAL	Indoor use	
OPERATING TEMPERATURE	-20 °C ÷ 50 °C	
OPERATING HUMIDITY	25% ÷ 95% RH non condensing	

STORAGE TEMPERATURE	-30 °C ÷ +80 °C
STORAGE HUMIDITY	5% ÷ 90% RH non condensing
ALTITUDE	≤ 2000 m
POLLUTION DEGREE	2
OVERVOLTAGE CATEGORY	II
EMC ENVIRONMENT	Commercial and light industrial
IP DEGREE	IPX0 enclosure, not for wet location
REGULATORY	
COMPLIANCE	EMC: CE 2014/30/EU Electromagnetic compatibility Directive Safety: CE 2014/35/EU Low Voltage Directive

Tab. 3.1: Technical specifications for the DT5560SE / R5560 / R5560SE.

3.1 Operating limits

The following operating limits must be respected:

Net class	Connector	Unit	Min	Max
Power	/	Voltage	90 Vac	250 Vac
USER IO	LEMO	Voltage	-0.1 V	3.6 V
Digital lines	Digital 2.5 V	Voltage (3.3 V I/O bank selected) Current	0 V	2.7 V 10 mA
	LVDS	Voltage Current Common Mode	0.5 V 0.9 V	3 V 10 mA 1.75 V
	I2C/Serial	Voltage	0 V	3.5 V
	RS485/RS422	Voltage	0 V	+5 V
Analog Differential Input	RJ45	Voltage Common Mode V+, V- Absolute Range	 0.5 V 0.5 V	2 Vpp 3.8 V 3.8 V
Analog SE Input	MCX	Voltage Common Mode V+, V- Absolute Range	 0.5 V 0.5 V	1.5 Vpp 3.8 V 3.8 V

Tab. 3.2: Operating limits of the connectors.



WARNING: the digital lines are directly connected to the FPGA I/Os. Violation of the maximum absolute ratings illustrated in this document will likely destroy the FPGA. There is no buffer or protection on this line, which is necessary to preserve the possibility to operate at different voltages and with both single-ended and differential signals.



THIS DEVICE SHOULD BE INSTALLED AND USED BY SKILLED TECHNICIANS ONLY OR UNDER THEIR SUPERVISION

4 Packaging and compliancy

The DT5560SE is available as a desktop module housed in a 257 x 102 x 331 mm (W x H x D) aluminium case, with a handle bar adjustable in 30-degree increments by pressing the side mechanical button. The R5560/R5560SE is available as a 19" rack-mount module housed in an aluminium case, 2U high for the R5560 and 3U high for the R5560SE.

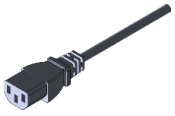
The unit is inspected by CAEN before the shipment, and it is guaranteed to leave the factory free of mechanical or electrical defects.

When receiving the unit, the user is strongly advised to inspect for any damage which may have occurred during transportation. Particularly, inspect for exterior damage like broken connectors and check that the panel is not scratched or cracked.

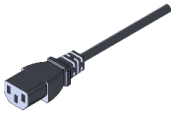
All packing material should be kept until the inspection has been completed. If damage is detected, the user must file a claim with the carrier immediately and notify CAEN.

Before installing the unit, make sure to read thoroughly the safety rules and installation requirements (see Chap. 16.3 and Chap. 8), then place the package content onto your bench.

The content of the delivered package depends on the model and standardly consists of the part list shown in **Tab. 4.1** for the DT5560SE and in **Tab. 4.2** for the R5560/R5560SE. All the official documentation, firmware updates, software tools, and accessories are available on www.caen.it at the product web page.

	Part	Description	Qty
	DT5560SE	32 channels 14-bit 125 MS/s Open FPGA Digitizer	x1
	Power supply cable	Standard C13 power supply cord	x1
	USB 2.0 cable	USB 2.0 type A-Mini B, I/O cable L = 1 MT	x2
	Ethernet cable	Ethernet CAT6 cable L = 2 MT	x1
	User guide	UM6952 – DT5560SE / R5560 / R5560SE User Manual	x1

Tab. 4.1: Delivered kit, DT5560SE.

	Part	Description	Qty
	R5560/R5560SE	128 Ch. 14 bit 125 MS/s Digitizer	x1
	Power supply cable	Standard C13 power supply cord	x1
	USB cable	USB 2.0 type A-B, I/O cable L = 2 MT	x1
	Ethernet cable	Ethernet CAT6 cable L = 2 MT	x4
	User guide	UM6952 – DT5560SE / R5560 / R5560SE User Manual	x1

Tab. 4.2: Delivered kit, R5560/R5560SE.



TO MANAGE THE PRODUCT, CONSULT THE OPERATING INSTRUCTIONS PROVIDED

It is recommended to:

- Inspect containers for damage during shipment. Report any damage to the freight carrier for possible insurance claims.
- Check that all the components received match those listed on the enclosed packing list. (CAEN cannot accept responsibility for missing items unless we are notified promptly of any discrepancies.)
- Open shipping containers; be careful not to damage contents.
- Inspect contents and report any damage. The inspection should confirm that there is no exterior damage to the unit such as broken knobs or connectors and that the front panel and display face are not scratched or cracked. Keep all packing material until the inspection has been completed.
- If damage is detected, file a claim with carrier immediately and notify CAEN service.
- If equipment must be returned for any reason, carefully repack equipment in the original shipping container with original packing materials if possible. Please contact CAEN service.
- If equipment is to be installed later, place equipment in original shipping container and store in a safe place until ready to install.



DO NOT SUBJECT THE ITEM TO UNDUE SHOCK OR VIBRATIONS



DO NOT BUMP, DROP OR SLIDE SHIPPING CONTAINERS



DO NOT LEAVE ITEMS OR SHIPPING CONTAINERS UNSUPERVISED IN AREAS WHERE UNTRAINED PERSONNEL MAY MISHANDLE THE ITEMS



USE ONLY ACCESSORIES WHICH MEET THE MANUFACTURER SPECIFICATIONS

4.1 SCI-Compiler license

The DT5560SE/R5560/R5560SE is compatible with SCI-Compiler (**[RD1]**, **[RD2]**), the Windows-based CAEN **firmware generator and compiler** for **easy FPGA programming**. It is an **automatic code generator** that, starting from a graphical block diagram, generates a VHDL piece of code that implements the required function. The software uses a prebuilt library set containing macroblocks implementing complex functions (MCA, Oscilloscope, Digitizer, TDC) that the user can connect to each other. SCI-Compiler also installs the **SciSDK**, the Software Development Kit to develop custom DAQ software in **Windows** and **Linux** (see Sec. 14.2).



Note: The SCI-Compiler full version works upon the purchase of a SCI-Compiler license (SW555 – SCI-Compiler PRO License), contact CAEN at info@caen.it for more information.



Note: The SCI-Compiler **Runtime License**, needed by the board to run a firmware generated by SCI-Compiler, is included with every DT5560SE/R5560/R5560SE unit. Without a Runtime License, a firmware generated by SCI-Compiler runs for 30 minutes only.



Note: A 30-day trial license of SCI-Compiler can be requested from the Licenses section of the Sci-Compiler portal; the trial does not include the Remote Customization Service (see **[RD1]**).

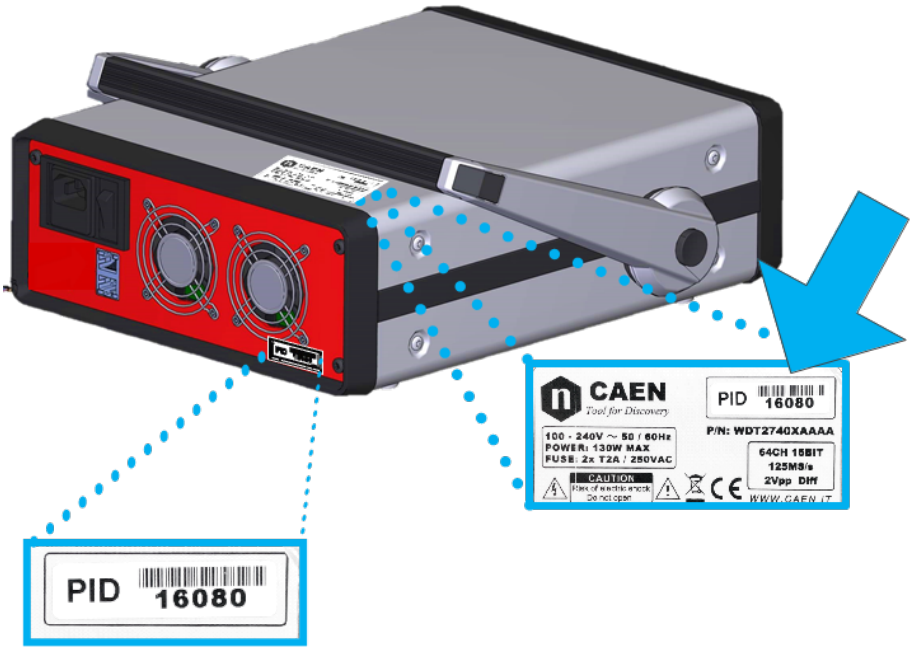
The SCI-Compiler license is a digital license, activated on the Sci-Compiler portal, reached from the My-CAEN+ area, with the coupon received by e-mail. Refer to **[RD1]** and **[RD2]** for the activation procedure and for more details.



Note: The user is not allowed to use the code generated by the SCI-Compiler on boards different from the DT5560SE/R5560/R5560SE. Even using a small part of the code generated by the SCI-Compiler on a custom design board or other products is an explicit violation of the license terms and it is an offense against CAEN S.p.A. and Nuclear Instruments S.R.L.

5 PID (Product Identifier)

PID is the CAEN product identifier, an incremental number greater than 10000 that is unique for each product. The PID is on a label affixed to the product (**Fig. 5.1**) and it is also stored in an on-board non-volatile memory readable via Touchscreen, Web interface or Readout Software (see Chap. 11, Chap. 12 and Chap. 15).



(a) Desktop unit (DT5560SE).



(b) Rack unit (R5560/R5560SE).

Fig. 5.1: PID location (the PID in the pictures and the device models are purely indicative). On the desktop unit the PID can be found on the top side of the module, together with the plate data of the module, and/or on the rear panel.

6 Power requirements

The DT5560SE/R5560/R5560SE is powered by an internal 100/240 V-12 V AC/DC stabilized power supply.

The power supply cord is included in the delivered kit and it is connected to equipment using an IEC C14 connector on the back panel. The power cord is a standard wall IEC C13.

7 Cooling management

The DT5560SE/R5560/R5560SE board can operate in the temperature range -20 ÷ +50 °C.

Air flow fans are installed onboard. The user must take care to provide a proper cooling to the board with external fan if the board is used in an enclosure or if the board is installed in a setup with poor air flow.

Excessive temperature will, in first instance, reduce the performance and the quality of the measurements and can also damage the board.

The R5560/R5560SE unit dissipates almost 100 W.

Note: Please do not close the rear fan holes to avoid unit overheating. If multiple R5560/R5560SE units are installed in a single rack tower, please consider external fans or a rack-mounted air conditioning system.

If the board is stored in a cold environment, please check for water condensation before powering on.

The board has not been tested for radiation hardness. High energy particles can be a source of soft errors and can damage the FPGA. If used in strong proton or neutron beams, arrange proper shielding or move the sensor away with a custom cable.

7.1 Cleaning the air vents

CAEN recommends occasionally cleaning the air vents on all vented sides of the board. Lint, dust, and other foreign matter can block the vents and limit the airflow. Be sure to unplug the board before cleaning the air vents and follow the general cleaning safety precautions.

8 Installing the device

The installation procedure depends on the form factor: the DT5560SE is a desktop unit, while the R5560/R5560SE is installed in a 19” rack.

8.1 Desktop unit (DT5560SE)

1. Connect the power supply input of the DT5560SE to the delivered C13 cord and connect the cord to the wall plug.
2. Set the rear switch to the ON position.

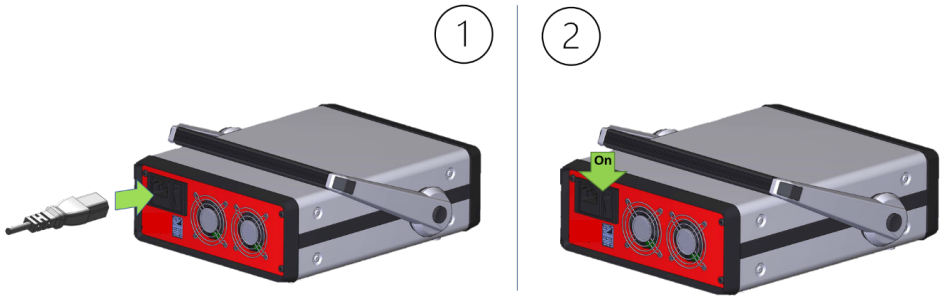


Fig. 8.1: Installing the DT5560SE.

8.2 Rack units (R5560/R5560SE)

1. Ensure you have the correct bolts needed to secure the device in your rack (bolts not included).
2. Carefully slide the system into an available rack space: 2U for the R5560, 3U for the R5560SE.
3. Screw the four locking bolts.
4. Connect the power supply cord to the AC power supply C14 inlet connector on the rear panel.

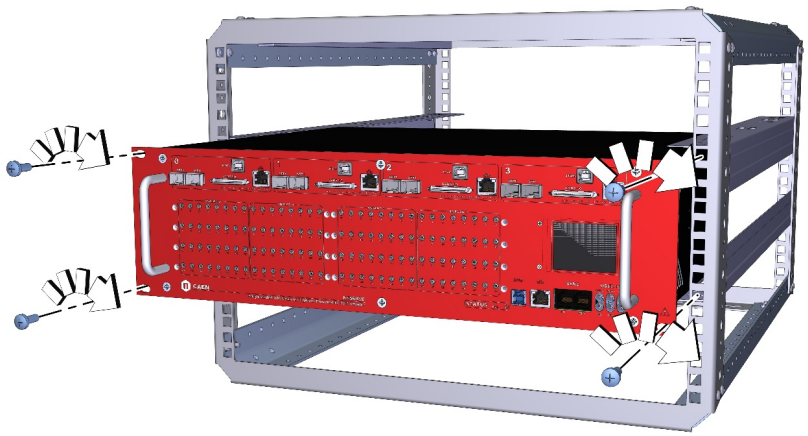


Fig. 8.2: Installing the device into a 19” rack.

8.3 Safety precautions

ONLY QUALIFIED PERSONNEL SHOULD PERFORM INSTALLATION OPERATIONS

DO NOT INSTALL THE EQUIPMENT SO THAT IT IS DIFFICULT TO OPERATE THE ON/OFF SWITCH OR TO ACCESS THE POWER CONNECTION ON THE REAR PANEL

IT IS RECOMMENDED THAT THE SWITCH OR CIRCUIT-BREAKER IS NEAR THE EQUIPMENT

THE SAFETY OF ANY SYSTEM THAT INCORPORATES THE DEVICE IS UNDER THE RESPONSIBILITY OF THE ASSEMBLER OF THE SYSTEM

Do not use the device and contact technical support if one of these situations is verified:

- Enclosure integrity is compromised
- Insulation of HV cord is damaged (if present)
- The indication LED or display is not performing as required (e.g. LED not working, display with incorrect graphic)
- Fans are not working (if present)

9 Panels description

9.1 Dimensions and panel views

9.1.1 DT5560SE

The DT5560SE unit is housed in a desktop aluminium case with an adjustable handle bar.



Fig. 9.1: DT5560SE front panel view and dimensions.



Fig. 9.2: DT5560SE rear panel view.

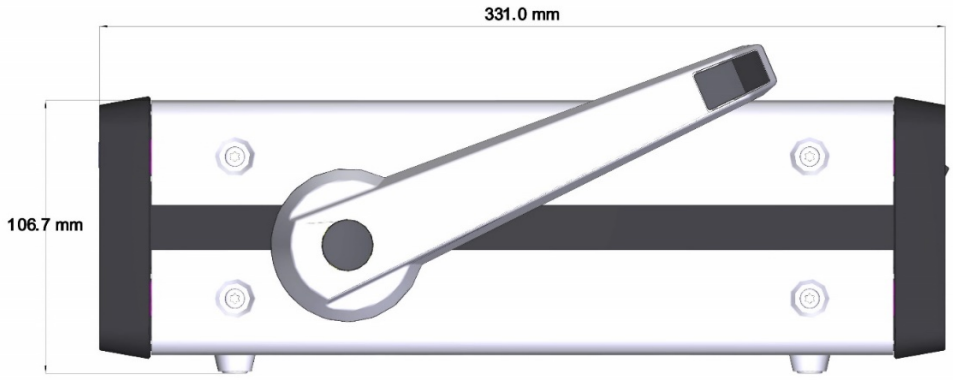


Fig. 9.3: DT5560SE side view and dimensions.

9.1.2 R5560

The R5560 unit is housed in a standard 2U 19” rack-mount aluminium box.

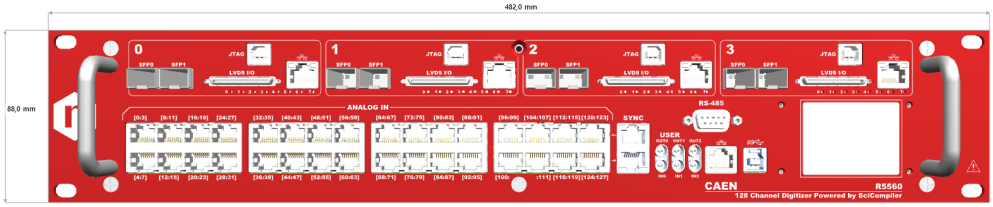


Fig. 9.4: R5560 front panel view and dimensions.

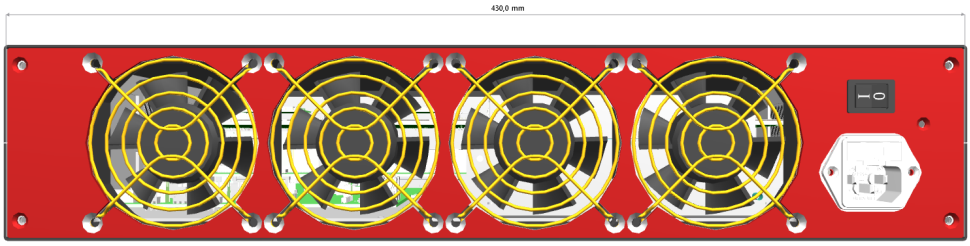


Fig. 9.5: R5560 rear panel view and dimensions.

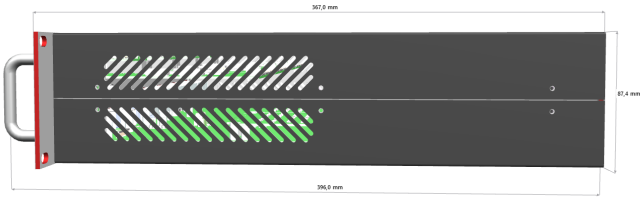


Fig. 9.6: R5560 side view and dimensions.

9.1.3 R5560SE

The R5560SE unit is housed in a standard 3U 19” rack-mount aluminium box.

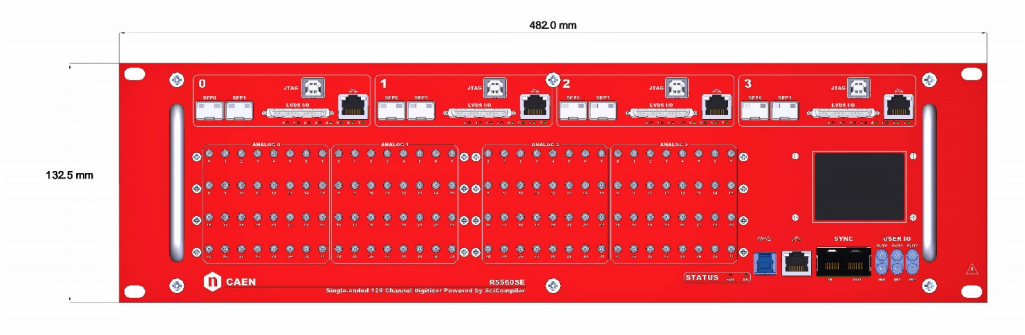


Fig. 9.7: R5560SE front panel view and dimensions.

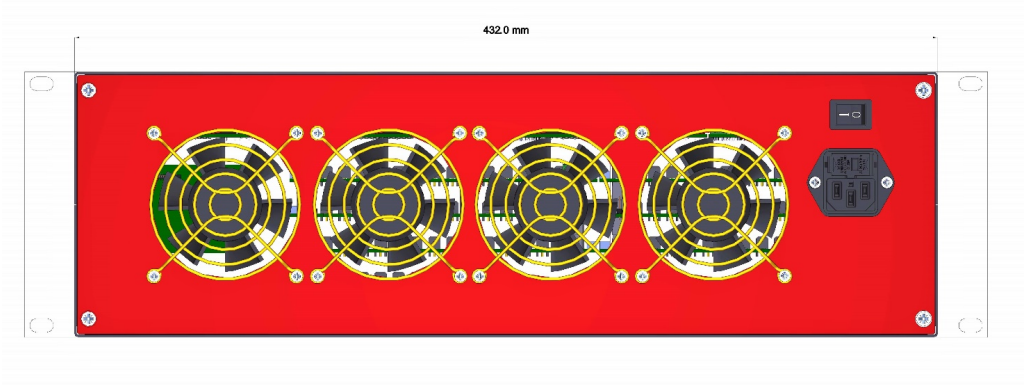


Fig. 9.8: R5560SE rear panel view and dimensions.

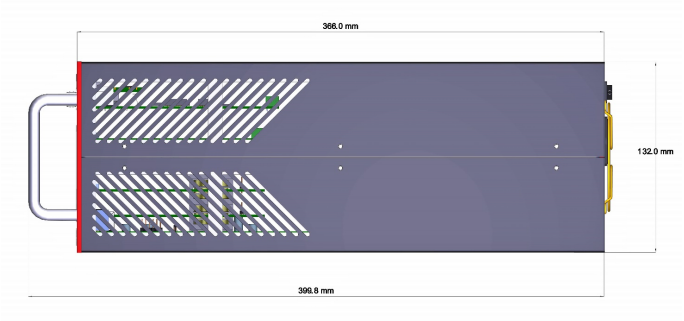
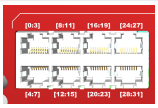


Fig. 9.9: R5560SE side view and dimensions.

9.2 Front panel

SINGLE-ENDED ANALOG INPUT – DT5560SE only		
	**FUNCTION** Input connectors from CH0 to CH31 receive the input analog signals. **ELECTRICAL SPECS** Input dynamics: [1.5 : 0.015] Vpp Input impedance: 50 Ω / 1 kΩ	**MECHANICAL SPECS** Series: LEMO

DIFFERENTIAL ANALOG INPUT – R5560 only

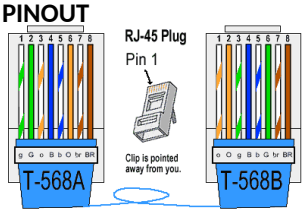


FUNCTION
8 x RJ45 connectors carrying 32 analog differential signals. Compatible with standard Ethernet cable CAT5e with standard Ethernet pinout.

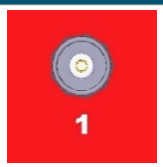
ELECTRICAL SPECS
Sign. type: differential (Analog 0...3 V Common Mode, 2 Vpp differential signal).
Coupling: DC.
Z_{diff}: 100 Ω.

MECHANICAL SPECS
Series: RJ45.
Suggested plug: Ethernet crimp plug.

PINOUT



SINGLE-ENDED ANALOG INPUT – R5560SE only



FUNCTION
Input connectors from CH0 to CH31 on four different sections. They receive the single-ended input analog signals.

ELECTRICAL SPECS
Input dynamics: [1.5 : 0.015] Vpp
Input impedance: 50 Ω / 1 kΩ

MECHANICAL SPECS
Series: MCX

JTAG



DT5560SE



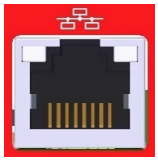
R5560/R5560SE

FUNCTION
Connector for direct FPGA access in order to perform fast firmware download and FPGA signal monitoring and probing.

ELECTRICAL SPECS
Standard: compliant with USB 2.0

MECHANICAL SPECS
Series:
- DT5560SE: mini-USB
- R5560/R5560SE: USB 2.0 – B

ETHERNET



FUNCTION
1 Gbps Ethernet interface:
- DT5560SE: 1 port
- R5560/R5560SE: 4 ports for high-speed readout, one for each DAQ section, plus 1 port for slow control

Slow control communication is not implemented in the default firmware.

ELECTRICAL SPECS
Not available

MECHANICAL SPECS
Series: RJ45

OPTICAL LINK



FUNCTION
Slot for SFP+ connector. A 10 Gbps optical transceiver can be inserted here. The DT5560SE has 2 slots, the R5560/R5560SE has 8, two per DAQ section.

SFP+ transceivers are not provided with the board kit. Optical link communication is not implemented in the default firmware.

ELECTRICAL SPECS
Not available

MECHANICAL SPECS
Series: SFP+ socket

LVDS I/O – R5560/R5560SE only

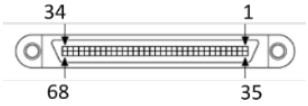


FUNCTION
68-pin connector carrying 32/64 differential input digital signals.

ELECTRICAL SPECS
Sign. type: differential (LVCMOS 2.5 V, LVDS, BLVDS).
Z_{diff}: 100 Ω

MECHANICAL SPECS
Series: VHDCI connector.
Type: 71430-0008.
Manufacturer: TE Connectivity.
Suggested plug: 68-pin VHDCI.

PINOUT



LEDs

FUNCTION
8 user-configurable LEDs.

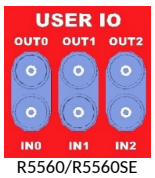
ELECTRICAL SPECS
Not available

MECHANICAL SPECS
Not available

USER IOs



DT5560SE



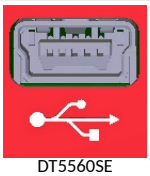
R5560/R5560SE

FUNCTION
General purpose digital I/O connectors. Their function is defined at firmware level. Every model has 3 input/output pairs (IN0...2, OUT0...2).

ELECTRICAL SPECS
NIM/TTL

MECHANICAL SPECS
Series: LEMO

USB


DT5560SE



R5560/R5560SE

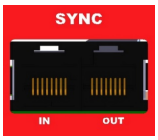
FUNCTION
USB connector for data readout: USB 2.0 on the DT5560SE, USB 3.0 on the R5560/R5560SE.

USB 3.0 communication is not implemented in the default firmware of the R5560/R5560SE.

ELECTRICAL SPECS
Standard: compliant with USB 2.0 (DT5560SE) or USB 3.0 (R5560/R5560SE)

MECHANICAL SPECS
Series:
- DT5560SE: mini-USB
- R5560/R5560SE: USB 3.0 – B

SYNC – R5560/R5560SE only



FUNCTION
2 x RJ45 connectors carrying 3 sync (CLOCK, T0, RUN) differential signals for each connector (IN-/OUT).

ELECTRICAL SPECS
Compatible with standard Ethernet cable.
Sign. type: differential (LVDS).
Coupling: DC.
Z_{diff}: 100 Ω

MECHANICAL SPECS
Series: RJ45.
Suggested plug: Ethernet crimp plug.
Suggested cable: CAT5e or CAT6 Ethernet cable.

RS-485 PREAMPLIFIER BOX – R5560 only



FUNCTION
RS-485 interface to remotely control the preamplifier gain/shaping time.

RS-485 communication is not implemented in the default firmware.

ELECTRICAL SPECS
Not available

MECHANICAL SPECS
Series: D-SUB9

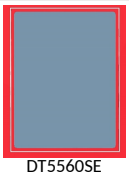
STATUS LEDs – DT5560SE/R5560SE only



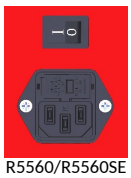
FUNCTION
FAIL: if ON, firmware not correctly uploaded on-board.
OK: if ON, board correctly operating.

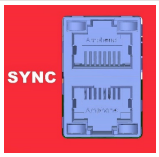
ELECTRICAL SPECS
Not available

MECHANICAL SPECS
Not available

DISPLAY		
 DT5560SE  R5560/R5560SE	**FUNCTION** 2.4” Touchscreen Display for board control and monitoring. **ELECTRICAL SPECS** Not available	**MECHANICAL SPECS** Not available

9.3 Rear panel

AC 220 V and ON/OFF switch		
 DT5560SE  R5560/R5560SE	**FUNCTION** Input connector for the main power supply (100-240 V) and ON/OFF switch. **ELECTRICAL SPECS** Not available	**MECHANICAL SPECS** Series: AC power supply C14 inlet connector

SYNC – DT5560SE only		
	**FUNCTION** 2 x RJ45 connectors carrying 3 sync (CLOCK, T0, RUN) differential signals for each connector (IN-/OUT). **ELECTRICAL SPECS** Compatible with standard Ethernet cable. Sign. type: differential (LVDS). Coupling: DC. Z_{diff}: 100 Ω	**MECHANICAL SPECS** Series: RJ45. Suggested plug: Ethernet crimp plug. Suggested cable: CAT5e or CAT6 Ethernet cable.

10 Functional description

The DT5560SE/R5560/R5560SE is a flexible board meant to be used as a fully programmable readout system for different applications. In the following we give a detailed description of the board hardware and its operation.

All three models are built around the same DAQ section, which digitizes and processes 32 analog channels: the DT5560SE hosts one of them, the R5560/R5560SE hosts four. Everything that is said below about a DAQ section therefore applies to every model; what changes is how many there are and what surrounds them.

10.1 System architecture

10.1.1 DT5560SE

The DT5560SE integrates a single System-on-Chip (SoC) with configurable FPGA and dual core ARM-A9 processor, hosted on the DAQ board, plus a base board carrying the connectors, the power supply circuits, the clock and sync logic and the touchscreen display. Two AFE boards condition the 32 single-ended analog inputs. There is no Slow Control and Sync Processor and no horizontal link, both of which exist on the R5560/R5560SE to coordinate its four DAQ sections.

10.1.2 R5560/R5560SE

The R5560/R5560SE integrates four System-on-Chip (SoC) devices with configurable FPGA and dual core ARM-A9 processors. The FPGA of each chip is devoted to the signal acquisition and real-time processing while the ARM processor can run C/Python applications that operate on the data pre-processed by the FPGA. This allows executing complex real-time algorithms that require operating on less data but with several operations. For these purposes, a large quantity of memory is available to store list mode data (up to 1 Gbyte per processor).

For example, it is possible to implement in the FPGA triggers and pulse shape discrimination algorithms for Neutron/Gamma discrimination. On the processor side, it is possible to implement, for instance, a Neural Network that performs Neutron/Gamma discrimination with a separation capability much higher than the algorithm operating in real-time in the FPGA. The Neural Network will never be able to sustain MHz rates, while the FPGA will easily manage this amount of data. When the FPGA is unable to distinguish a pulse, it is possible, for example, to transfer that particular pulse to the ARM processor for a more advanced analysis: 90% of the events will be processed in real-time by the FPGA while the remaining 10% will be queued in a buffer for further investigation by the ARM processor.

Each ARM processor controls one dedicated 1 Gbps Ethernet link, allowing extremely easy development of custom communication interfaces. The development of the FPGA processing algorithm can be done with SCI-Compiler, while the development of the ARM part of the firmware can be done in Eclipse using the available tools provided with the Virtual Machine.

The R5560/R5560SE is built on several sub-boards:

- **Base board**: it hosts all connectors, the main power supply circuits, the system monitor and the logic for synchronization. A SoC manages the slow control, the interface between the DAQ modules, the generation and distribution of the clock and sync signals. The board also controls a touchscreen display with a simple GUI to configure and control the system.
- **4 DAQ boards**: this is the programmable core of the system. Each one digitizes and processes in real-

time 32 independent channels. Each section hosts a large and powerful SoC based on a Zynq processor (7030 or 7035 model), 14-bit 125 MSPS ADCs, a 1 Gbps Ethernet link, two optical links, LVDS I/Os and an integrated USB 2.0 JTAG.

- **8 AFE boards (R5560SE only)**: this board performs signal conditioning in order to convert the single-ended inputs into differential ones. It controls the signal shaping, gain, offset and input impedance.

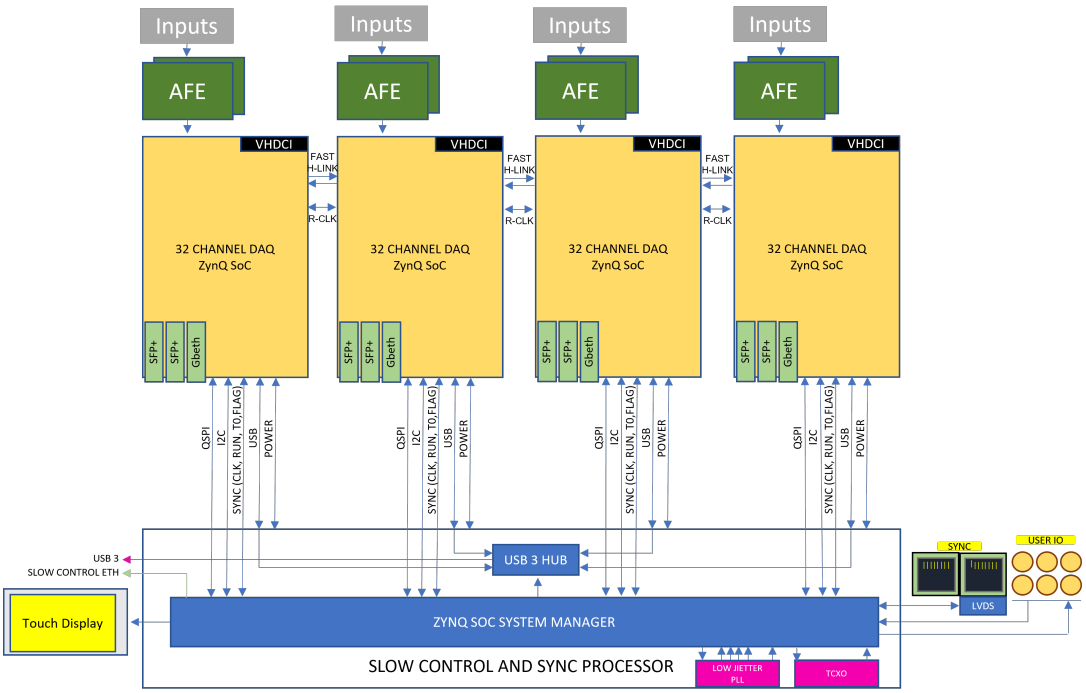


Fig. 10.1: The complete block diagram of the R5560SE. The R5560 works the same without the AFE boards.

Fig. 10.1 shows the full scheme of the R5560SE. Each DAQ board is an independent unit (with independent firmware) processing up to 32 analog channels. The analog signals come from the AFE board as a differential signal with 2 Vpp dynamic and 1 V common mode. The ADCs convert independently each channel and provide the digital streams as input to the SoC. The SoC manages the real-time processing and high-speed communication with external facilities through Gigabit Ethernet or optical link. Each DAQ section can exchange data with nearby sections (DX, SX) through a dedicated horizontal link (H-LINK). The bidirectional H-LINK offers 12.5 Gbps bandwidth and allows a significant calculation load reduction in the readout facility.

Note: Please find more details about the H-LINK management at `https://github.com/NuclearInstruments/r5560-daq-horizontal-interconnections`.

The Slow Control and Sync Processor (SCSP) is a SoC that manages the synchronization between multiple R5560/R5560SE units and an external system. Moreover, the SCSP is connected with a slow speed bus to all modules (I2C, 100 Mbps QSPI bus). This link allows the SCSP to collect data from each of the four DAQ sections, configure the memory-mapped registers and read all memory-mapped FIFOs or memories instantiated in the firmware. The SCSP has an Ethernet connector and exposes a server that allows readout of data and set/get of any register from a single connection. The bandwidth is limited to 100 Mbps even if the link is a 1 Gbps link, and it is meant to be used for debug purposes.

A USB 3.0 hub interconnects together all the SoCs, offering a 400 Mbit/s readout link per section on a single USB 3.0 connector.

A touchscreen display allows monitoring the status of the system, environmental parameters, and setting and reading the Ethernet configuration of all DAQ boards and of the SCSP board.

10.2 DAQ section

The DT5560SE is organized in a single DAQ section featuring 32 analog inputs; the R5560/R5560SE is organized in four DAQ sections, each one featuring 32 analog inputs for a total of 128 readout channels. Each section hosts 4 ADCs operating in simultaneous sampling at 125 MS/s, 14 bit.

The front panel clearly identifies the analog channels, grouping them with the relative digital signals. On the R5560/R5560SE the different sections can easily communicate with each other thanks to fast horizontal links.

Each DAQ board hosts a Xilinx Zynq-7000 (Z-7030 on the DT5560SE and on the R5560A/R5560SEA, Z-7035 on the R5560B/R5560SEB), a powerful System-on-Chip that uses a large programmable logic for massive real-time signal processing and a dual core 1 GHz ARM processor for data post-processing. The user can implement here the needed custom firmware, possibly exploiting the SCI-Compiler tool. The firmware is stored on a 1 GB flash memory while 1 GB DDR3 works as a primary memory for Linux and as cache for the data to be transferred from the FPGA to the LAN. The ARM microcontroller features all the software required to control and configure the system and to implement Ethernet communication.

On each DAQ board, there are also 2 optical links, capable of reaching up to 10 Gbit/s data transfer, for a total of 2 on the DT5560SE and 8 on the R5560/R5560SE. The optical link protocol can be configured by the user to interface the unit with external readout systems. The DAQ section also hosts a USB JTAG connector, in order to allow for on-the-fly firmware uploading and USB-to-UART bridge access. In this way it is possible to directly access the Linux console of the Zynq. The Zynq also has a USB interface connected to a USB connector (on the base board, in the bottom part of the instrument) that emulates a LAN over USB, in order to be able to read out and configure the board.

Each DAQ section is connected to the base board through 6 digital signals: 3 input sync-in and 3 output sync-out. These lines connect the FPGA of the Zynq to the FPGA on the base board. The user can configure the Zynq signals to be connected to LEMO I/Os or to the SYNC signal connectors.

The DAQ section is the part of the device in charge of the analog-to-digital conversion. The ADCs accept the analog signals coming from the input connectors. In the case of the R5560, the differential analog input directly feeds the ADCs, while in the DT5560SE/R5560SE each differential input of the ADCs is driven by the output of the analog frontend (AFE) section (see Sec. 10.2.2), that, in turn, accepts single-ended analog input signals.

The ADCs receive the clock either from the on-board generator or from the base board. The base board allows selecting different clock sources, for example, from a LEMO connector, dedicated SYNC connector or internal base board clock.

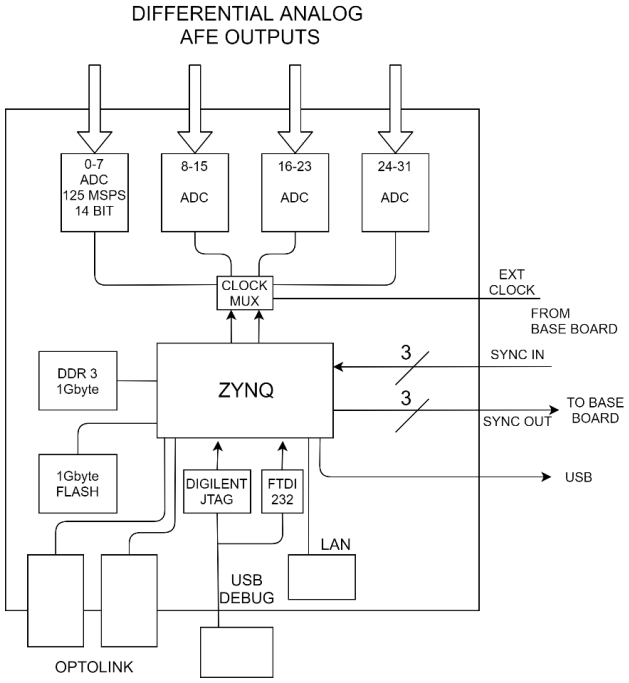


Fig. 10.2: Block diagram of a single DAQ section. Signals interconnecting the DAQ with the base board and AFE boards are shown. This diagram is identical on all models of the family.

10.2.1 R5560 differential inputs

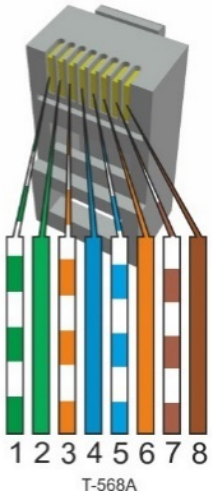
The R5560 has four groups of 8x RJ45 shielded connectors to carry analog signals, for a total of 128 differential analog inputs. Differential signals are used to minimize the cross-talk between channels and the noise, even with long cables carrying the analog inputs from the detectors. The pinout of the analog connector is shown in **Fig. 10.3**. This allows using standard CAT5e pre-crimped cables to connect differential signals to the board.

Pin cable	Color	Board ch/polarity
1	White-Green	2+
2	Green	2-
3	White-Orange	4+
4	Blue	3+
5	White-Blue	3-
6	Orange	4-
7	White-Brown	1+
8	Brown	1-

Tab. 10.1: Analog RJ45 connector cable pinout.

The R5560 inputs have the following characteristics:

- Bandwidth: 60 MHz
- Gain: 1 V/V
- Differential input dynamic: 2 Vpp
- Any common mode voltage is accepted if within the absolute maximum rating



Pin	Description	10base-T	100Base-T	1000Base-T
1	Transmit Data+ or BiDirectional	TX+	TX+	BI_DA+
2	Transmit Data- or BiDirectional	TX-	TX-	BI_DA-
3	Receive Data+ or BiDirectional	RX+	RX+	BI_DB+
4	Not connected or BiDirectional	n/c	n/c	BI_DC+
5	Not connected or BiDirectional	n/c	n/c	BI_DC-
6	Receive Data- or BiDirectional	RX-	RX-	BI_DB-
7	Not connected or BiDirectional	n/c	n/c	BI_DD+
8	Not connected or BiDirectional	n/c	n/c	BI_DD-

Fig. 10.3: RJ45 analog connector pinout: the pinout for the first group of channels [1:4] is shown. The same ordering can be applied to the subsequent groups.

- Absolute input dynamic: 0.5 V ... 3.8 V
- Differential input impedance: 100 Ω
- Single ended impedance: 50 Ω
- 50 μV integrated noise

As previously said, the R5560 accepts differential analog input signals, which have several advantages:

- Immunity to common mode noise
- Immunity to channel cross-talk
- Immunity to RF noise
- Immunity to ground-loops
- Immunity to ADC clock coupling
- With limited signal swing, it is possible to achieve a larger input dynamic

Differential signals can be directly connected to the R5560 RJ45 input connectors, according to the input parameters reported in **Tab. 10.2**. Both V_P and V_N must be connected.

Symbol	Description	Min	Max
V_{CM}	Input common mode	0.5	3.5
V_{DIFF}	Input differential signal	0	2 Vpp
V_P	Input positive terminal	0.5	3.5
V_N	Input negative terminal	0.5	3.5
R_{IN}	Input impedance		100

Tab. 10.2: Differential parameters for the R5560 analog input.

10.2.2 Analog frontend (DT5560SE/R5560SE only)

The DT5560SE/R5560SE features an advanced programmable analog frontend. There are two AFE sections for each DAQ section, managing the single-ended analog inputs and routing the signals to the ADCs of that DAQ section: 2 in total on the DT5560SE and 8 on the R5560SE. The first AFE of a section manages the analog input channels from 0 to 15, the second the channels from 16 to 31.

The analog frontend converts the input single-ended analog signal to a differential one, that fits at best the input dynamic range of the ADC converter. There are several stages inside the AFE board, as shown in **Fig. 10.4**.

- The first block allows the selection of the **input impedance** between 50 Ω and 1 kΩ.
- The second block allows dividing the signal by 5, if needed.
- The third block allows shaping the input signal, selecting among 1, 10, 30 μs shaping time or bypassing the shaper using DC coupling. The output of the shaper is connected to the offset block.
- The **offset** block allows adding or subtracting from the input signal baseline any value between -2 V and +2 V.
- The **gain** stage allows selecting the gain to apply to the input signal between 1 and 100. Possible gain values are distributed on a logarithmic scale and can be selected via software. The output of the gain stage is a differential signal that is connected to the ADC.

The analog input channels are grouped in pairs. Each pair of channels shares the configuration of the input impedance, division and gain, while the offset can be selected independently on each channel. The shaper can be configured at instrument level, meaning that the shaping time is the same on all channels.

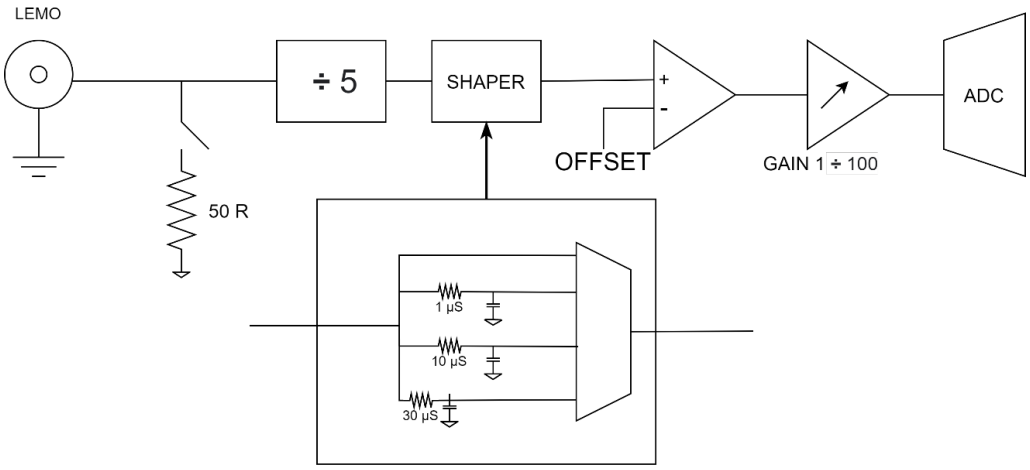


Fig. 10.4: Block diagram of an AFE section.

10.3 Digital stage

On every model, the general purpose digital I/O LEMO connectors can drive NIM/TTL signals to be used for different purposes. The function of the LEMO I/Os is defined at firmware level and can be customized by the user by exploiting open FPGA features via SCI-Compiler.

10.3.1 VHDCI connectors (R5560/R5560SE only)

On the R5560/R5560SE the digital stage is also available on the VHDCI connectors (LVDS I/Os) on the front panel.

Each VHDCI connector carries:

- 32 differential lines supporting the standards LVDS, BLVDS (or PSEUDO LVDS)
- 58 single ended CMOS 2.5 V from `DATA1_P/_N` to `DATA29_P/_N`
- 6 single ended CMOS 1.8 V from `DATA30_P/_N` to `DATA32_P/_N`
- 4x ground pins

The pinout of the digital VHDCI connector is shown in **Fig. 10.5**.

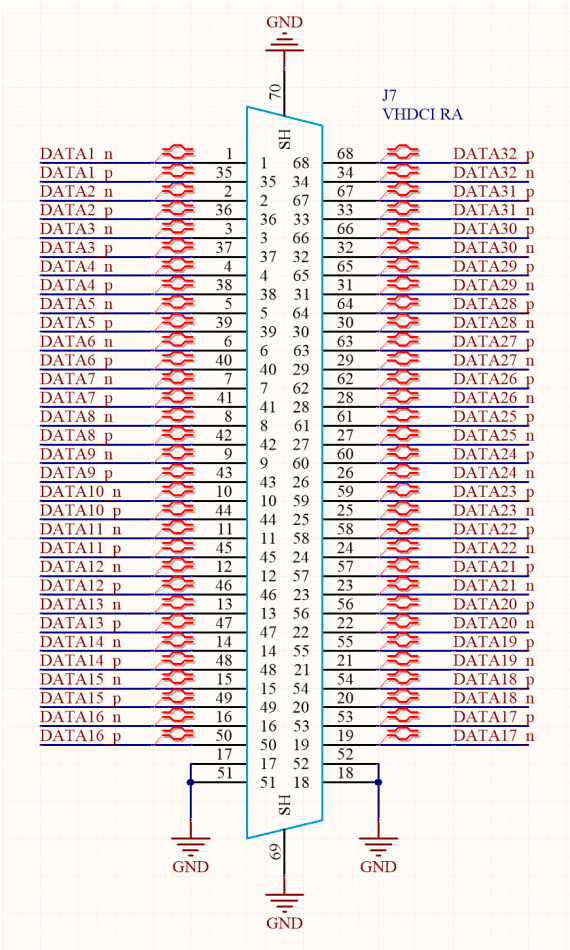


Fig. 10.5: Pinout of the LVDS digital connectors of the R5560/R5560SE.

Digital I/Os on the VHDCI connector can be used for several applications, for example:

- Fast parallel readout (protocol supported by SCI-Compiler)
- Input for triggers/veto
- Input from photon counting detector / ASIC
- Readout of digital ASIC, both with serialized output or parallel output

- Control of the readout process of analog ASIC

Using SCI-Compiler, it is possible to select the I/O standard and the pin direction (IN/OUT). Pin direction and signal standard must be selected at configuration time and cannot be changed in real-time.

To ensure fast communication (up to 200 MHz DDR, 12.8 Gbps) no buffer or over-voltage protection is present between FPGA and I/O. Short circuits or over-voltage will damage the FPGA.

In single ended mode the last 6 signals (DATA30, 31 and 32) are 1.8 V ONLY. CMOS 2.5 V or CMOS 3.3 V will damage the FPGA.

10.4 Data readout

The DT5560SE/R5560/R5560SE is designed to provide the maximum flexibility in experiments requiring hundreds of readout channels. Both the wide range of available communication protocols and the scalable architecture are designed to help users in building a multi-board system and in integrating it in existing facilities.

There are several ways to read out data from the board. Not all of them exist on every model: the DT5560SE offers the DAQ section Ethernet port, the USB 2.0 port and the optical links, while the R5560/R5560SE adds the slow control Ethernet port, the USB 3.0 readout, the optical daisy chain and the parallel LVDS readout.

10.4.1 Readout using DAQ section Ethernet port

Each DAQ section has a dedicated Ethernet link connected to the PS (processor side) of the SoC. This Ethernet port is fully managed by the user application that implements a custom protocol using TCP/IP or UDP sockets. A basic application is provided as an example to customize the protocol (see Chap. 16 for more details).

On the R5560/R5560SE a multi-port 10 Gbps or an Ethernet switch is required in order to read out the four sections.

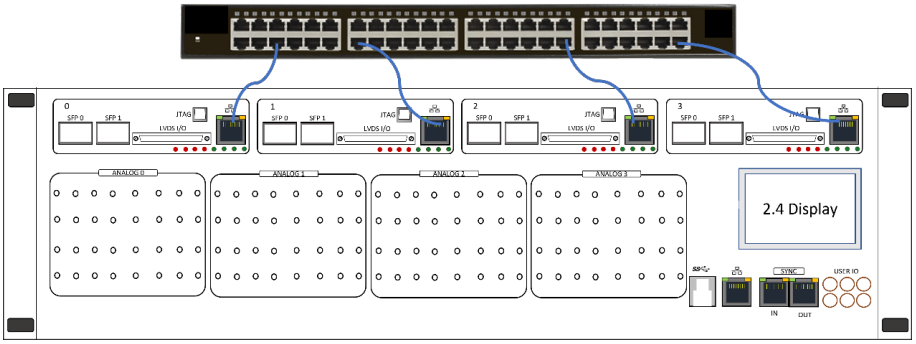


Fig. 10.6: Readout using the DAQ section Ethernet port, one port per section, through an external Ethernet switch.

10.4.2 Readout using slow control Ethernet port (R5560/R5560SE only)

The slow control Ethernet port allows implementing a low-speed readout. With a single Ethernet port it is possible to read out data from all channels and set/get registers. Please consider that everything can be

read out using this system, except the fast FIFO managed by the DMA.

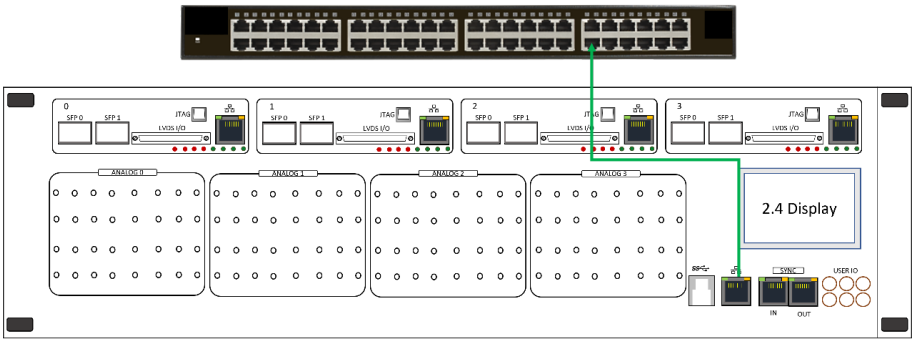


Fig. 10.7: Readout using the slow control Ethernet port on the SCSP board.

10.4.3 Readout using the USB 3.0 (R5560/R5560SE only)

The SoC exposes a virtual Ethernet card on the USB 3.0 connector using the RNDIS protocol. Connecting the USB 3.0 cable, the user will enumerate 5 new Ethernet interfaces on the computer. Each interface is a virtual Ethernet card with an IP address (configurable on the display). Max speed 300 Mbit/s per section.

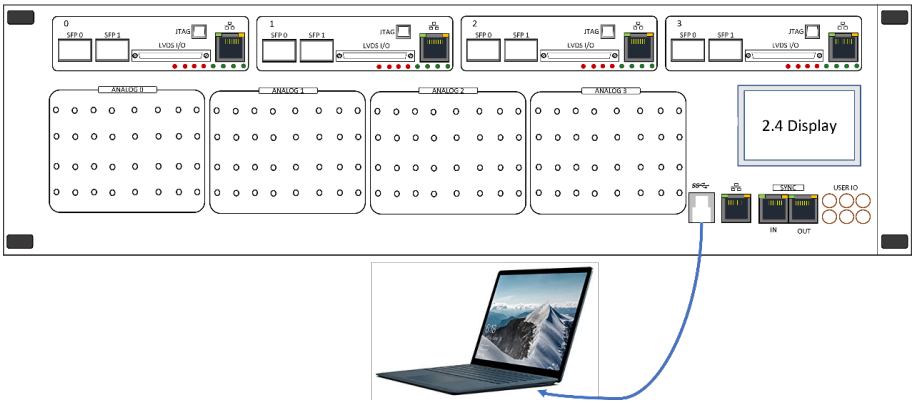


Fig. 10.8: Readout using the USB 3.0 connector.

10.4.4 Readout using optical link

The optical link allows obtaining the maximum bandwidth from the board. On the R5560/R5560SE, aggregating the 8 links at 12.5 Gbps, it is possible to reach a maximum readout speed of 100 Gbps.

A free open-source protocol is available and can be used to interface with external facilities (see Chap. 16 for more details).

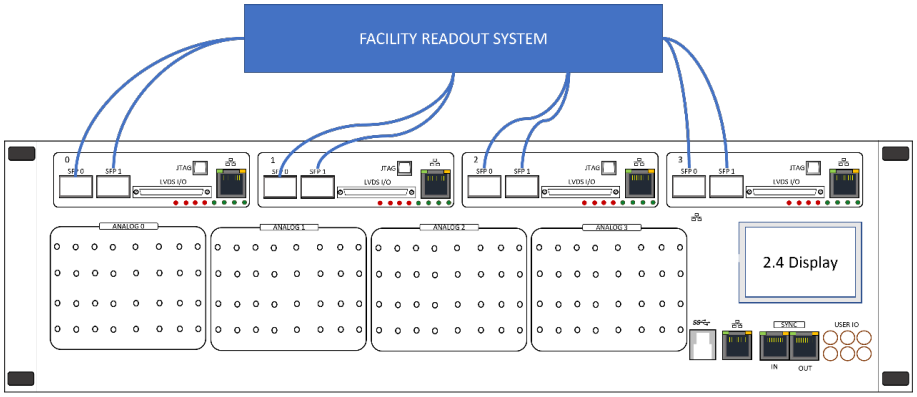


Fig. 10.9: Readout using the optical links, one link pair per DAQ section, connected to the facility readout system.

10.4.5 Optical daisy chain (R5560/R5560SE only)

Multiple R5560/R5560SE units can be daisy-chained in order to reduce the number of optical links. In daisy-chain mode the total bandwidth of the chain must be less than 12.5 Gbps.

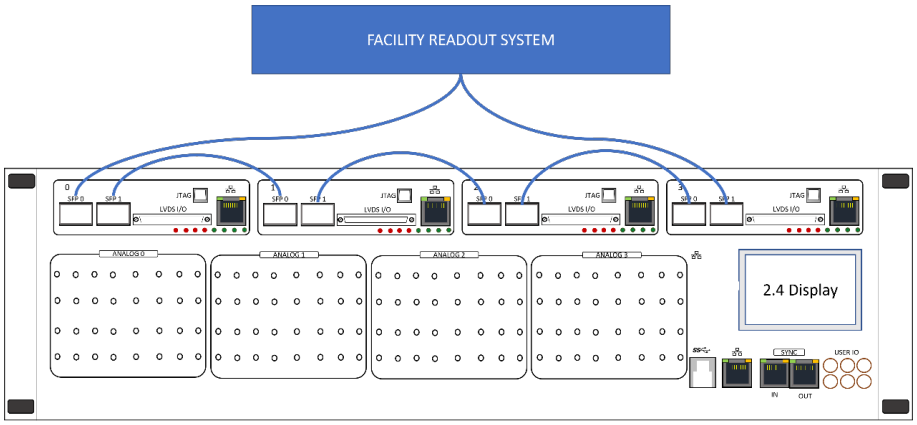


Fig. 10.10: Optical daisy chain readout topology.

10.4.6 Parallel readout (R5560/R5560SE only)

The R5560/R5560SE can be read out in parallel using the LVDS interface. The LVDS interface offers a 32-bit wide bus. The user must develop their own protocol using the available SDK.

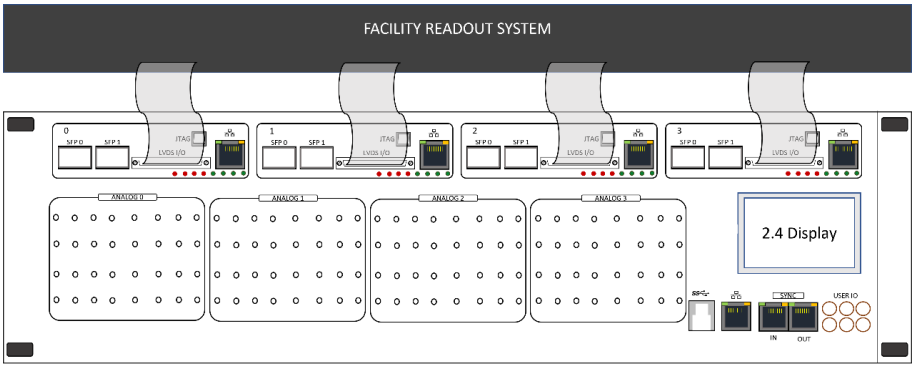


Fig. 10.11: Parallel readout using the LVDS interface of each DAQ section.

10.5 Clock distribution

The DT5560SE/R5560/R5560SE has a very versatile clocking network (see **Fig. 10.12**). It is possible to use different clock sources for the ADCs:

- the internal clock, generated on the base board
- the internal clock, generated on the DAQ board
- external clock

The DAQ board selects the 125 MHz clock to be provided to the ADCs between its internal clock and the clock provided by the base board. This is done by a selector (SEL) controlled by the FPGA. The FPGA is driven by the firmware loaded in the Zynq, overwriting any setting that has been configured via the web interface or via the display. The SEL signal can be configured using SCI-Compiler. When SEL is 0, the default value of the clock is provided to the ADC by the Zynq itself. When SEL is 1, the Zynq is bypassed, and the clock is provided directly by the base board.

The base board accepts and routes to the DAQ a clock signal, following these available options:

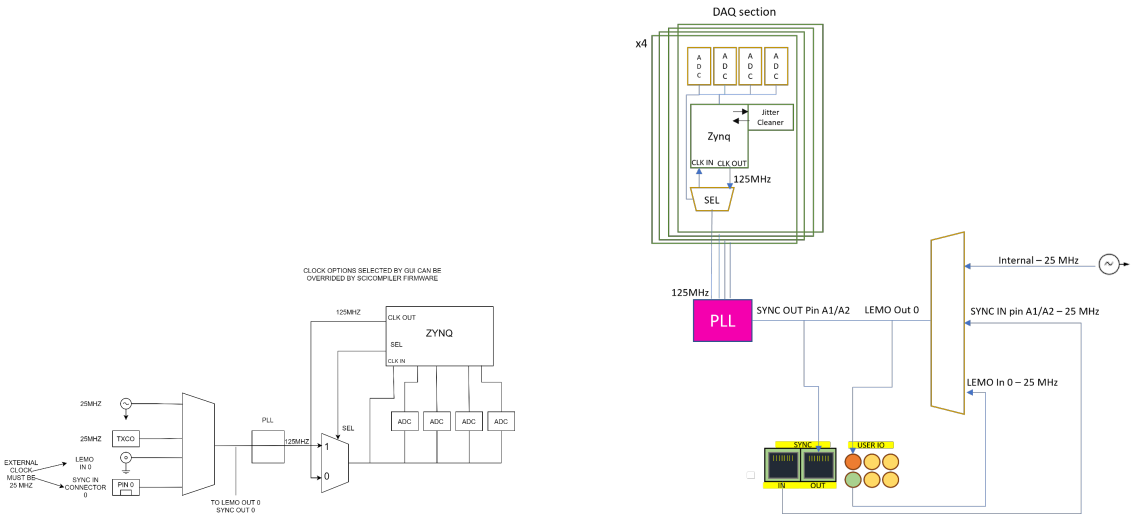
- 25 MHz internal oscillator
- LEMO IN 0 connector
- pin A1/A2 of the SYNC IN connector (see **Fig. 10.13**)

If the clock is provided to the base board externally by the LEMO 0 or the SYNC connector, the frequency must be 25 $\pm$ 0.5 MHz; using a different frequency requires an external VCO/PLL.

The clock signal fed into the base board is converted by the PLL, passing from a 25 MHz single-ended into a 125 MHz LVDS clock.

On the R5560/R5560SE the PLL has 5 outputs with 300 fs jitter: four of them are connected to the clock input of the four DAQ boards, while the fifth is connected to the SCSP.

The 25 MHz input clock can be routed to the LEMO OUT 0 or to the pin A1/A2 of the SYNC OUT connector (see **Fig. 10.13**), which is on the rear panel of the DT5560SE and on the front panel of the R5560/R5560SE.



(a) DT5560SE. **(b)** R5560/R5560SE.

Fig. 10.12: Scheme of the clock distribution. The R5560/R5560SE feeds four DAQ sections and the SCSP from the same PLL; the DT5560SE feeds its single DAQ section.

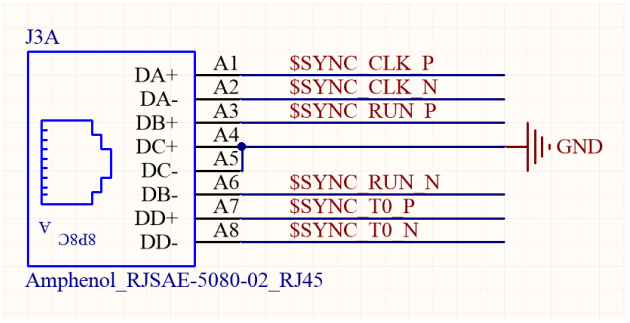


Fig. 10.13: Pinout of the SYNC RJ45 connector.

10.6 Synchronization among multiple boards

The DT5560SE/R5560/R5560SE has been designed to operate in large readout systems. Multiple boards can be connected in such a way that they can operate in simultaneous sampling with a common time reference, to be able to accurately timestamp events.

Synchronization operates at two levels:

- **Online**: a system manager timestamping logic with a common clock and reset.
- **Offline**: the event builder software in the DAQ uses the timestamp to merge events coming from different boards in a single event.

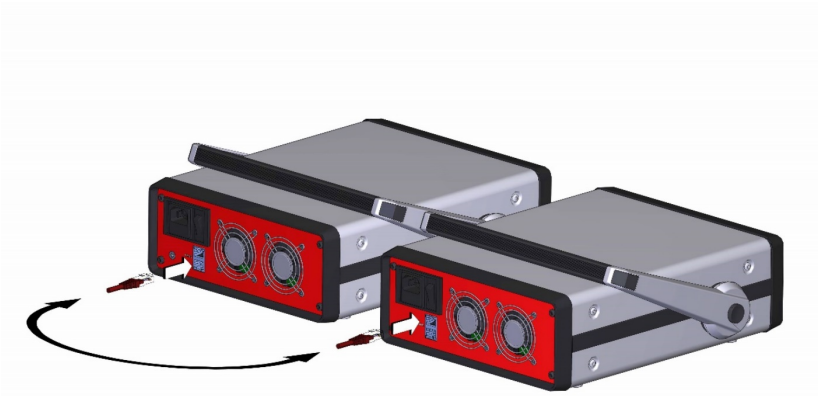
Synchronization requires the following signals:

- **Common clock**: ensures that all ADCs operate in phase exactly at the same frequency. It also guarantees that the internal time counter and TDC will be clocked at the same frequency, and they will be synchronous.
- **Run**: is a common signal generated by the master board and indicates the start of a run. Global timestamp and global event counter will be reset on the rising edge of this signal. All events will be times-

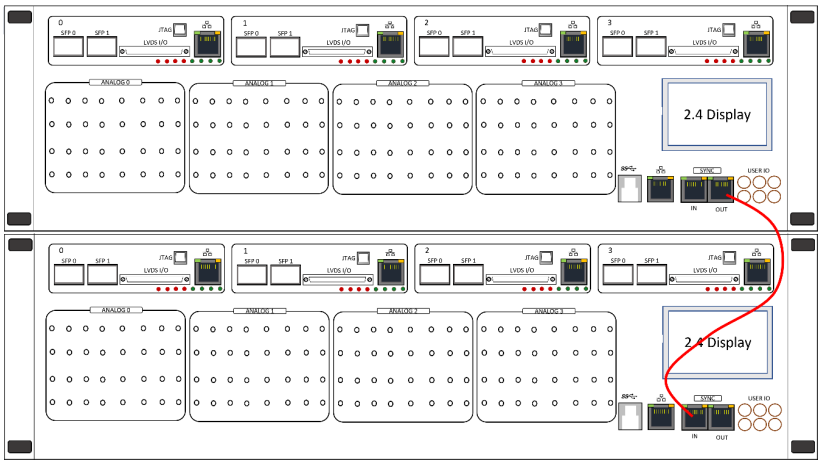
tamped with counters zeroed by this signal. The event builder in the readout system will merge different events coming from multiple boards using this unique timestamp.

- **T0/flag**: is a user signal that can be used to reset a secondary timestamp counter or to mark an event. This signal is optional, and it is used to make a measurement with respect to a periodic zero signal (e.g. a beam signal). The function of T0 is configurable in SCI-Compiler.

The synchronization between boards can work using a single RJ45 cable that connects, in daisy chain, multiple instruments through the SYNC IN/OUT port on the front panel. The cable carries all synchronization signals, including CLK (25 MHz), RUN and T0 on LVDS differential signals. A standard Ethernet CAT5e cable can be used for board-to-board synchronization. In order to set the boards to properly operate using this single-cable synchronization, refer to Chap. 11 and Chap. 12.



(a) DT5560SE: the SYNC IN/OUT connectors are on the rear panel, slave on the left, master on the right.



(b) R5560/R5560SE: the SYNC IN/OUT connectors are on the front panel, slave at the bottom, master at the top.

Fig. 10.14: Synchronization connection between two boards using a single Ethernet cable. The slave board receives the synchronization signals on its SYNC IN connector from the master board SYNC OUT.

As an alternative to the distribution of the needed signals through the SYNC IN/OUT connectors, on the front panel there are also three USER IN and three USER OUT LEMO connectors. The function can be selected through the configuration web page of the SCSP among:

- Clock reference

- T0
- Start of run
- Flag/Beam
- Next frame (useful in imaging applications, where it is necessary to accumulate counts for a fixed period)

10.7 DAQ datapath (R5560/R5560SE only)

R5560/R5560SE boards can be read out using multiple interfaces: Ethernet, USB, optical link and H-LINK. All these connection interfaces should be able to access the same data in the PL. To do this, a router is used to commit a R/W request from one of the interfaces to the PL and complete the R/W operation.

SCI-Compiler works with the logic of memory mapped resources. Every endpoint in the FPGA (register, digitizer, oscilloscope, frame transfer, list transfer, etc.) is mapped at a particular address of the PL in an address space 32 bit wide. Each endpoint has a memory size multiple of a power of two: a register, for example, has a memory size of 1, a FIFO of 1, an oscilloscope of 0xFFFF.

The DAQ is always a slave of a master system. The master system could be the PC client that connects via Ethernet to the server running on the CPU (PS) of the Zynq, or the facility readout system in more complex systems. The master starts a transfer operation with a short packet addressing the slave board and specifying the operation, the address in the slave and, for write operations, the payload. The router catches the packet and checks the board address. If the board address matches the board id or the general call (0xFFF), the packet is processed. If the packet comes from an optical link and does not match the board ID, or the packet is a general call (0xFFF), the packet is forwarded to the TX optical link and to the H-LINK.

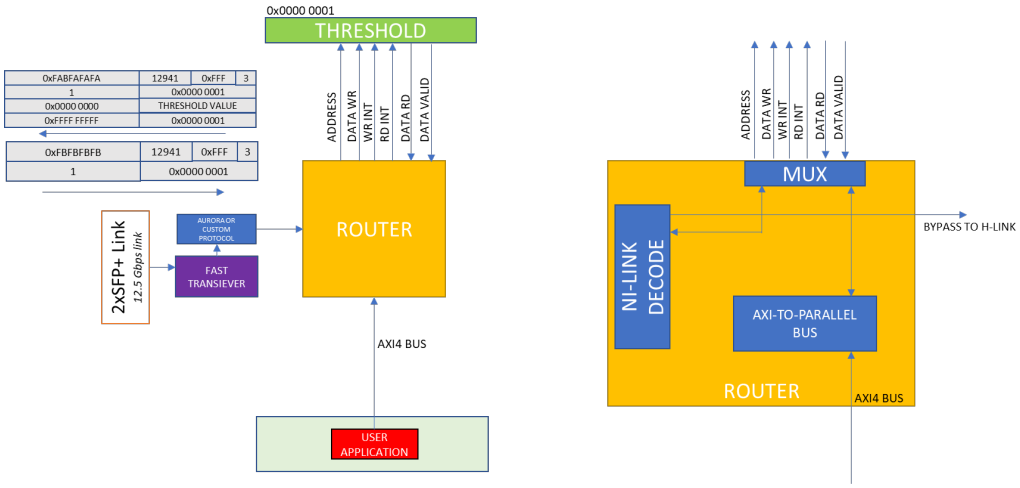


Fig. 10.15: Example of the router operation for data transfer.

The router also manages the H-LINK. The H-LINK appears in the user programmable logic as four FIFOs (two TX and two RX). The user has a TX/RX pair of FIFOs to send/receive data from the SX side and a pair for the DX side. The user can just push data in the correct FIFO and the framework will transfer data to the nearby DAQ board.

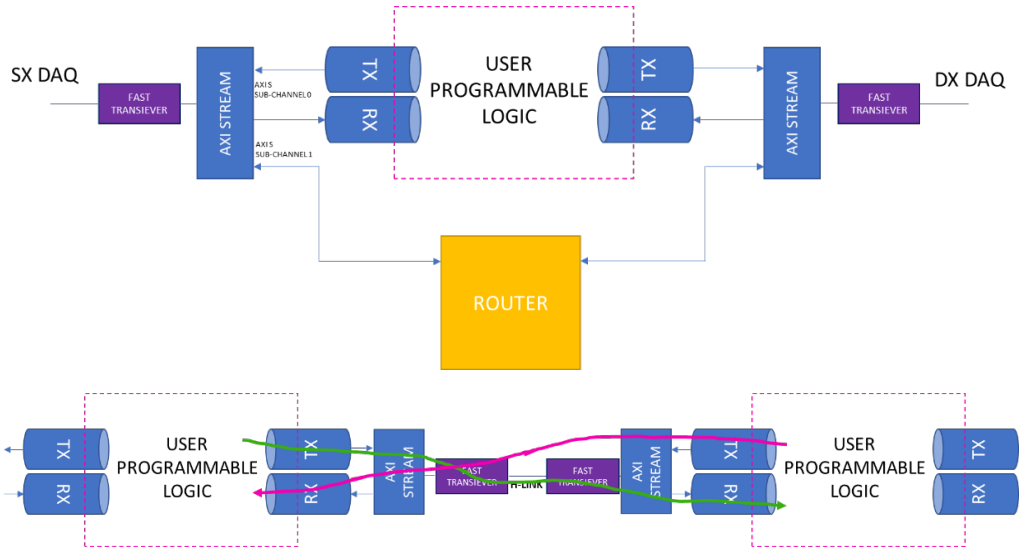


Fig. 10.16: Data transfer between two DAQ sections, using the router or the H-LINK.

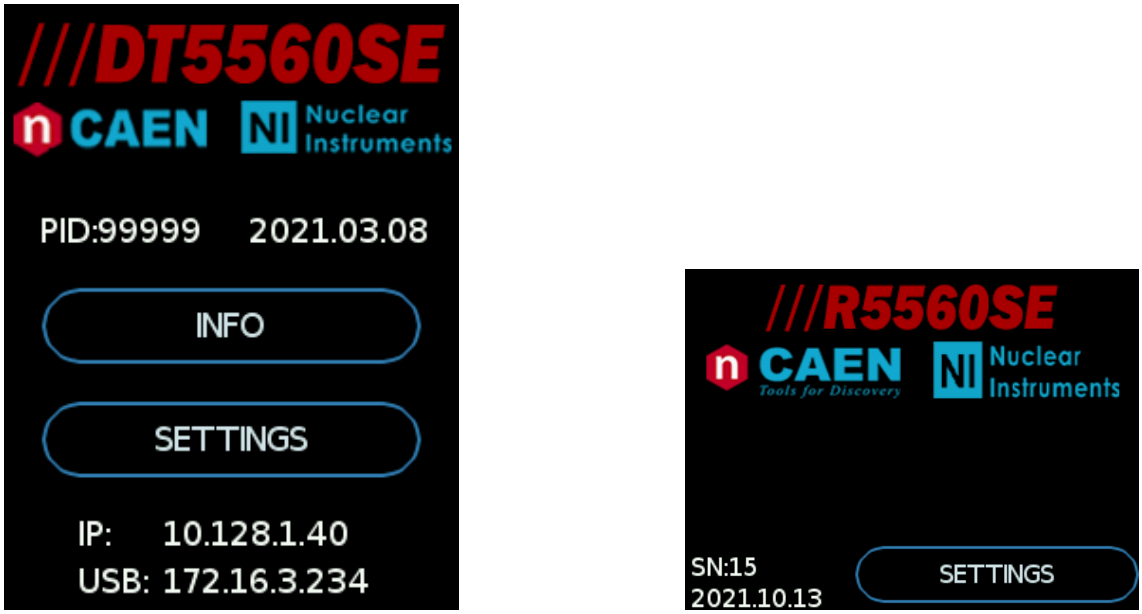
11 Touchscreen display guide

The DT5560SE/R5560/R5560SE features a 2.4” Touchscreen display for basic board configuration.

DO NOT USE SHARP METAL OBJECTS, SUCH AS TWEEZERS OR SCREWDRIVERS, OR POINTED OBJECTS, SUCH AS PENS OR PENCILS, TO TOUCH THE DISPLAY

The main window shows the PID (Product Identification Number) of the board and the current app version installed. On the DT5560SE it also shows the IP address of the Ethernet and USB interfaces.

By tapping on the **SETTINGS** button, it is possible to access a series of tabs where the user can configure the IP address of the board and select the routing configuration for signals like the Clock source and SYNC signals.



(a) DT5560SE.

(b) R5560/R5560SE.

Fig. 11.1: The main page of the Touchscreen display.

11.1 Ethernet settings

11.1.1 DT5560SE

In order to change the IP configuration:

1. Tap on **SETTINGS** from the main page and use the arrows on the top bar to navigate to the **ETHERNET** tab.
2. It is possible to enable the DHCP in order to automatically obtain the IP from the DHCP server in the network, or it is possible to assign the IP manually by tapping on the keyboard icon on the right.

3. Assign an IP in a valid range for your network in order to be able to reach the instrument from your computer. For example, if your computer IP address is 10.128.1.1, type an IP that ranges from 10.128.1.2 to 10.128.1.254 and the mask (NM) must be 255.255.255.0.
4. The user will be notified if the IP is invalid or if the configuration has been successfully applied. A reboot of the board is needed for the changes to become effective.

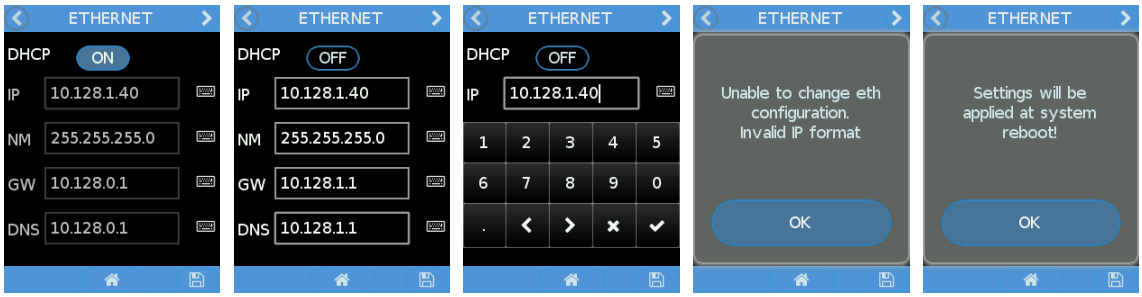
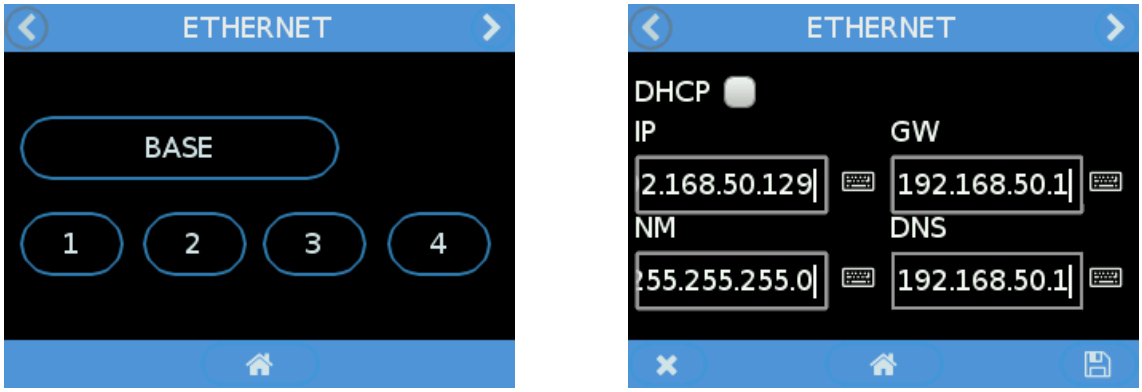


Fig. 11.2: The sequence on the Touchscreen display to set the Ethernet configuration of the DT5560SE.

11.1.2 R5560/R5560SE

The **ETHERNET** tab allows the user to configure the IP address of the baseboard and of each DAQ section. To change the IP configuration:

1. Tap on **SETTINGS** from the main page and use the arrows on the top bar to navigate to the **ETHERNET** tab. Then tap on **BASE** or on the DAQ section number to set the IP address of the baseboard or of the corresponding DAQ section, respectively.
2. Assign an IP, Gateway (GW), Netmask (NM) and DNS in a valid range for your network, so that the instrument can be reached from your computer. For example, if your computer IP address is 10.128.1.1, type an IP that ranges from 10.128.1.2 to 10.128.1.254, and the mask (NM) must be 255.255.255.0.
3. It is possible to enable DHCP to automatically obtain the IP from the DHCP server in the network, or to assign the IP manually by tapping on the keyboard icon on the right of each field.
4. Press the **Save** button in the bottom right corner of the screen.
5. The user will be notified if the IP has been successfully applied. A reboot of the board is needed for the changes to become effective.



(a) Baseboard or DAQ section selection.

(b) IP address configuration.

Fig. 11.3: The Ethernet tabs on the Touchscreen display of the R5560/R5560SE.

11.2 Signal routing settings

In the **SETTINGS** menu it is also possible to configure the routing of some internal digital signals of the board, in order to configure clock distribution and synchronization. Refer to Sec. 10.5 of this manual to better understand the clocking strategy of the board.

11.2.1 DT5560SE

Tap on **SETTINGS** from the main page and use the arrows on the top bar to navigate to the **CLOCK**, **DAQ SYNC IN**, **SYNC OUT** and **LEMO OUT** tabs.

In the **CLOCK** tab it is possible to select the **clocking source**, choosing among **INTERNAL**, **LEMO IN 0** and **SYNC IN 0**. If **LEMO IN 0** is selected, the clock must be a 25 MHz TTL signal; if **SYNC IN 0** is selected it must be an LVDS signal.

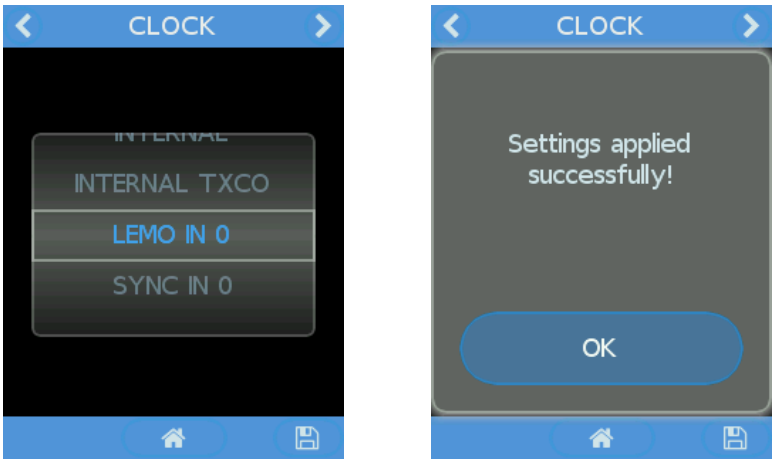


Fig. 11.4: The CLOCK tab of the DT5560SE.

The **DAQ SYNC IN** tab is used to select the source for the synchronization signals used by the board. Synchronization needs three independent signals to be routed to the DAQ board (CLOCK, RUN, T0). It is possible to select between **LEMO IN** and **SYNC IN** connectors source for each of the three available lines. For standard synchronization using the SYNC IN/OUT connectors daisy chain (see **Fig. 10.14**), choose the **SYNC IN** option for each of the slave boards.



Fig. 11.5: The DAQ SYNC IN tab of the DT5560SE.

The **SYNC OUT** tab is used to set the signals carried on the RJ45 SYNC OUT connector placed on the rear panel of the instrument. This connector carries three independent signals that allow the synchronization of two or more boards. It is possible to select between **LEMO IN**, **SYNC IN**, **DAQ SYNC OUT** and **CLOCK 25 MHz** for each of the three available lines. For standard synchronization using the SYNC IN/OUT connectors daisy chain, choose the **DAQ SYNC OUT** option.



Fig. 11.6: The SYNC OUT tab of the DT5560SE.

The **LEMO OUT** tab is used to select which signals are to be routed on the front panel USER Outputs; it is possible to select between **LEMO IN**, **SYNC IN**, **DAQ SYNC OUT** and **CLK 25 MHz**.



Fig. 11.7: The LEMO OUT tab of the DT5560SE.

11.2.2 R5560/R5560SE

Tap on **SETTINGS** from the main page and use the arrows on the top bar to navigate to the **CLOCK INPUT**, **DAQ SYNC**, **SYNC OUT**, **LEMO OUT** or **PULSER** tabs.

In the **CLOCK INPUT** tab, it is possible to select the clocking source, choosing among **INTERNAL**, **LEMO IN 0** or **SYNC CONNECTOR**. If **LEMO IN 0** is selected, the clock must be a 25 MHz TTL signal. If **SYNC CONNECTOR** is selected, it must be a 25 MHz LVDS signal connected to pin A1/A2 of the SYNC IN RJ45 connector. For standard synchronization using the SYNC IN/OUT connectors daisy chain, choose the **SYNC CONNECTOR** option for each of the slave boards.

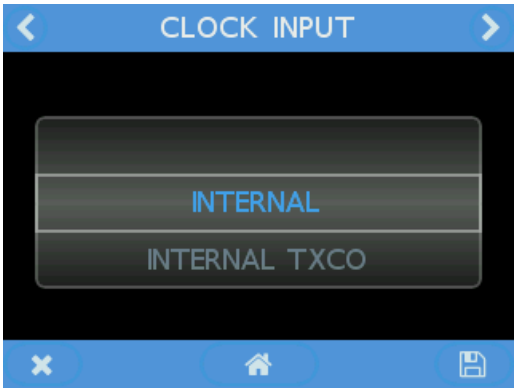
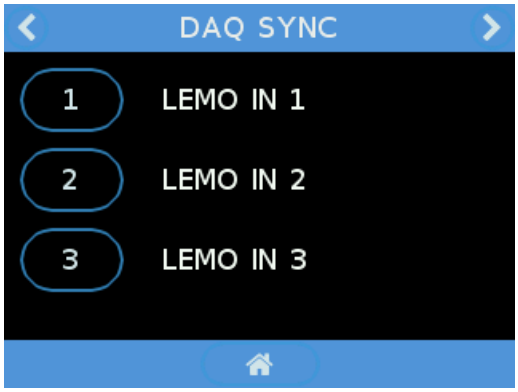


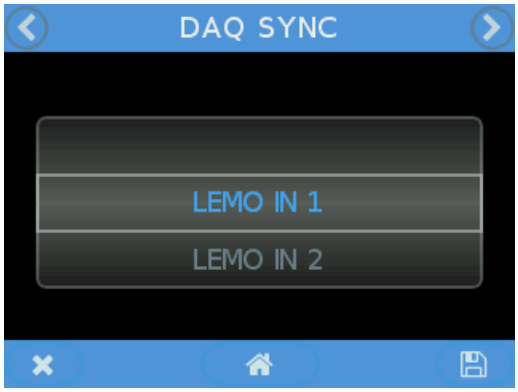
Fig. 11.8: The CLOCK INPUT tab of the R5560/R5560SE.

The **DAQ SYNC** tab is used to select the source for the synchronization signals used by the board. Synchronization needs three independent signals to be routed to the DAQ board (CLOCK, RUN, T0). It is possible to select between **LEMO IN**, **SYNC IN** and **Internal Pulser** source for each of the three available lines. For standard synchronization using the SYNC IN/OUT connectors daisy chain, choose the **SYNC IN** option for each of the slave boards. If using the Internal Pulser source for one of the three signals, the remaining two can be driven via register using the **SYNC REG** option.

The **SYNC OUT** tab is used to set the signals carried on the RJ45 SYNC OUT connector placed on the front panel of the instrument. This connector can carry three independent signals that allow the synchronization of two or more boards. It is possible to select between **LEMO IN**, **SYNC IN** and **INTERNAL PULSER** for each of the three available lines. For standard synchronization using the SYNC IN/OUT connectors daisy chain, choose the **SYNC-IN** option.



(a) Line selection.

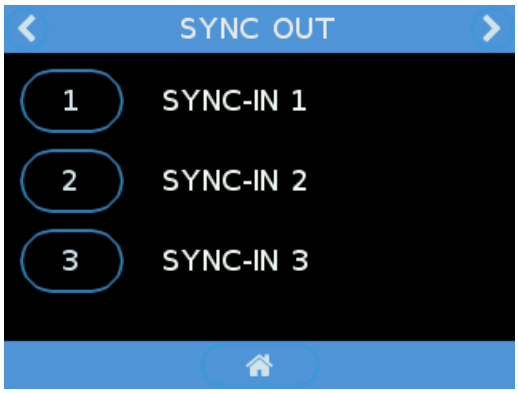


(b) Source selection for a line.

Fig. 11.9: The DAQ SYNC tab of the R5560/R5560SE.



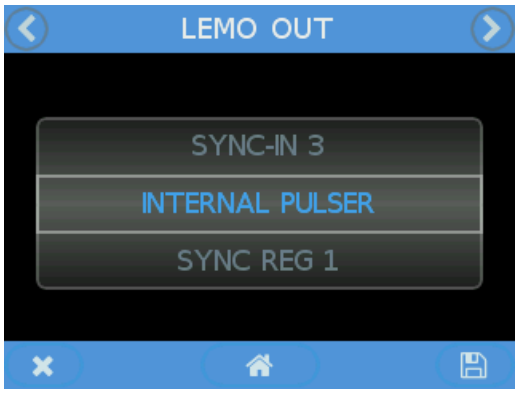
(a) Source selection for a line.



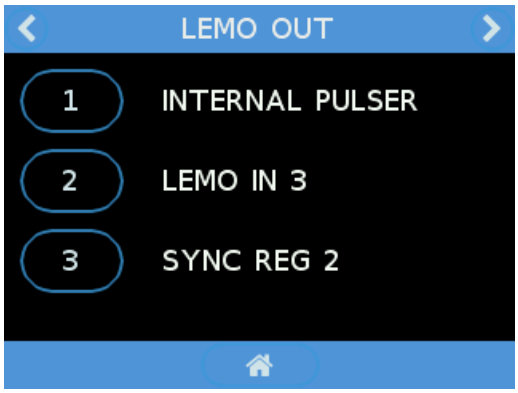
(b) Line selection.

Fig. 11.10: The SYNC OUT tab of the R5560/R5560SE.

The **LEMO OUT** tab is used to select which signals are routed on the front panel three USER Outputs; it is possible to select between **LEMO IN**, **SYNC IN**, **INTERNAL PULSER** and **SYNC REG**.



(a) Source selection for a line.



(b) Line selection.

Fig. 11.11: The LEMO OUT tab of the R5560/R5560SE.

The **PULSER** tab is used to set the internal pulser to be used in combination with **CLOCK INPUT** = **INTERNAL**. Frequency and width of the pulser can be chosen in the dedicated fields.

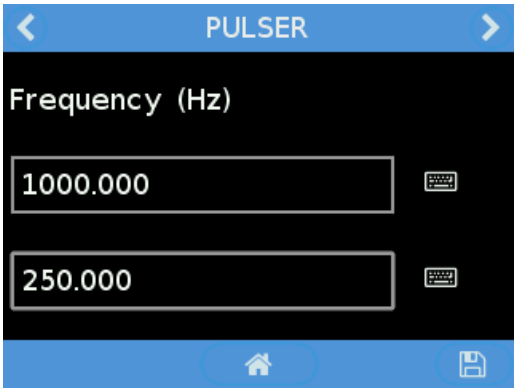


Fig. 11.12: The PULSER tab of the R5560/R5560SE.

11.3 Info and status

11.3.1 DT5560SE

In the **INFO** menu, the user can view the current running firmware and monitor the temperature measured in different points of the hardware.

The DT5560SE can store in its memory several different firmware, and it is possible to retrieve one of the loaded firmware:

- Tap on the **CHANGE** button: a scroll-down menu showing the available firmware is shown
- Select the firmware of interest and press the **Save** button in the bottom right corner of the screen
- the DT5560SE will ask you to reboot in order to load the firmware. Tap OK and the board will automatically reboot
- After reboot is complete, the new firmware is loaded and the board is operational



Fig. 11.13: The sequence on the Touchscreen display of the DT5560SE to retrieve a pre-loaded firmware.

In the **TEMPERATURE** tab, it is possible to monitor the temperature measured in different parts of the instrument. If the temperature exceeds the maximum limit, the instrument will automatically shut down.

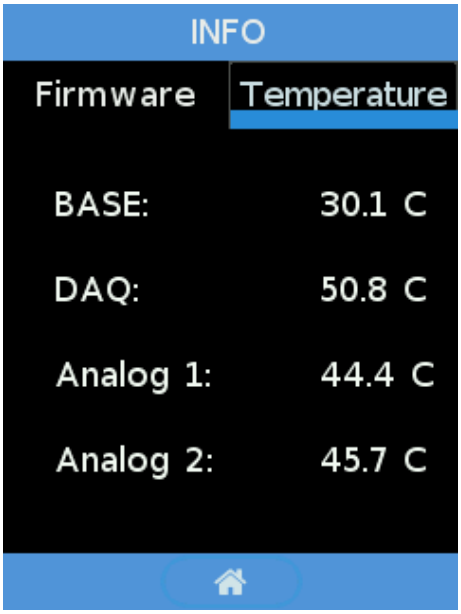
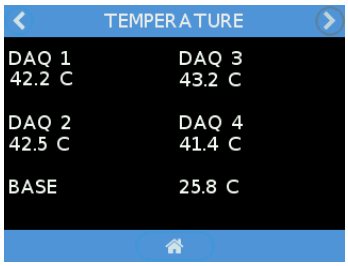


Fig. 11.14: The temperature tab of the DT5560SE.

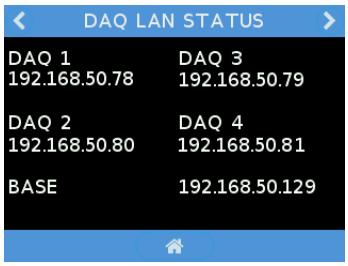
11.3.2 R5560/R5560SE

From the **TEMPERATURE**, **DAQ LAN STATUS** and **DAQ INFO** tabs, it is possible to check:

- the DAQ boards and baseboard temperature;
- the DAQ boards and baseboard IP address;
- the name of the firmware loaded on each DAQ board.



(a) TEMPERATURE tab.



(b) DAQ LAN STATUS tab.



(c) DAQ INFO tab.

Fig. 11.15: The TEMPERATURE, DAQ LAN STATUS and DAQ INFO tabs of the R5560/R5560SE.

11.4 Cleaning the Touchscreen

In order to clean the touchscreen, wipe the screen with a towelette designed for cleaning monitors, or with a clean cloth moistened with water. Do not use sprays or aerosols directly on the screen: the liquid may seep into the housing and damage a component. Never use solvents or flammable liquids on the screen.

12 Web interface

The DT5560SE/R5560/R5560SE comes with an embedded web interface that allows the user to configure the analog front end parameters, upload new firmware and upgrade the instrument OS.

The DT5560SE, having a single DAQ section, exposes one web interface. The R5560/R5560SE exposes two: one on the baseboard, which oversees the whole unit and its four DAQ sections, and one on each DAQ section, reachable on that section's own Ethernet port.

To connect to the web interface:

- Connect the instrument to the PC via Ethernet or USB. On the R5560/R5560SE it is possible to connect the baseboard or the single DAQ boards individually using the dedicated connection ports of each section.
- If using Ethernet, set your PC to be in the same network as the instrument.
- Open a web browser and type the IP or USB address as shown on the instrument display, according to the chosen physical connection (IP refers to Ethernet connection).

12.1 Baseboard web interface

12.1.1 Connection

To access the baseboard Web Interface, connect to the board by using the Ethernet port as shown below.

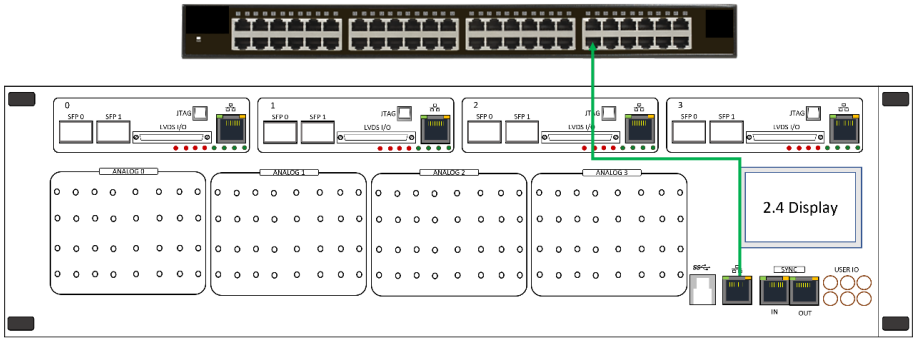


Fig. 12.1: Connection scheme to access the baseboard Web interface, shown on an R5560SE; the scheme is the same on the R5560. The baseboard port can be connected to the network or directly to a PC Ethernet port.

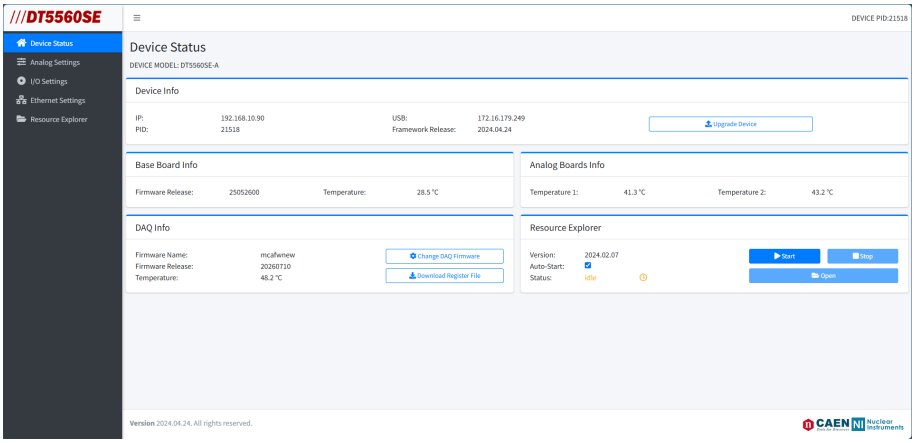
12.1.2 Device status and firmware upgrade

In the **Device Status** (Home) window it is possible to monitor:

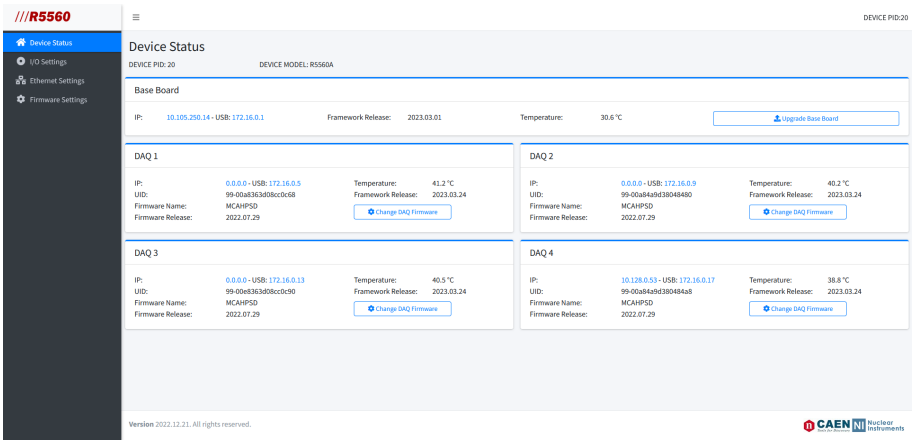
- the board IP address and USB address
- the board PID
- the temperature of the base board
- the baseboard firmware release (i.e., the OS of the system)

- the IP of each DAQ board (R5560/R5560SE only)
- the firmware loaded on each DAQ board (name and release), i.e., the pulse processing algorithm
- the temperature of the DAQ board
- the temperature of the analog boards (AFE section of the DT5560SE/R5560SE)
- the status of the Resource Explorer (DT5560SE; on the R5560/R5560SE it is shown in the DAQ section web interface), see Sec. 12.3

The **DAQ Info** section of the Home page also has the **Download Register File** button, which downloads a JSON file with the dump of the registers of the DAQ board. On the R5560/R5560SE the button is on the Home page of the DAQ section web interface (see Sec. 12.2.2), one for each DAQ section.



(a) DT5560SE.

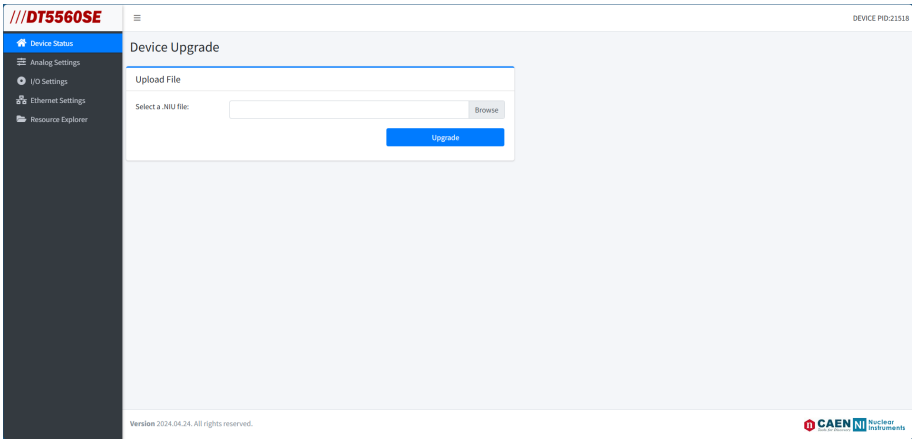


(b) R5560/R5560SE, baseboard interface.

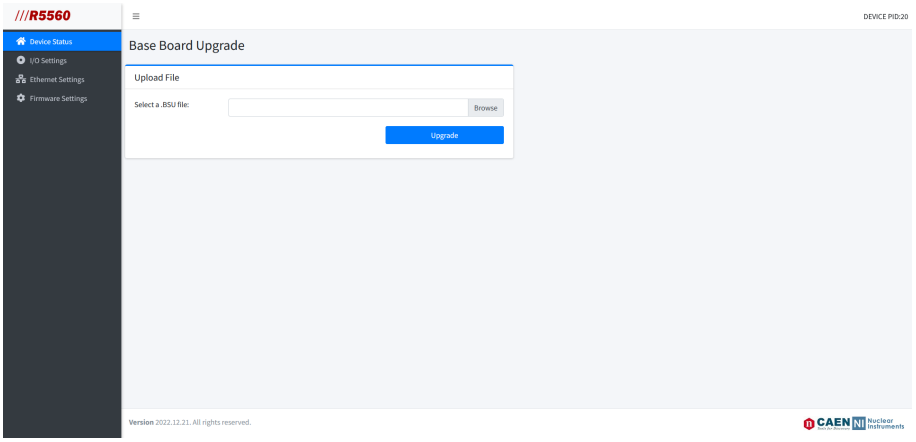
Fig. 12.2: The Home page of the Web interface.

If a newer version of the OS running on the ARM processor is released on the CAEN website, it is possible to upgrade the device by clicking on the **Upgrade** button (**Upgrade Base Board** on the R5560/R5560SE) in the base board section. The opening page will ask the user to select the file previously downloaded from the CAEN website (a `.bsu` file on the R5560/R5560SE). Press the **Upgrade** button to perform the loading of the new OS version and follow the given instructions.

From the Home page of the web interface it is also possible to change and upload a new firmware generated by SCI-Compiler. In order to upload a new firmware, press the **Change DAQ Firmware** button in the DAQ Info section. The opening page will ask you to select a firmware file (`.niu`) from your computer. Press



(a) DT5560SE.



(b) R5560/R5560SE, baseboard interface.

Fig. 12.3: Upgrade of the OS of the board from the Web interface.

Upload to load the firmware on the device. The firmware will be copied in the persistent memory of the board.

Note: If upgrading the firmware via USB, follow the procedure below to avoid Windows USB peripheral recognition problems: change the firmware and let the unit reboot automatically as per the standard process; disconnect the USB cable; reboot the module manually; reconnect the USB and the unit should be correctly seen in the Device Manager as Remote NDIS Compatible Device.

It is possible to activate the firmware by selecting its name in the **Manage Firmware** menu and pressing the **Install** button.

It is also possible to remove an existing firmware by selecting its name in the scroll-down menu and pressing the **Delete** button.

For any of these operations, follow the instructions given in the subsequent opening windows and always check if a successful info message appears.

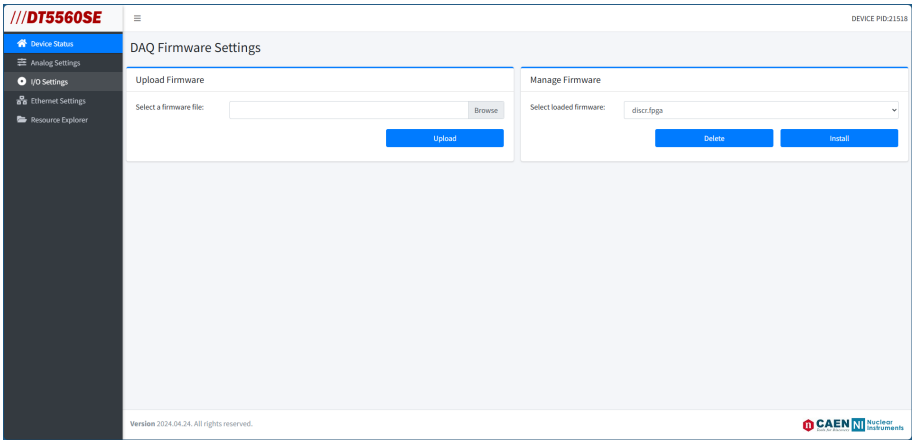


Fig. 12.4: Upgrade of the DAQ firmware from the Web interface of the DT5560SE.

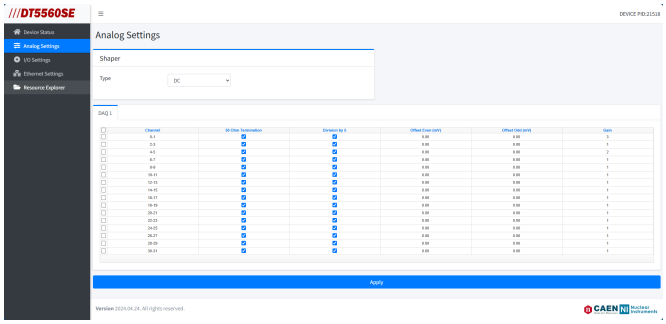
12.1.3 Analog settings

The **Analog Settings** window is used to configure the AFE boards by selecting the relevant parameters for the analog input stage of each DAQ section. According to the description of the analog stage given in Sec. 10.2.2, it is possible to configure the following parameters:

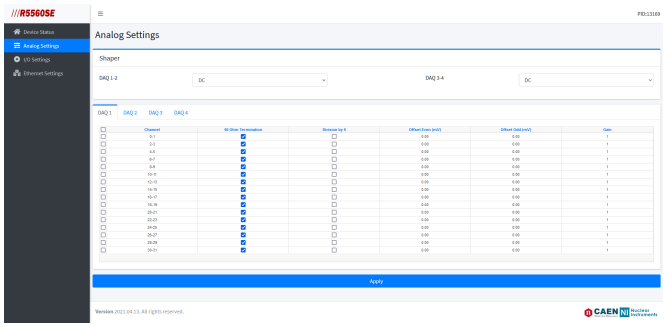
- analog coarse gain (hereinafter **Gain**)
- offset
- input impedance (50 Ω / 1 kΩ)
- Division by 5
- Shaper

Gain, impedance and division can be configured for each pair of channels, the offset independently for each single channel (odd or even channel of the pair), while the shaper is common for all channels on the DT5560SE and for all channels of DAQ1-2 and DAQ3-4 sections separately on the R5560/R5560SE. Always press **Apply** button to store the settings and check if a successful info message appears.

Note: This page has no effect on the R5560, whose differential inputs feed the ADCs directly and have no analog frontend to configure.



(a) DT5560SE.



(b) R5560SE, baseboard interface.

Fig. 12.5: The Analog Settings page of the Web interface.

12.1.4 I/O settings

The **I/O settings** page is used to configure the clock and synchronization signals routing. Refer to Sec. 10.5 of this manual to better understand the clocking strategy of the board, and to Sec. 10.6 for the daisy chain of multiple units.

On the **DT5560SE** it is possible to set:

- Clock source (**CLOCK INPUT**) choosing among INTERNAL, LEMO IN 0 and SYNC IN 0. If LEMO IN 0 is selected, the clock must be a 25 MHz TTL signal, if SYNC IN 0 is selected it must be an LVDS signal.
- The source for the synchronization signals used by the board (**DAQ SYNC IN**). Synchronization needs three independent signals to be routed to the DAQ board (CLOCK, RUN, T0). It is possible to select between LEMO IN and SYNC IN connectors source for each of the three available lines. For standard synchronization using SYNC IN/OUT connectors daisy chain, please choose SYNC IN option for each of the slave boards.
- The signals carried on the RJ45 **SYNC OUT** connector placed on the rear panel of the instrument. This connector carries three independent signals that allow the synchronization of two or more boards. It is possible to select between LEMO IN, SYNC IN, DAQ SYNC OUT and CLOCK 25 MHz for each of the three available lines. For standard synchronization using SYNC IN/OUT connectors daisy chain, please choose DAQ SYNC OUT option.
- The signals to be routed on the front panel **LEMO OUT**; it is possible to select between LEMO IN, SYNC IN, DAQ SYNC OUT, and CLK 25 MHz.

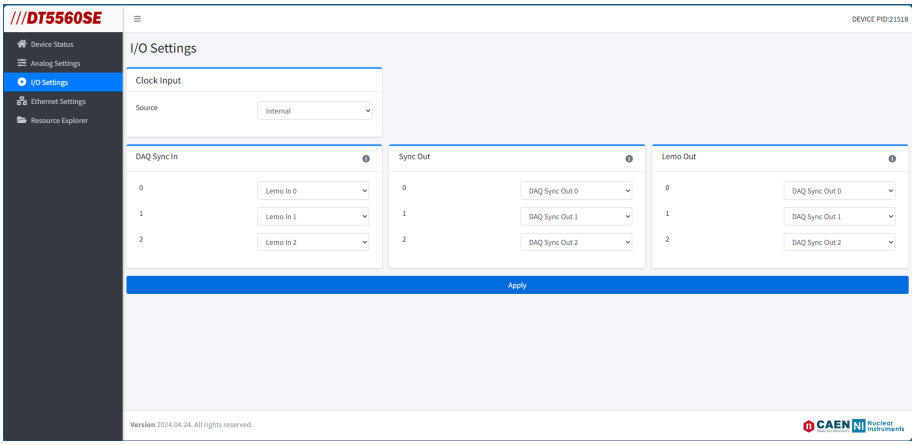


Fig. 12.6: The I/O Settings page of the Web interface of the DT5560SE.

On the **R5560/R5560SE** it is possible to set:

- Clock source (**CLOCK INPUT**) choosing among INTERNAL, LEMO IN 0 or SYNC CONNECTOR. If LEMO IN 0 is selected, the clock must be a 25 MHz TTL signal. If SYNC CONNECTOR is selected it must be a 25 MHz LVDS signal connected to pin A1/A2 of the SYNC IN RJ45 connector. For standard synchronization using SYNC IN/OUT connectors daisy chain, please choose SYNC CONNECTOR option for each of the slave boards.
- Internal pulser to be used in combination with CLOCK INPUT = Internal: frequency and width of the pulser can be chosen in the dedicated fields.
- The source for the synchronization signals used by the board (**DAQ SYNC IN**). Synchronization needs three independent signals to be routed to the DAQ board (CLOCK, RUN, T0). It is possible to select between LEMO IN, SYNC IN and Internal Pulser source for each of the three available lines. For standard synchronization using SYNC IN/OUT connectors daisy chain, please choose SYNC IN option for each of the slave boards. If using Internal Pulser source for one of the three signals, the remaining two can be driven via register using the option SYNC REG.
- The signals carried on the RJ45 **SYNC OUT** connector placed on the front panel of the instrument. This connector can carry three independent signals that allow the synchronization of two or more boards. It is possible to select between LEMO IN, SYNC IN, and INTERNAL PULSER for each of the three available lines. For standard synchronization using SYNC IN/OUT connectors daisy chain, please choose SYNC-IN option.
- The signals to be routed on the three front panel **LEMO OUT**; it is possible to select between LEMO IN, SYNC IN, INTERNAL PULSER and SYNC REG.

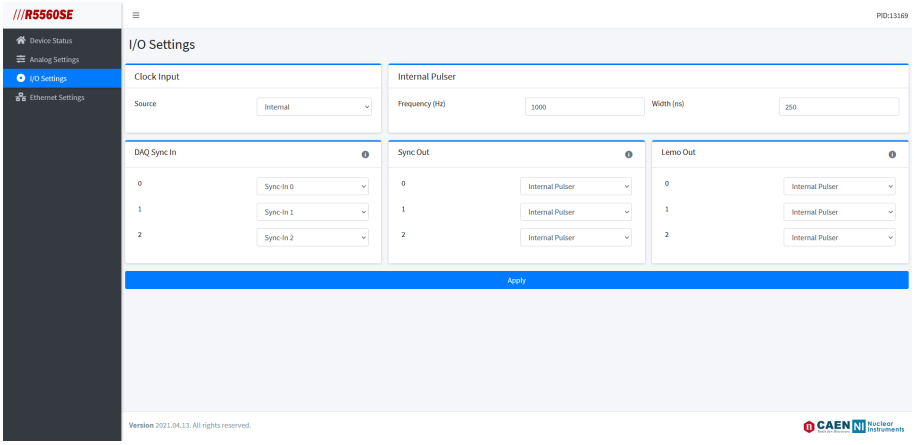


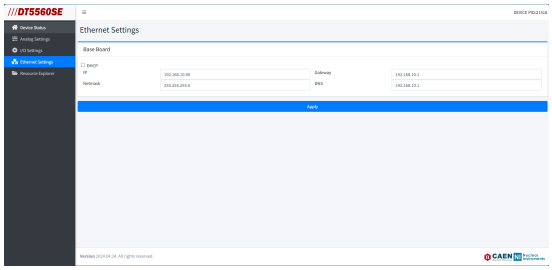
Fig. 12.7: The I/O Settings page of the baseboard Web interface of the R5560/R5560SE.

12.1.5 Ethernet settings

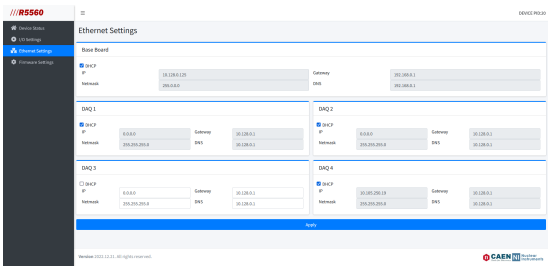
The **Ethernet Settings** window allows the user to configure the IP address of the board, and on the R5560/R5560SE of the baseboard and of each DAQ section. It is possible to enable the DHCP in order to automatically obtain the IP from the DHCP server in the network or it is possible to manually assign the IP by clicking on the keyboard icon on the right.

Assign an IP in a valid range for your network in order to be able to reach the instrument from your computer. For example, if your computer IP address is 10.128.1.1, type an IP that ranges from 10.128.1.2 to 10.128.1.254 and the mask (NM) must be 255.255.255.0.

After modifying the Ethernet settings, press the **Apply** button and follow the instructions given in the opening window.



(a) DT5560SE.



(b) R5560/R5560SE, baseboard interface.

Fig. 12.8: The Ethernet Settings page of the Web interface.

12.1.6 Firmware settings (R5560/R5560SE only)

The **Firmware Settings** window allows the user to manage the FPGA firmware to be loaded into the DAQ sections. It is possible to load the same firmware for all sections or different files for each section. Once the DAQs of interest are selected, it is possible to load firmware already installed onboard (choose from the drop-down menu and press **Install**) or to upload a new firmware file, for instance a firmware generated by SCI-Compiler.

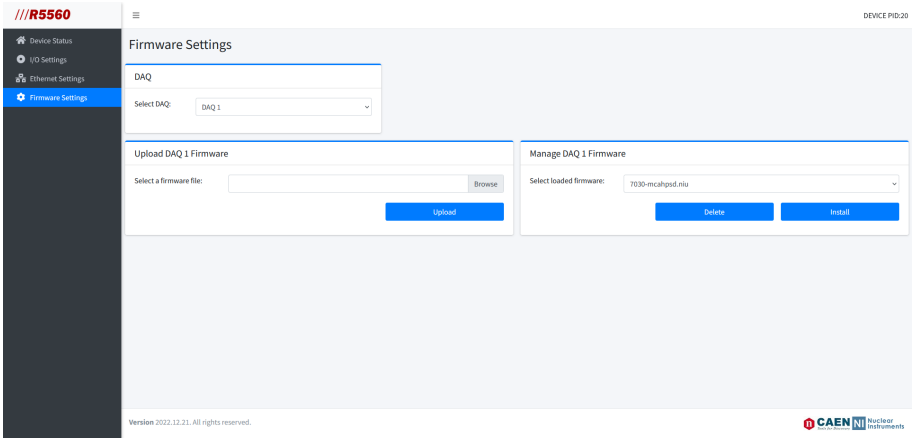


Fig. 12.9: The Firmware Settings page of the baseboard Web interface.

12.2 DAQ section web interface (R5560/R5560SE only)

12.2.1 Connection

To access the specific Web Interface of a DAQ section, connect to the board by using the Ethernet ports as shown below.

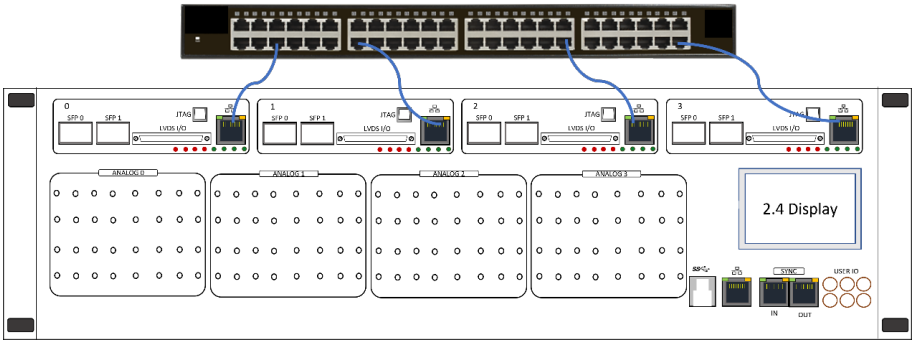


Fig. 12.10: Connection scheme to access the DAQ section Web interface. Each port can be connected independently to the network or directly to a PC Ethernet port.

12.2.2 Device status and firmware upgrade

In the **Device Status** (Home) window it is possible to monitor:

- the DAQ board IP address
- the FPGA UID
- the temperature of the DAQ board
- the Framework release (i.e., the OS of the system)
- the firmware loaded on the DAQ board (name and release), i.e., the pulse processing algorithm
- Resource Explorer status

- Connection information for base and all DAQs

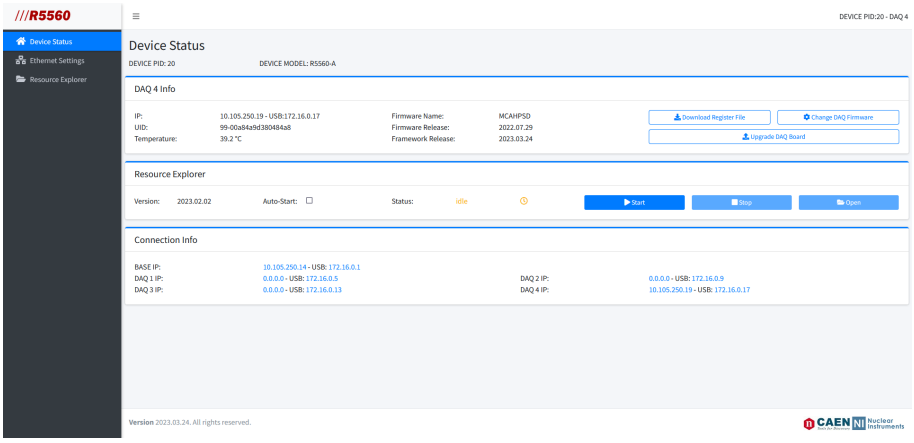


Fig. 12.11: The Home page of the DAQ section Web interface.

If a newer version of the OS running on the ARM processor is released on the CAEN website, it is possible to upgrade the device by clicking on **Upgrade DAQ Board**. The opening page will ask the user to select the file previously downloaded from the CAEN website. Press the **Upgrade** button to perform the loading of the new OS version and follow the given instructions.

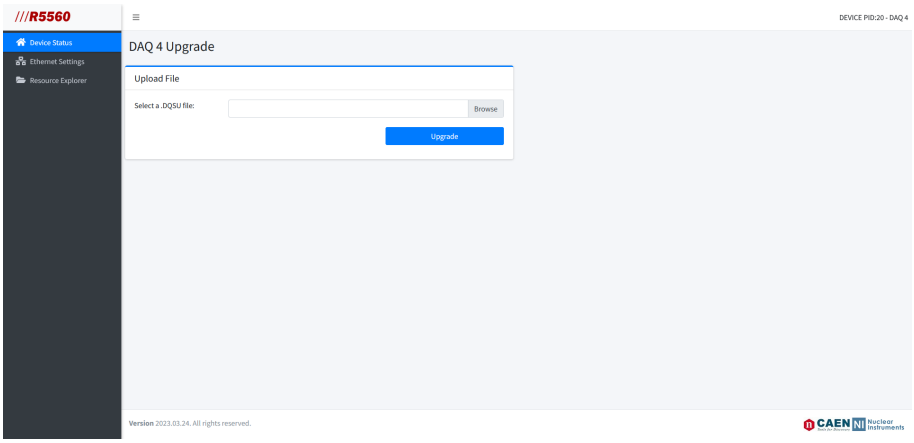


Fig. 12.12: Upgrade of the OS of the board from the DAQ Web interface.

It is also possible to change and upload a new FPGA firmware, for instance one that has been generated by SCI-Compiler. To upload a new firmware, press the **Change DAQ Firmware** button in the DAQ section. In the opening page, it is possible to load a firmware already installed onboard (choose from the drop-down menu and press **Install**) or to upload a new firmware file (`.niu`). The firmware will be copied in the persistent memory of the board.

Note: If upgrading the firmware via USB, follow the procedure below to avoid Windows USB peripheral recognition problems: change the firmware and let the unit reboot automatically as per the standard process; disconnect the USB cable; reboot the module manually; reconnect the USB and the unit should be correctly seen in the Device Manager as Remote NDIS Compatible Device.

It is possible to activate the firmware by selecting its name in the **Manage Firmware** menu and pressing the **Install** button.

It is also possible to remove an existing firmware by selecting its name in the scroll-down menu and pressing the **Delete** button.

For any of these operations, follow the instructions given in the subsequent opening windows and always check if a successful info message appears.

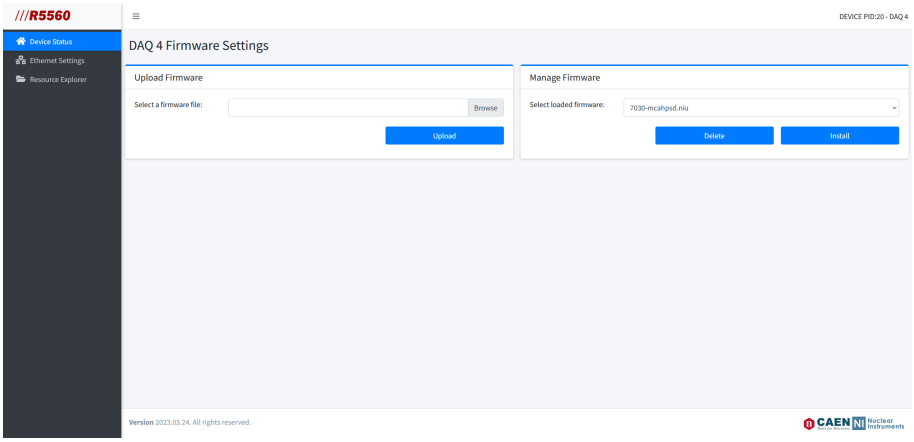


Fig. 12.13: Upgrade of the firmware from the DAQ section Web interface.

12.2.3 Analog settings (R5560SE only)

The **Analog Settings** window is used to configure the relevant parameters for the analog input stage of the DAQ section, as described in Sec. 12.1.3. On this page the shaper is common for all channels of the DAQ section.

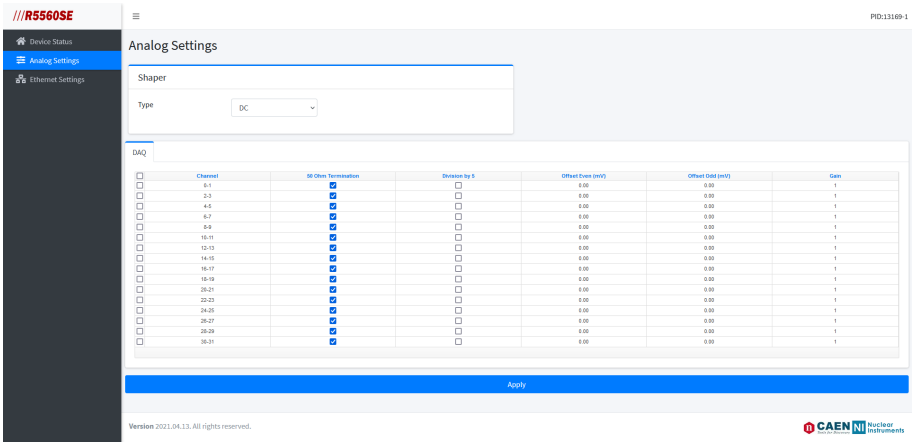


Fig. 12.14: The Analog Settings page of the DAQ section Web interface.

12.2.4 Ethernet settings

The **Ethernet Settings** window allows the user to configure the IP address of the DAQ section. It is possible to enable the DHCP in order to automatically obtain the IP from the DHCP server in the network or it is possible to manually assign the IP by clicking on the keyboard icon on the right.

Assign an IP in a valid range for your network in order to be able to reach the instrument from your computer. For example, if your computer IP address is 10.128.1.1, type an IP that ranges from 10.128.1.2 to 10.128.1.254 and the mask (NM) must be 255.255.255.0.

After modifying the Ethernet settings, press the **Apply** button and follow the instructions given in the opening window.

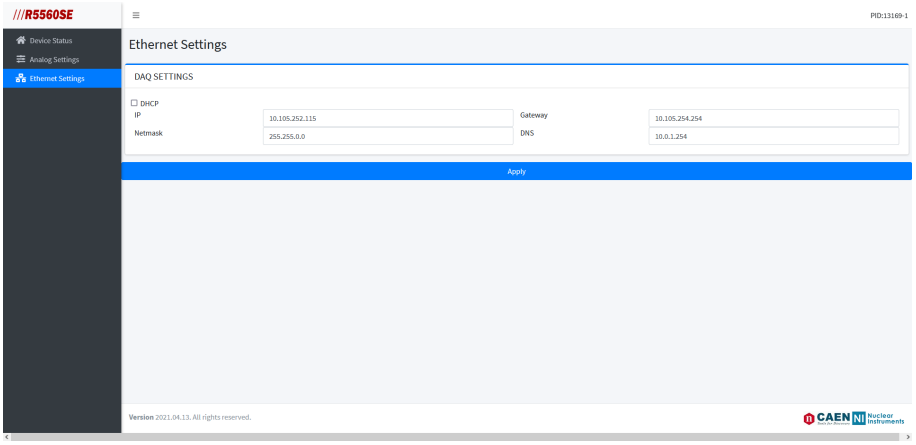


Fig. 12.15: The Ethernet Settings page of the DAQ section Web interface.

12.3 Resource Explorer (beta)

The **Resource Explorer** is available in the web interface of the DT5560SE and in the DAQ section web interface of the R5560/R5560SE (see Sec. 12.2). It replicates the same functionalities as the Resource Explorer tool available in SCI-Compiler.

SCI-Compiler offers a simple built-in tool, called Resource Explorer, to connect one of the supported boards and test the features of the FPGA firmware with no need to write any software code. This tool manages all Local Bus readout blocks placed in the firmware diagram, therefore it gives the possibility to read and write the configuration registers (i.e., Signal Processing parameters) and shows the readout instruments like Spectrum, Oscilloscope, etc. in a GUI, so that it is possible to change the signal processing parameters and test the results of acquisition live, with no need to write any software code. In practice, the Resource Explorer is a first-use debug interface to test the firmware and check the behaviour of the Processing algorithm specified through the block diagram.

Refer to **[RD2]** for more details.

12.3.1 Starting the Resource Explorer

The Resource Explorer is started from the Resource Explorer panel of the web interface (**Fig. 12.16**): press the **Start** button (play icon) and then go to the dedicated Resource Explorer tab, or click on **Open**. The panel also shows the version of the Resource Explorer, the **Auto-Start** option and the **Status**, which is **idle** when the Resource Explorer is not running.

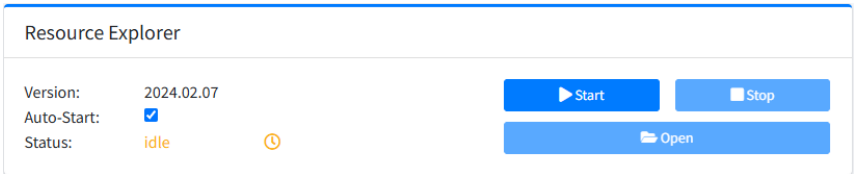


Fig. 12.16: The Resource Explorer panel of the web interface.

When the **Start** button is pressed and the Resource Explorer has started correctly, the Status turns to **running**, in green (**Fig. 12.17**).



Fig. 12.17: Status of the Resource Explorer once started.

At the first opening the resource tree on the left is not loaded (**Fig. 12.18**): press **Get Tree**, expand the **MMCComponents** folder and drag & drop the element of interest (for example the oscilloscope of a channel) from the tree on the left to the panel.

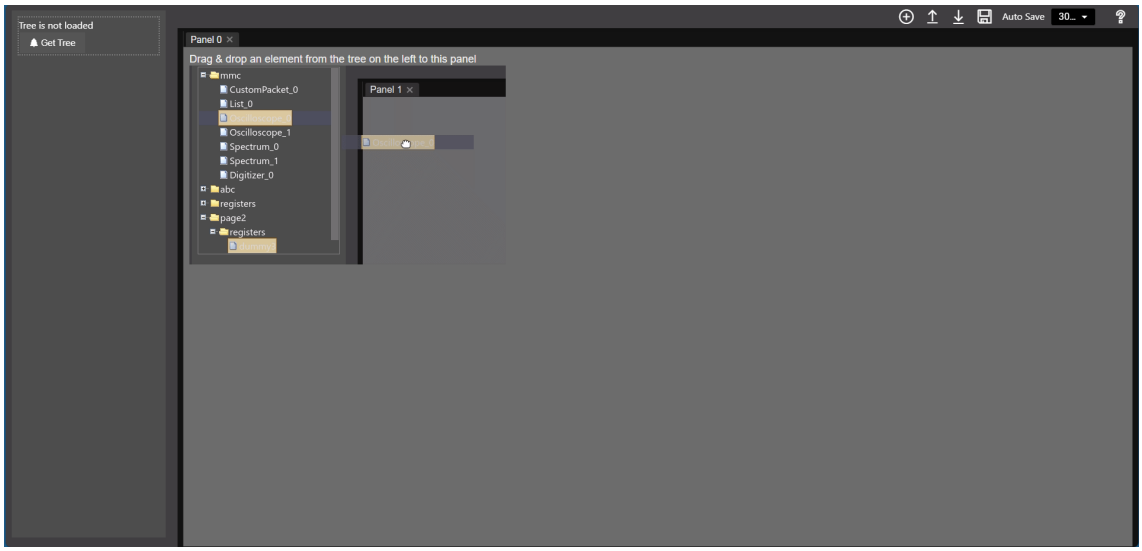
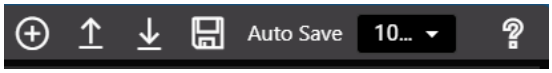


Fig. 12.18: The Resource Explorer at the first opening: the tree is loaded with Get Tree, then an element is dragged from the tree to the panel.

12.3.2 Panels

The elements are shown in **panels**, each one with its own tab in the upper left corner (Panel 0, Panel 1, ...). To create a new panel, press the **+** button of the top bar (**Fig. 12.19a**), type the name of the panel in the **Add new Panel** window and press **Add** (**Fig. 12.19b**). Once added, the panel must be dragged to be actually opened: its tab then appears in the upper left corner, and the element to be displayed can be dragged into it from the tree. **Fig. 12.20** shows a panel with the oscilloscope of a channel, before the acquisition is started.



(a) Top bar.



(b) Add new Panel window.

Fig. 12.19: Creation of a new panel.

The other controls of the top bar are, from left to right:

- **Import a previous layout from file**;
- **Download this layout**, to save the current layout of the panels to a file;
- the save button and the **Auto Save** menu, with intervals from 10 s to 5 min (**Fig. 12.21a**);
- the **?** button, which shows the versions of the frontend (FE) and of the backend (BE) of the Resource Explorer (**Fig. 12.21b**).

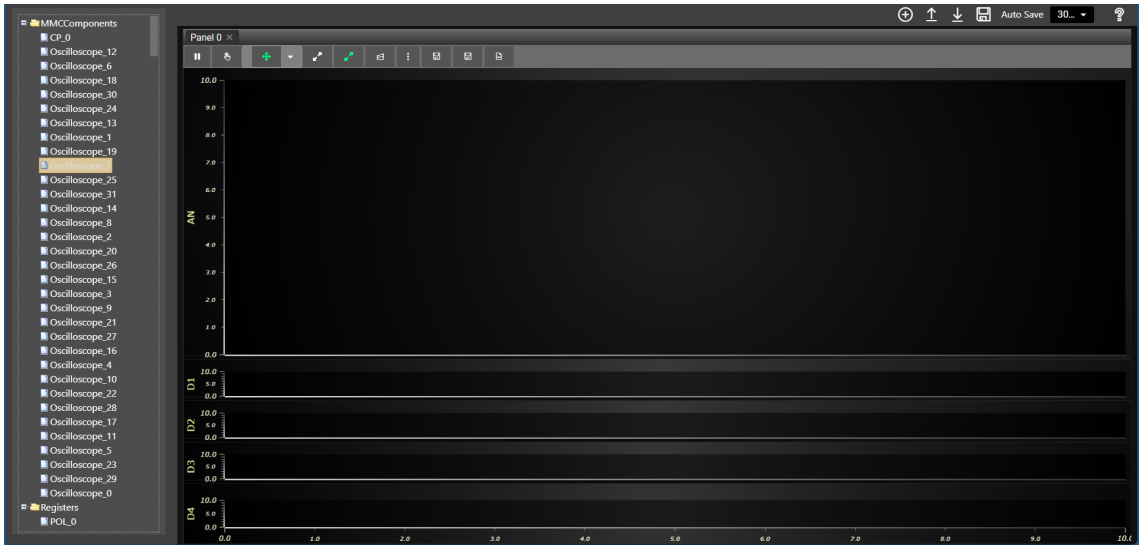
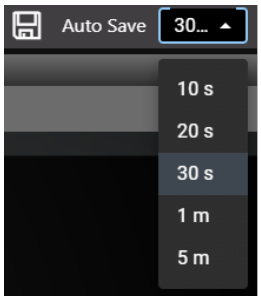


Fig. 12.20: A panel with the oscilloscope of a channel, acquisition not started yet.



(a) Auto Save menu.



(b) Versions of frontend and backend.

Fig. 12.21: The Auto Save menu and the version information of the top bar.

12.3.3 Panel toolbar

Each panel has its own toolbar (**Fig. 12.22a**). From left to right:



(a) Panel toolbar.

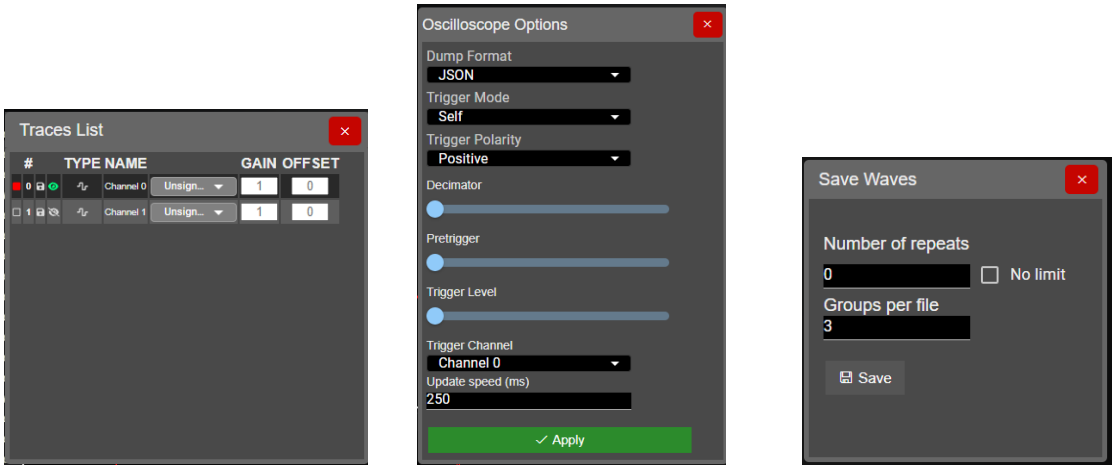


(b) Zoom.

Fig. 12.22: The panel toolbar and the zoom submenu.

- **Start/Stop**: starts and stops the acquisition;
- **Acquire one frame**: stops the continuous acquisition and shows one acquisition at a time, a new one every time the button is pressed;
- **Zoom**: manages the zoom; the arrow next to it opens the submenu of **Fig. 12.22b**, to zoom on the horizontal axis, on the vertical axis or on both;

- **Reset Zoom**;
- **Autoscale**;
- **Show Traces List**: opens the list of the traces of the panel (**Fig. 12.23a**), where each trace can be shown or hidden and its gain and offset can be set;
- **Show Options**: opens the options of the instrument (**Fig. 12.23b**), that manage the trigger (mode, polarity, level and channel), the decimator, the pretrigger, the dump format of the saved data and the update speed of the plot; press **Apply** to make the changes effective;
- **Save current waves**: saves the waves currently displayed;
- **Set multiple saves**: opens the **Save Waves** window (**Fig. 12.23c**), where it is possible to set the number of repeats (or no limit) and the number of groups per file, then press **Save**;
- **Save chart as an image file**.



(a) Traces List. **(b)** Oscilloscope Options. **(c)** Save Waves.

Fig. 12.23: The windows opened from the panel toolbar.

Fig. 12.24 shows the oscilloscope of **Fig. 12.20** during the acquisition: the analog trace (AN) and the digital traces (D1 to D4).

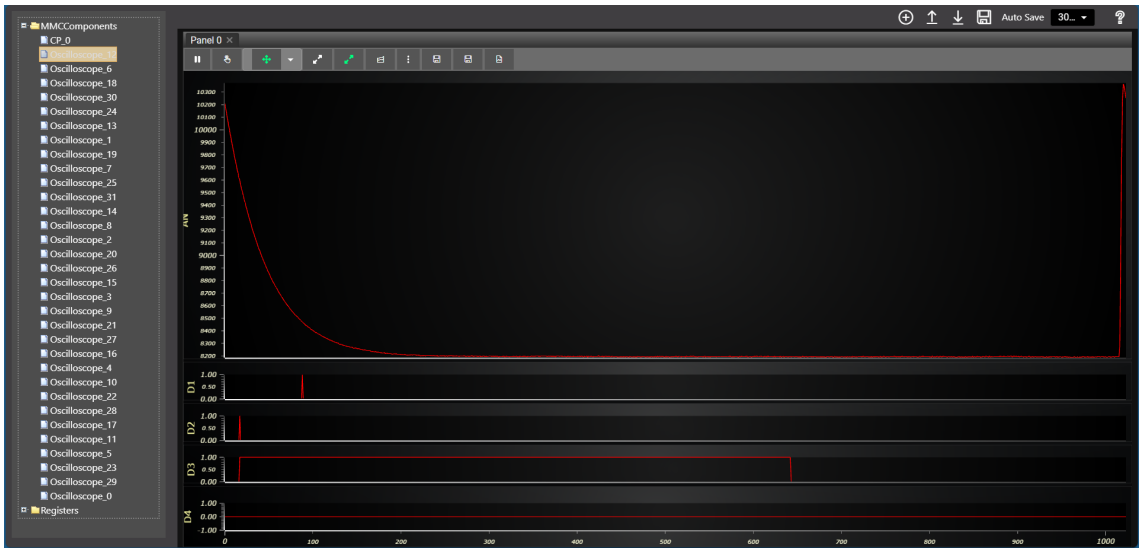


Fig. 12.24: The oscilloscope of a channel during the acquisition.

13 Firmware developing

The DT5560SE/R5560/R5560SE is a programmable platform and it is designed in order to encourage the user to develop their own custom firmware using **SCI-Compiler** to generate and compile the firmware code.

A full working **default firmware** is provided: this is a fully featured solution and it is developed not as a basic example to start developing with SCI-Compiler but as a full DAQ readout system firmware. The board comes with the default firmware already uploaded. In any case the firmware can be downloaded from the CAEN website and easily installed on the board with the OpenHardware – Firmware Upgrader tool (see Sec. 13.3).

The **default firmware** implements the typical features of Waveform Recording Digitizers and QDC algorithms. It is fully managed by the free-downloadable and open-source Open Hardware Readout Software. In more detail, the default firmware implements the following pulse processing features:

- **Waveform digitization** of all analog channels
- Leading edge or derivative trigger
- 32 independent channels **trapezoidal filter** for energy calculation on each DAQ section, for a total of 32 on the DT5560SE and 128 on the R5560/R5560SE
- Timestamping with 8 ns resolution
- Different readout modes: independent channels, frame trigger (OR of all channels trigger), external trigger
- External veto and trigger on programmable LEMO GPIO
- List readout mode
- Rate meter with dead time calculation on each channel

Note: The default firmware supports readout via Ethernet or USB. No support for the optical link is included in the default firmware.

13.1 Firmware design: the processing core

The DT5560SE/R5560/R5560SE processing core is a Xilinx Zynq-7000 SoC (Z-7030 or Z-7035), whose programmable logic is a Kintex-7 class FPGA. The FPGA can be programmed in several ways:

- **VHDL/Verilog**: this is the typical language for developing processing systems in FPGAs. It is based on basic operations (logic/arithmetic/conditional processing/sequential elements) connected to each other to build components. Components are then connected together in order to create more complex designs.
- **C**: a C/C++ program is automatically converted in a VHDL design by a software tool.
- **Design with high level blocks**: the user connects together a series of pre-designed building blocks in order to obtain the desired processing system.

SCI-Compiler is a set of very high-level blocks which implement the functionalities of the most common instrumentation used in physics experiments: digitizer, MCA, TDC, Time Over Threshold, trigger, scaler, etc.

SCI-Compiler is designed to work at best with the DT5560SE/R5560/R5560SE and CAEN suggests using this firmware generator and compiler tool, since it focuses the attention only on the functional blocks of the application to be implemented and does not require a deep knowledge of the device in use.

The user is free to develop a firmware in VHDL without using the SCI-Compiler. We suggest using the default framework to correctly initialize the board components. Framework source code is partially encrypted in order to protect our IP and avoid unauthorized copy of the board. The framework code is available inside the installation folder of SCI-Compiler.

SCI-Compiler generates VHDL code and Vivado Project. If the user wants to add custom functionalities that are not implementable with SCI-Compiler, it is possible to develop some parts of the design in VHDL and then compile as a standard Vivado Project.

The DT5560SE/R5560/R5560SE can be used for different applications, requiring different firmware (for example multichannel analog digitization, readout of mixed signal ASICs, direct readout of shaped signals).

In the following, we describe the default firmware.

13.2 Default firmware

The DT5560SE/R5560/R5560SE has a default firmware pre-programmed from factory. The default firmware implements a 32-channel digitizer for each DAQ section, with **waveform recording** and **Pulse Height Analysis**.

The default firmware is designed to work with Open Hardware Readout Software. It is fully developed in SCI-Compiler (see **Fig. 13.1**) and can be easily modified by the user. The default firmware project is available in the SCI-Compiler example section.

The default firmware implements an independent trapezoidal energy filter on each channel for energy calculation. It is based on a derivative trigger followed by a trapezoidal energy filter and gives in output a packet containing the timestamp and the energy of the input signal. The trapezoidal filter is a filter able to transform the typical exponential decay signal generated by a charge sensitive preamplifier into a trapezoid whose flat top height is proportional to the amplitude of the input pulse (that is to the energy released by the particle in the detector) (see **Fig. 13.2**). The trapezoid plays almost the same role of the shaping amplifier in a traditional analog acquisition system. There is an analogy between the two systems, both have a “shaping time” constant. For both, a long shaping time gives a better resolution but has higher probability of pile-up. Both are AC coupled with respect to the output of the preamplifier whose baseline is hence removed, but both have their own output DC offset and this constitutes another baseline for the peak detection.

The output data packet contains a header, a word for the energy, two words for the timestamp and a word for the channel ID.

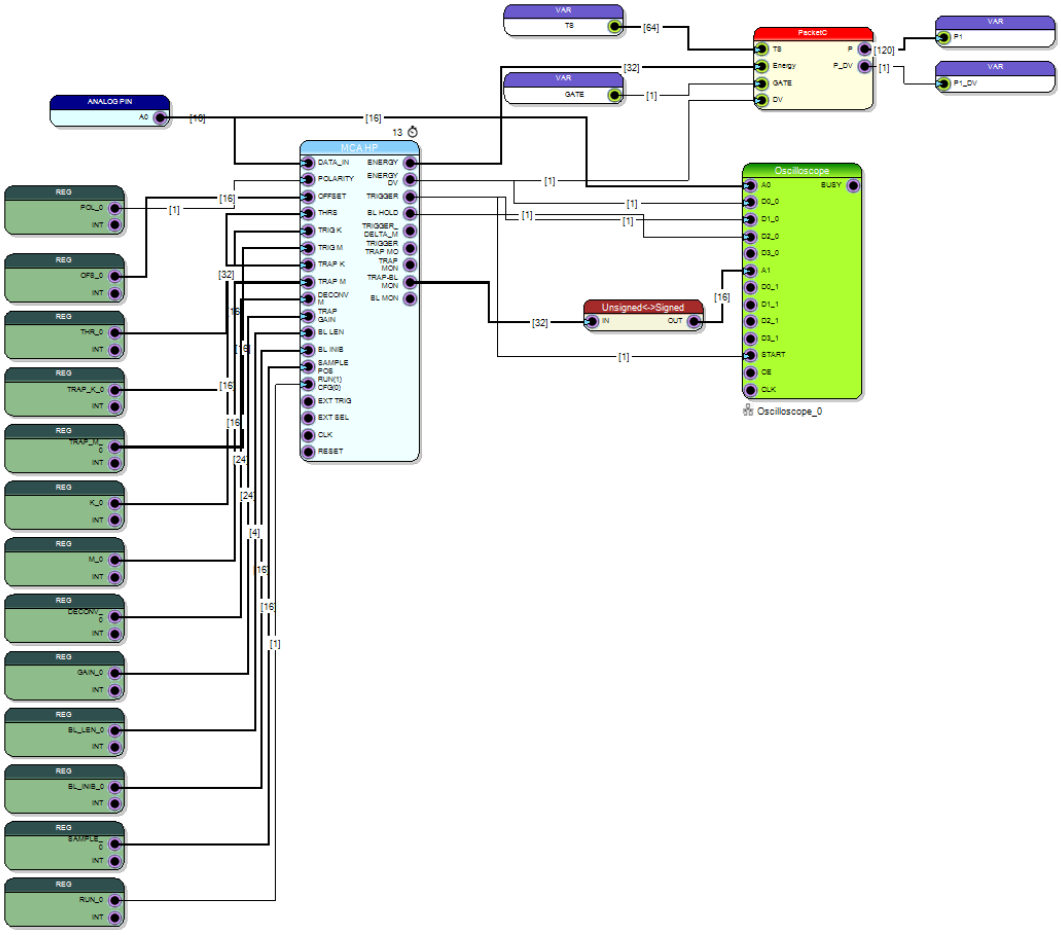


Fig. 13.1: SCI-Compiler scheme for single channel data processing in the default firmware.

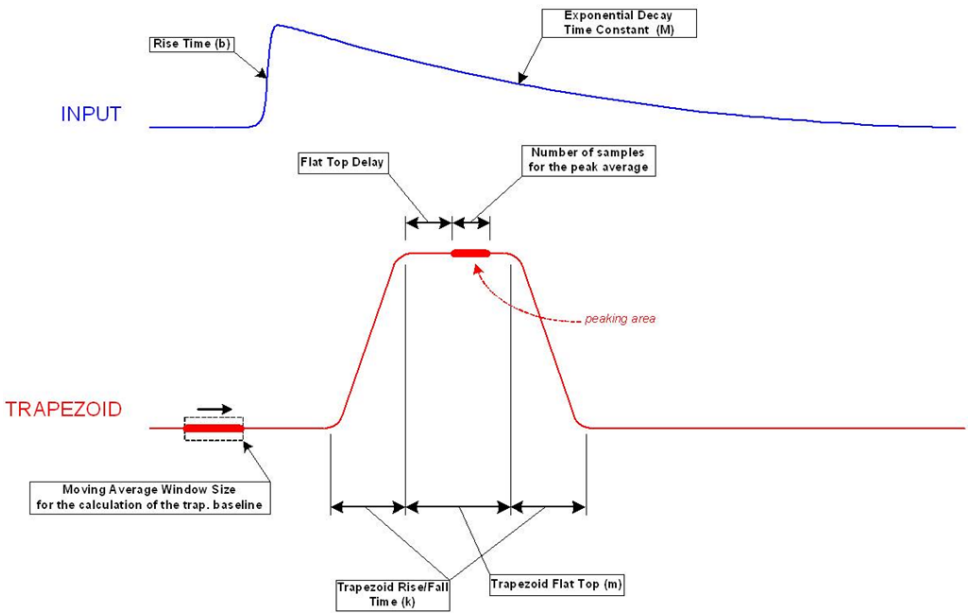


Fig. 13.2: The trapezoid method used in the board default firmware.

13.3 Firmware upgrade (Legacy)

Note: The OpenHardware Firmware Upgrader is a discontinued software: it remains available on the CAEN website as Legacy. The firmware can be upgraded from the Web Interface (see Sec. 12.1.2).

Apart from using the Web Interface (see Sec. 12.1.2) it is possible to change the flash firmware using the firmware upgrade tool (OpenHardware – Firmware Upgrader), included both in SCI-Compiler and Open Hardware Readout Software setup. The firmware upgrade tool interfaces with the FPGA bootloader via Ethernet in order to load the firmware in the flash memory of the board.

Note: On the R5560/R5560SE each DAQ section has its own flash memory and its own IP address, so each section must be upgraded individually.

In order to upgrade the firmware, follow the steps below:

1. Power on and connect the board to the PC through the Ethernet or USB port.
2. Browse to `C:\OpenHardware\FirmwareUpgrader`.
3. Open the **OpenHardware Firmware Upgrader**, which is automatically installed with SCI-Compiler or Open Hardware Readout Software.
4. A wizard will guide you in the firmware upgrade process. **Select your board model** among the available listed hardware (`DT5560SE` for the desktop unit) and press **Next**.

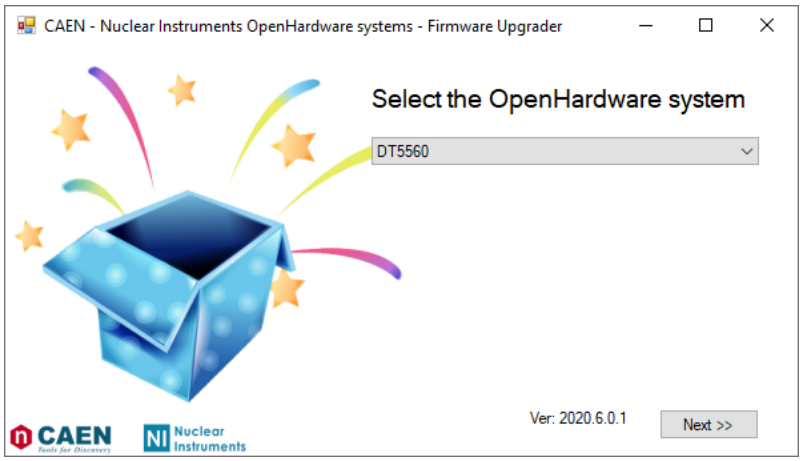


Fig. 13.3: Hardware selection in the OpenHardware Firmware Upgrader.

5. Insert the IP address of the board section to be upgraded and press **Next**.
6. **Load the file** containing the firmware code (.fpga in case of the default firmware available on the CAEN website) and press the "**Start**" button.
7. When the firmware is fully loaded a message invites the user to **power cycle the board** in order to load the new firmware.

THERE IS NO WAY FOR THE SYSTEM TO RECOGNIZE IF A FIRMWARE IS CORRECT FOR A PARTICULAR HARDWARE. FOR EXAMPLE, A FIRMWARE DESIGNED FOR ANOTHER BOARD CAN BE LOADED ON THIS BOARD BUT IT CAN DAMAGE IT. USER MUST CHECK THE HARDWARE TARGET FOR A SPECIFIC FIRMWARE BEFORE LOADING IT ON THE FLASH MEMORY



Fig. 13.4: IP address of the board to be upgraded.

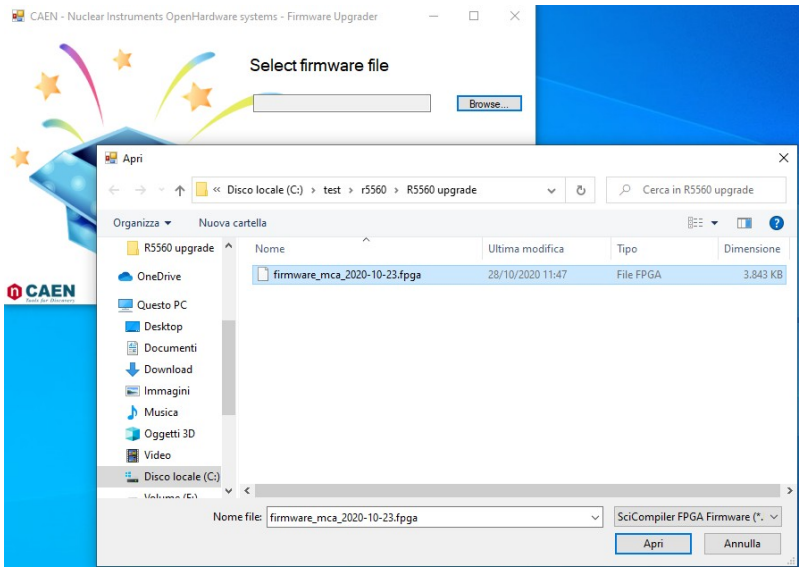


Fig. 13.5: Loading the firmware file.

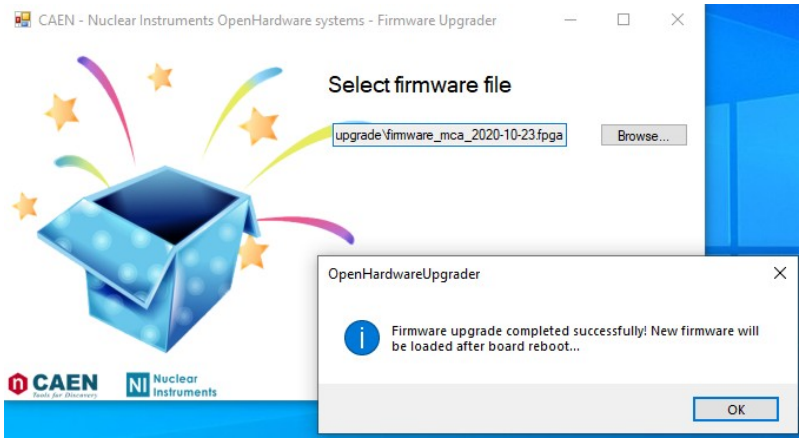


Fig. 13.6: Firmware upgrade completed.

Note: The latest release of the default firmware is available on the CAEN website at the relevant product page. The user can load it on the board using the OpenHardware – Firmware Upgrader, following the procedure described above.

13.4 Operating system upgrade (Legacy)

Note: The OpenHardware Firmware Upgrader is a discontinued software: it remains available on the CAEN website as Legacy. The operating system can be upgraded from the Web Interface (see Sec. 12.1.2).

It is possible to change the operating system file of the DAQ section (of each of the four DAQ sections on the R5560/R5560SE) using the firmware upgrade tool (OpenHardware – Firmware Upgrader), included both in SCI-Compiler and Open Hardware Readout Software setup. The firmware upgrade tool interfaces with the onboard ARM CPU via Ethernet in order to load the system file in the board memory.

In order to upgrade the Operating System, follow the steps below:

1. Power on and connect the board to the PC through the Ethernet port. On the R5560/R5560SE an Ethernet port is available on each of the 4 sections of the board, and each section should be upgraded individually since it has a different IP address.
2. Open the **OpenHardware Firmware Upgrader**, which is automatically installed with SCI-Compiler or Open Hardware Readout Software.
3. A wizard will guide you in the firmware upgrade process. **Select your board model** among the available listed hardware and press **Next** (see **Fig. 13.3**).
4. Insert the IP address of the board (on the R5560/R5560SE, of the section to be upgraded) and press **Next** (see **Fig. 13.4**).
5. Select the file type **Linux Image (*.rsys)**.
6. **Load the file** containing the Operating System (*.rsys on the CAEN website if new updates are available) and press the "**Start**" button.

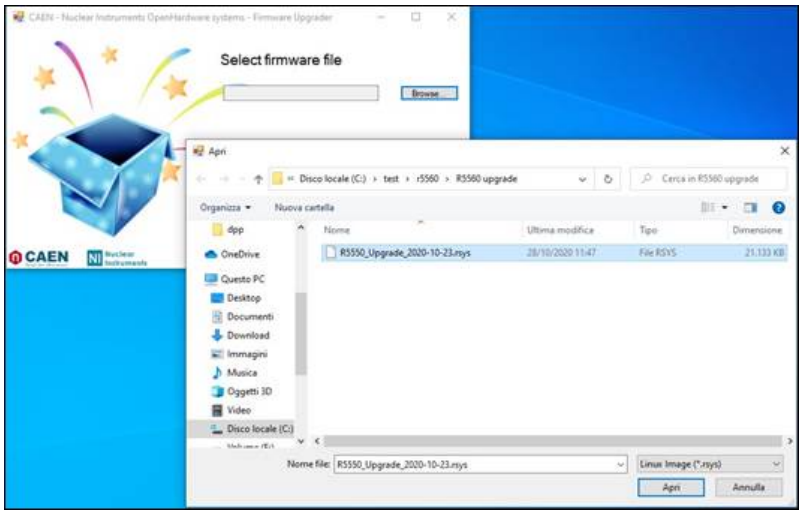


Fig. 13.7: Loading the Operating System file.

7. When the file is fully loaded a message invites the user to **power cycle the board** in order to load the new Operating System.

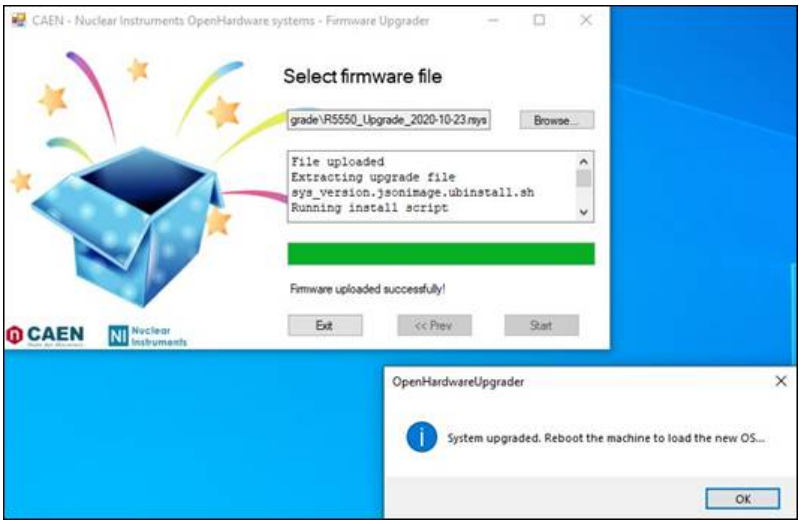


Fig. 13.8: Operating System upgrade completed.

14 Drivers & libraries

The DT5560SE/R5560/R5560SE does not require any driver to operate with Ethernet link and it is compatible with any OS. To interface with USB, an RNDIS driver is required in Windows OS, while it will work as is in modern Linux OS.

14.1 Drivers installation

14.1.1 DT5560SE

In order to install the USB drivers for correct operation in Windows OS, follow the steps below:

1. Connect the mini-USB connector of the DT5560SE to the PC.
2. Open the Device Manager of your computer. You will find an RNDIS device or an unknown device with yellow exclamation mark label.
3. Right click on the device → update driver → Browse my computer → Let me pick from a list → Network Adapters. Select Microsoft as manufacturer and Remote NDIS Compatible Device as model. Authorize the installation and the driver should be installed automatically.

14.1.2 R5560/R5560SE

To install the USB drivers for correct operation in Windows OS, follow the steps below:

1. Connect the USB 3.0 connector of the board to the PC and Windows will locate 6 new devices and a root hub:
   - 5x RNDIS compatible Ethernet card
   - 1x Serial Adapter

   The serial adapter does not require any driver.
2. Open the Device Manager of your computer. You will find 5 peripherals with yellow exclamation mark label or an RNDIS device or an unknown device.
3. Right click on each one → update driver → Browse my computer → Let me pick from a list → Network Adapters. Select Microsoft as manufacturer and Remote NDIS Compatible Device as model. Authorize the installation and the driver should be installed automatically.

14.2 Software development kit

Libraries for C/C++/C#/Python are available for custom firmware generated by SCI-Compiler under the form of a Software Development Kit called SciSDK. The SciSDK is installed together with the SCI-Compiler setup and is compatible with any board supported by SCI-Compiler.



Note: The SciSDK can be downloaded from `https://gitlab.sci-compiler.com/sdk/sci-sdk`. For specific documentation of the functions and parameters of the SciSDK, refer to the SCI-Compiler online Help or to `https://sci-sdk.sci-compiler.com/`.

Note: Alternative installation methods for the SciSDK are given in the online documentation.

The SDK is directly managed by SCI-Compiler when developing a custom firmware. When a memory mapped device (i.e. register, oscilloscope, readout list, ...) is added to SCI-Compiler, it generates SDK functions to directly access the endpoint of the memory mapped device. The auto-generated code is converted into a C library that can be compiled on Windows, Linux and macOS.

A Visual C++ project is generated in Windows. After compilation, the project generates a DLL that can be used in any programming language like Visual Studio (C/.NET), LabVIEW, Matlab, Java, etc.

In Linux, a gcc makefile is used to compile a .so dynamic library that can be installed in the system. The .so library can be used in other custom application developed in C, Qt studio or ROOT.

The SCI-Compiler SDK uses low-level Ethernet raw protocol to implement the communication with the device. Please refer to the automatically generated documentation when you compile a SCI-Compiler project for more information about connection and operations.

14.3 Libraries for the default firmware (Legacy)

Note: The library described in this section is Legacy: the supported library for the development of custom software is the SciSDK (see Sec. 14.2).

Note: A library to manage the default firmware, with open-source software example code, is available at `https://github.com/NuclearInstruments/r5560_sdk`. This repository is published by Nuclear Instruments. For new developments the recommended path is the SciSDK described in Sec. 14.2.

It contains the functions to start and close the connection with the device, to write and read data to/from registers, to write and read array of data to/from buffers and to read arrays of data from FIFO buffers.

Libraries compatible with the default firmware are also available in the installation folder of the Open Hardware Readout Software. See Chap. 16 for more details and examples of the libraries usage in C codes.

Note: In the reference below, `<BOARD>` stands for the model name: `DT5560SE_ConnectTCP` and `tDT5560SE_Handle` on the DT5560SE, `R5560SE_ConnectTCP` and `tR5560SE_Handle` on the R5560SE. All the other functions, whose names start with `NI_`, are identical on every model.

All the functions require a connection handle to specify the board to communicate with. The handle is defined in the following way:

```
typedef struct {
    int Csocket;
    int connected;
    uint32_t __IICBASEADDRESS;
    uint32_t __IICBASEADDRESS_STATUS;
    SOCKET_TYPE socketType;
} t<BOARD>_Handle;
```

All the functions return an integer indicating the communication status:

- 0: ok
- -1: no connection
- -2: socket error
- -3: connection drop
- -4: allocation error
- -5: out of limit error
- -6: timeout error

14.3.1 Start connection

Starts the Ethernet connection to the device. The arguments are the IP address passed as a string, the connection port and the connection handle to the board which will be returned by the function.

```
int <BOARD>_ConnectTCP(char *ipaddress, uint32_t port,
                       t<BOARD>_Handle *handle);
```

Argument	Direction	Type
ipaddress	Input	char
port	Input	uint32_t
handle	Input/output	t<BOARD>_Handle

Tab. 14.1: Arguments of the Start Connection function.

Example:

```
t<BOARD>_Handle handle;
<BOARD>_ConnectTCP("192.168.50.150", 8888, &handle);
```

In the code, <BOARD> is replaced by the model name, as explained at the beginning of Sec. 14.3. The same example, written for each model:

- **DT5560SE**:

  ```
  tDT5560SE_Handle handle;
  DT5560SE_ConnectTCP("192.168.50.150", 8888, &handle);
  ```

- **R5560SE**:

  ```
  tR5560SE_Handle handle;
  R5560SE_ConnectTCP("192.168.50.150", 8888, &handle);
  ```

14.3.2 Close connection

Disconnects the device specified by the connection handle passed as argument.

```
int NI_CloseConnection(t<BOARD>_Handle *handle);
```

Argument	Direction	Type
handle	Input/output	t<BOARD>_Handle

Tab. 14.2: Arguments of the Close Connection function.

Example:

```
t<BOARD>_Handle handle;
NI_CloseConnection(&handle);
```

14.3.3 Write register

Writes a value in a register of the FPGA in the device. The arguments are the data to be written in the register, the register address to be written and the board connection handle.

```
int NI_WriteReg(uint32_t data, uint32_t address,
                t<BOARD>_Handle *handle);
```

Argument	Direction	Type
data	Input	uint32_t
address	Input	uint32_t
handle	Input/output	t<BOARD>_Handle

Tab. 14.3: Arguments of the Write Register function.

Example:

```
t<BOARD>_Handle handle;
NI_WriteReg(1, 0xA, &handle);
```

14.3.4 Read register

Reads a value stored in a register of the FPGA in the device. The arguments are a pointer returning the data read from the register, the register address to be read and the board connection handle.

```
int NI_ReadReg(uint32_t *data, uint32_t address,
               t<BOARD>_Handle *handle);
```

Argument	Direction	Type
data	Input/output	uint32_t
address	Input	uint32_t
handle	Input/output	t<BOARD>_Handle

Tab. 14.4: Arguments of the Read Register function.

Example:

```
t<BOARD>_Handle handle;
uint32_t data;
NI_ReadReg(&data, 0xA, &handle);
```

14.3.5 Write data

Writes an array of values in a register of the FPGA in the device. The arguments are the data array to be written in the register, the number of data to be written, the register address to be written, the board connection handle and the number of data written in the register returned by the function.

```
int NI_WriteData(uint32_t *data,
                 uint32_t count,
                 uint32_t address,
                 t<BOARD>_Handle *handle,
                 uint32_t *written_data);
```

Argument	Direction	Type
data	Input/output	uint32_t
count	Input	uint32_t
address	Input	uint32_t
handle	Input/output	t<BOARD>_Handle
read_data	Input/output	uint32_t

Tab. 14.5: Arguments of the Write Data function.

Example:

```
t<BOARD>_Handle handle;
uint32_t data[10] = {0, 2, 10, 1, 0, 1, 3, 1, 2, 0};
uint32_t written_data;
NI_WriteData(&data, 10, 0xA, &handle, &written_data);
```

14.3.6 Read data

Reads an array of values from a buffer of the FPGA in the device. The arguments are the preallocated array of data to be read from the buffer, the number of data to be read, the buffer address to be read, the board connection handle and the number of data read from the buffer returned by the function.

```
int NI_ReadData(uint32_t *data,
                uint32_t count,
                uint32_t address,
                t<BOARD>_Handle *handle,
                uint32_t *read_data);
```

Argument	Direction	Type
data	Input/output	uint32_t
count	Input	uint32_t
address	Input	uint32_t
handle	Input/output	t<BOARD>_Handle
written_data	Input/output	uint32_t

Tab. 14.6: Arguments of the Read Data function.

Example:

```
t<BOARD>_Handle handle;
uint32_t data[100];
uint32_t read_data;
NI_ReadData(&data, 100, 0xA, &handle, &read_data);
```

14.3.7 Read FIFO

Reads data from a FIFO buffer of the FPGA in the device. The arguments are the pre-allocated array of data to be read from the buffer, the number of data to be read, the buffer address to be read, the address of the register containing the FIFO status, the type of bus mode (0 for register access like, 1 for blocking streaming access like and 2 for non-blocking streaming access like), the timeout of the read operation in milliseconds, the board connection handle and the number of data read from the buffer returned by the function.

```
int NI_ReadFifo(uint32_t *data,
                uint32_t count,
                uint32_t address,
                uint32_t fifo_status_address,
                BUS_MODE bus_mode,
                uint32_t timeout_ms,
                t<BOARD>_Handle *handle,
                uint32_t *read_data);
```

Argument	Direction	Type
data	Input/output	uint32_t
count	Input	uint32_t
address	Input	uint32_t
fifo_status_address	Input	uint32_t
bus_mode	Input	BUS_MODE
timeout_ms	Input	uint32_t
handle	Input/output	t<BOARD>_Handle
read_data	Input/output	uint32_t

Tab. 14.7: Arguments of the Read FIFO function.

Example:

```
t<BOARD>_Handle handle;
uint32_t data[100];
uint32_t read_data;
NI_ReadFifo(&data, 100, 0xA, 0xB, 1, 1000, &handle, &read_data);
```

15 Open Hardware Readout Software

Open Hardware Readout Software is **free and open-source** software developed for Windows OS to operate **in conjunction with the default firmware** of the DT5560SE/R5560/R5560SE, in order to provide a ready-to-use solution.

Moreover, CAEN provides free and open source libraries as a Software Development Kit (see Chap. 14 for more details).

The software implements typical features needed to acquire and process data in nuclear spectroscopy and particle physics:

- Waveform monitor for all the channels in free running mode or with leading edge trigger
- List mode readout (energy, time) and data saving with channel independent trigger
- Energy Spectrum plot for all channels
- Bidimensional heatmap visualization for imaging, with configurable detector shape
- Spectrum fitting and energy calibration
- Data saving with waveform dump on file

The software is distributed both as compiled application and as source code. The source code is written in VB.NET and C# and it can be easily customized by the user to adapt to a custom firmware and for any other need. In order to recompile the Open Hardware Readout Software, a free version of Visual Studio .NET 2015 or later must be installed on the user's PC.

The following guide and screenshots refer to Open Hardware Readout Software version 2026.9.3.1, with a single DAQ section connected.

15.1 Software installation

Open Hardware Readout Software is compliant with Windows 10, 64-bit.



Note: The software is standalone and does not require the prior installation of any library.

In order to install the Open Hardware Readout Software, follow the steps below:

1. Download the software package from the product page of your board on the CAEN website (**login required**).
2. Unzip and run the executable: the setup wizard starts.
3. Choose the destination folder and **press "Next"**. By default, the software is installed in `C:\OpenHardware\OHReadout`.

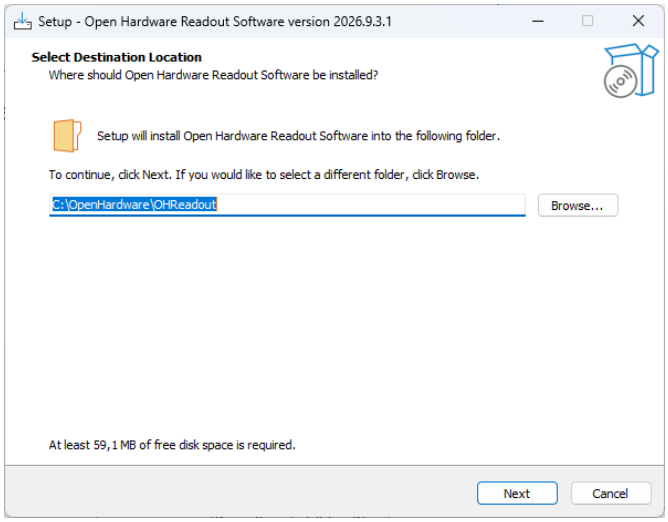


Fig. 15.1: Choice of the destination folder.

4. Choose the Start Menu folder where the program's shortcuts are created (by default **Open Hardware Platforms**) and **press "Next"**. Select **Don't create a Start Menu folder** to skip the shortcuts.

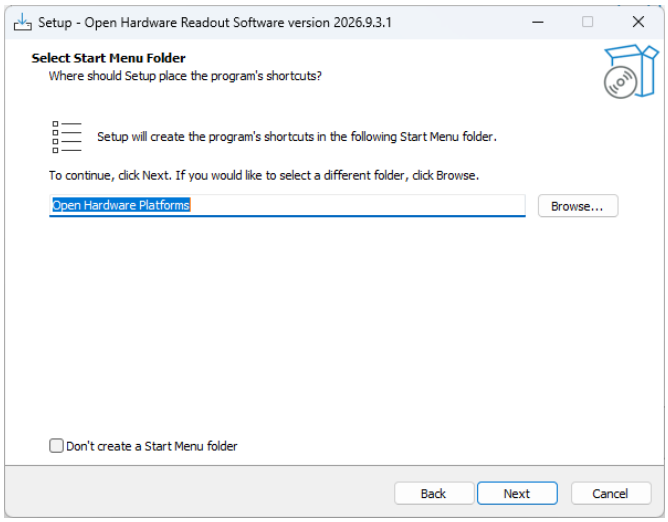


Fig. 15.2: Choice of the Start Menu folder.

5. Check the destination location and the Start Menu folder in the summary, then **click "Install"** to start the installation. Press **Back** to change any setting.

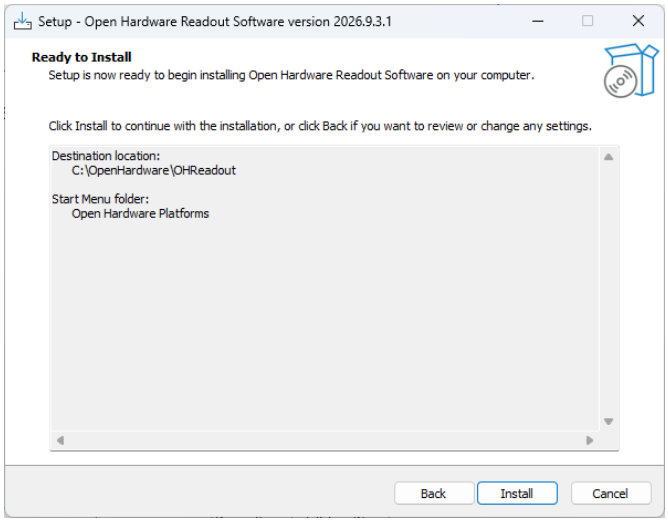


Fig. 15.3: Summary of the settings before the installation.

6. Wait while the setup copies the files.

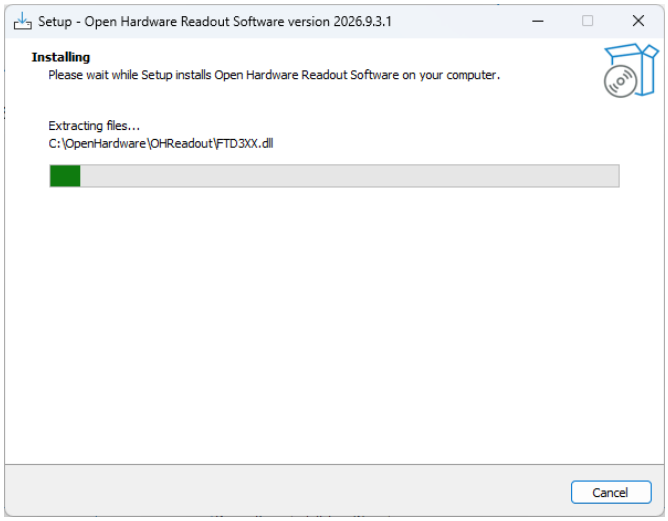


Fig. 15.4: Installation in progress.

7. At the end of the installation, leave the two driver options selected (**Install DT1260 Drivers (FTDI D2XX)** and **Install DT5550X Drivers (FTDI D3XX)**) and **press "Finish"**.

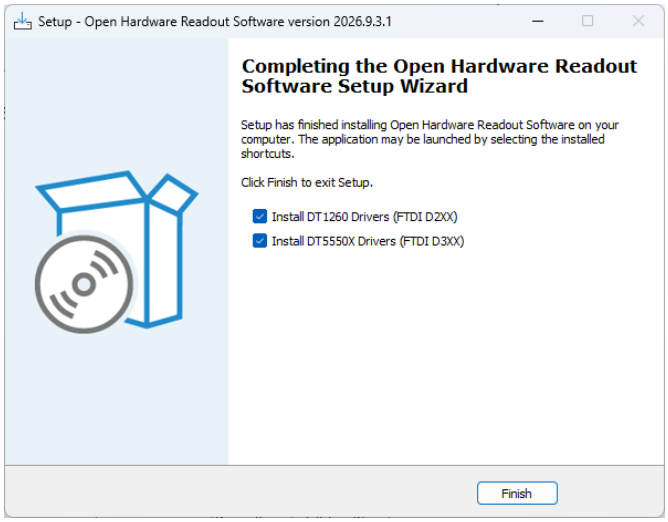


Fig. 15.5: Completion of the software installation, with the selection of the drivers to install.

8. Once Open Hardware Readout Software installation is complete, the Wizard will ask to extract and install FTDI USB Drivers (needed to operate with other compatible boards). Press "**Extract**" to continue.

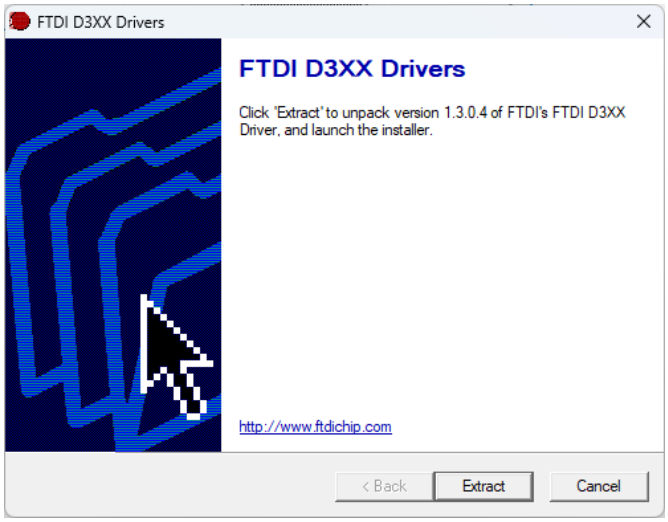
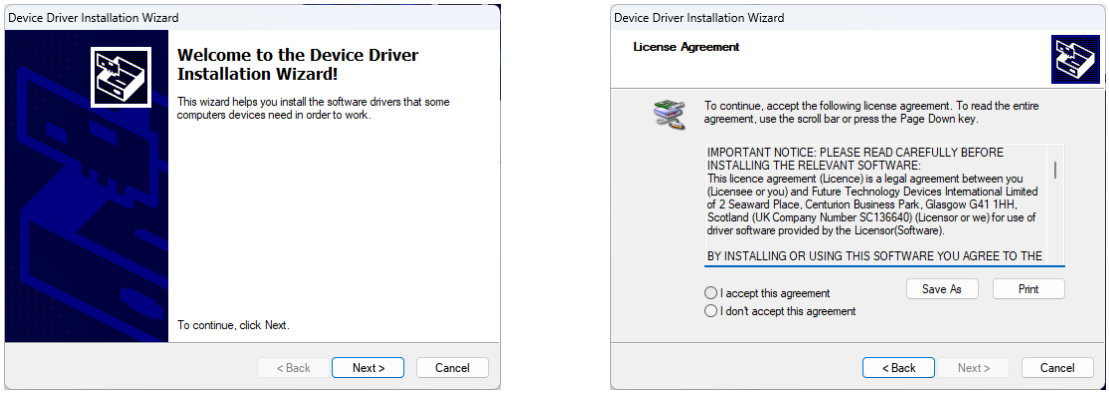


Fig. 15.6: Extraction of the FTDI USB Drivers.

9. **Press "Next"** in the following window to continue and **accept the License Agreement**.



(a) Driver installation wizard.

(b) License agreement.

Fig. 15.7: Installation of the FTDI USB Drivers.

10. The driver will then be installed. **Press "Finish"** to exit the Wizard.
11. Now it is possible to launch the software.

Note: The software is open source: it is possible to find the software VB.NET source code and the default firmware project (developed with SCI-Compiler) in the installation folder, to allow user customization.

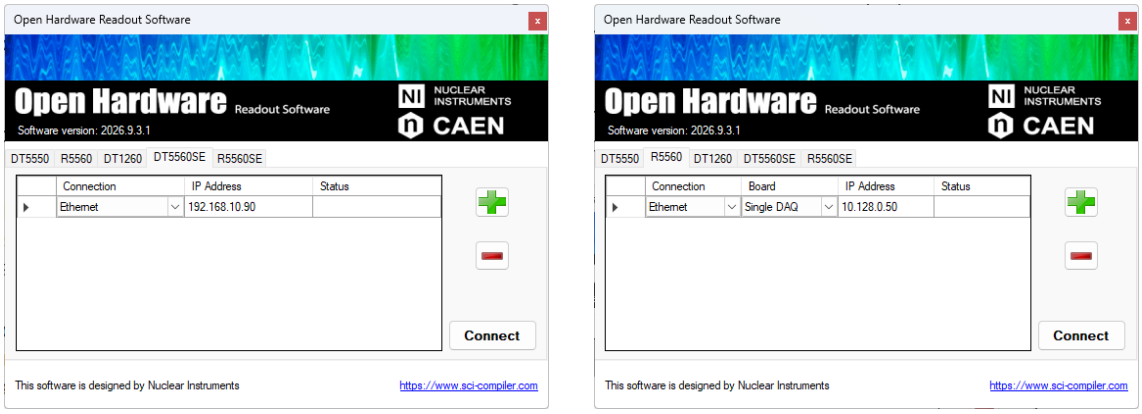
15.2 Board connection

After launching the software, the "Connection" window will open. By selecting the tab of your board the user can connect it by clicking on the 'Add' button. In the new row that appears in the table, it is possible to choose the connection type and to insert the IP address (Ethernet or USB) of the selected board type. On the R5560/R5560SE each of the four DAQ sections is connected as a separate entry, since each section has its own IP address.

- Connection: it is possible to choose between 'Ethernet' and 'USB'. Both connections work upon IP address protocol.
- IP Address: insert the IP address corresponding to the board to be connected and to the relevant connection interface. The Ethernet and USB addresses are shown on the Touchscreen display of the board: on the main page for the DT5560SE, in the DAQ LAN Status tab for the R5560/R5560SE (see Chap. 11).

After the connection parameters have been set, the user can press the "Connect" button to initialize the software GUI. In the 'Status' field of each board, the connection status is displayed. If the connection cannot be established for one or more of the inserted boards, an error message box will appear.

The 'Remove' button deletes the selected row of the table.



(a) DT5560SE: a single board entry.

(b) R5560/R5560SE: an entry for a single DAQ section.

Fig. 15.8: The “Connection” window at start-up of Open Hardware Readout Software.

15.3 Software GUI description

After successful connection, the main window will appear. The window is divided into the following areas:

- Control bar: here there are all the buttons needed to control the acquisition process.
- Main working area: the “MCA Settings”, “Oscilloscope”, “Spectrum” and “Statistics” tabs are shown here.
- Imaging module: the 2D image of the channel energies is reconstructed in the two heatmaps (one for the single frame view and the other for the cumulative view).
- Log Area: all messages related to the user actions are displayed in the history log.
- DAQ Status: a panel that takes the place of the Log Area when its tab, next to the “Log File” tab at the bottom of the window, is selected.

All panels can be undocked and rearranged also outside the main window in order to organize the software layout on multiple monitors.

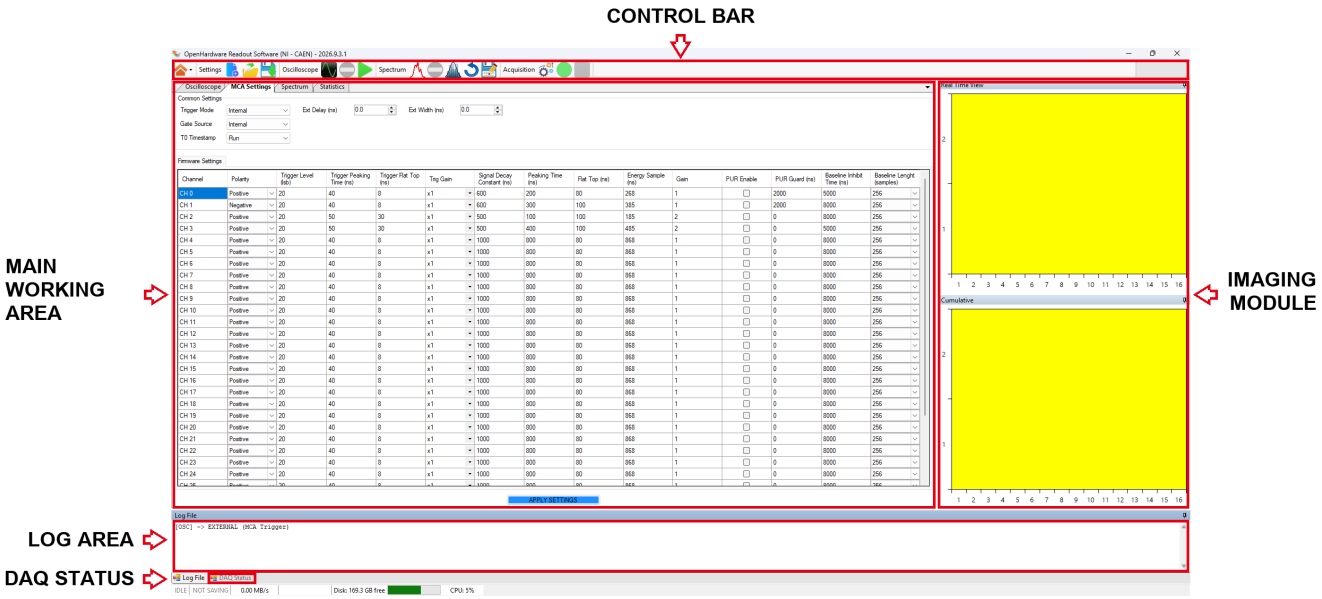


Fig. 15.9: The main window of Open Hardware Readout Software. The main areas are highlighted.

15.3.1 Control bar

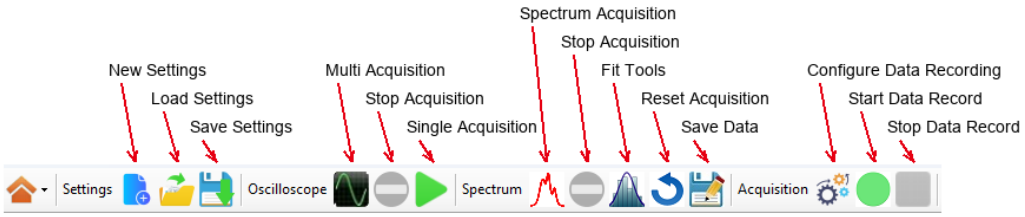


Fig. 15.10: The Control Bar of Open Hardware Readout Software.

The “Control Bar” contains the following buttons:

- 
  - **Spectrum**: change the visualization features of the spectrum (semi-logarithmic or linear on Y axis, rebinning on the X axis, graphical plot type)
  - **View**: select the zoom type
  - **File**: export and print plots
- 
  - **New Settings**: reset all settings to their default values
  - **Load** settings from file
  - **Save** Settings to file
- 
  - **Multi Acquisition**: start to acquire waves until stop acquisition button is pressed
  - **Stop Acquisition**: abort waves acquisition

  - **Single Acquisition**: acquire a single waveform every time the button is pressed

- 

  - **Spectrum Acquisition**: start spectrum and image acquisition process
  - **Stop Acquisition**: abort spectrum acquisition
  - **Fit Tools**: open the fitting tool window (see Sec. 15.4)
  - **Reset Acquisition**: reset spectrum and image
  - **Save Data**: save on disk the spectrum for each channel and the cumulative images

- 

  - **Configure Data Recording**: open the data dump window, the same one opened by Start Data Record, to configure the saving of data (waves or energy) event by event
  - **Start Data Record**: open the data dump window to configure and start saving data (waves or energy) event by event
  - **Stop Data Record**: stop current data dump process

15.3.2 MCA Settings tab

The MCA Settings Tab is used to configure all digitizer parameters. It is divided into two areas, as shown in **Fig. 15.11**:

- **Common Settings (red box)**: the trigger mode, the gate source and the T0 timestamp source, common to all channels
- **Channel Specific Settings (green box)**: this is divided into two tabs:
  - Firmware settings is a table with a row for each channel containing all configuration parameters for that channel, i.e. signal polarity, all the derivative trigger parameters, all the trapezoidal filter parameters and all the baseline calculation parameters.
  - AFE Settings (DT5560SE/R5560SE only) is a table with a row for each couple of channels, containing the configuration parameters of the analog frontend stage, i.e. the input impedance, the offset, shaping and analog coarse gain.

The user must push the button **APPLY SETTINGS**, at the bottom of the window, in order to let the settings become effective. It is suggested to stop an ongoing acquisition before applying new settings.

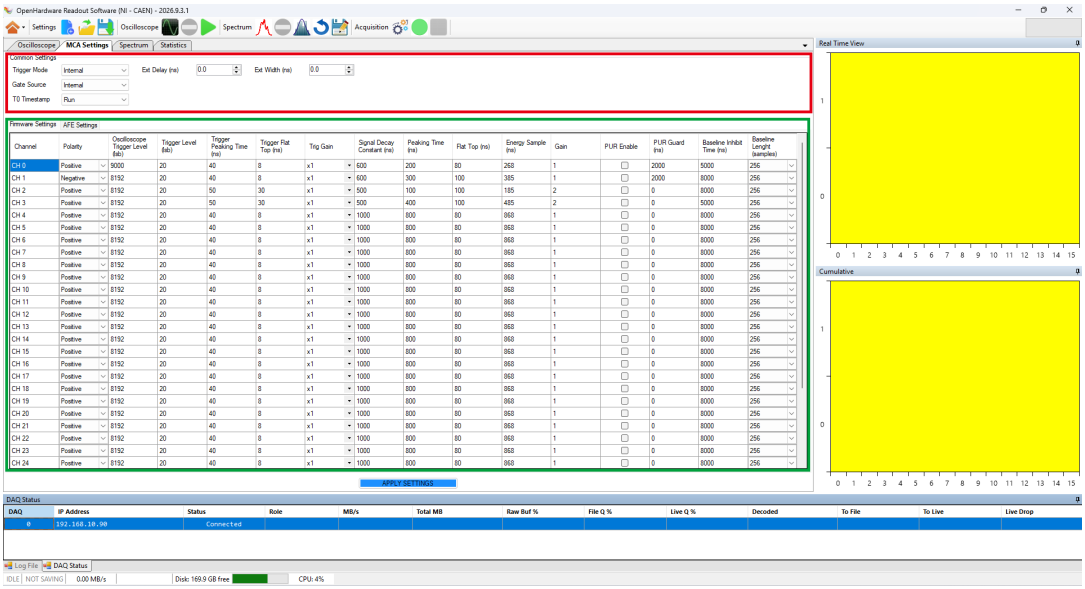


Fig. 15.11: The MCA Settings Tab of Open Hardware Readout Software.

15.3.2.1 Common settings

These settings (see red box in **Fig. 15.11**) are common to all the channels of the board.

- **Trigger Mode**: select the trigger mode among **Internal**, **EXT GATE**, **EXT VETO** and **EXT TRIGGER**.
- **Ext Delay (ns)** and **Ext Width (ns)**: set the delay and the width applied to the external signal used by the EXT GATE, EXT VETO and EXT TRIGGER modes.
- **Gate Source**: select the source of the gate among **Internal**, **LEMO 2 Level**, **LEMO 2 Edge** and **LEMO 0 Level (T0)**.
- **T0 Timestamp**: select the source of the T0 that resets the timestamp between **RUN** and **RUN + LEMO 0**.

Note: In the EXT GATE, EXT VETO and EXT TRIGGER modes the external signal must be connected to the **LEMO IN 1** input (USER IO connectors, see Sec. 9.2).

15.3.2.2 Firmware settings

These settings (see green box in **Fig. 15.11**) are used to configure the specific parameters of each channel, used by the MCA firmware component:

- Polarity: Select between positive and negative polarity according to the input channel.
- Oscilloscope Trigger level (lsb): set the absolute trigger threshold taken into consideration by the oscilloscope acquisition if Trigger Mode = Analog Signal in the Oscilloscope Settings of the Oscilloscope tab (it is the same parameter as the Thrs column of the channel table of that tab).
- Trigger Level (lsb): Set the derivative trigger level threshold. It is taken into consideration always for Spectrum acquisition, while for Oscilloscope acquisition only if Trigger Mode = MCA Trigger.
- Trigger Peaking Time (ns): Set the peaking time of the trapezoidal trigger signal (between 16 ns and 800 ns).
- Trigger Flat Top (ns): Set the flat top of the trapezoidal trigger signal.

- Trig Gain: Set the gain applied to the trigger signal.
- Signal Decay Constant (ns): Set the trigger inhibition time after a trigger.
- Peaking Time (ns): Set the peaking time of the energy trapezoidal filter (between 16 ns and 4 μs).
- Flat Top (ns): Set the flat top of the energy trapezoidal filter.
- Energy Sample (ns): Set the position where the value of the energy trapezoidal filter is sampled to measure the signal energy (should be peaking time + 0.8*Flat Top).
- Gain: Multiplication factor for the energy trapezoidal filter result. This allows recovering resolution avoiding throwing away the least significant bits of the energy filter. A low gain reduces the resolution of the measure while a too high gain will saturate the spectrum dynamic.
- PUR Enable: Enable the pile-up rejection.
- PUR Guard (ns): Set the time window in which the firmware looks for pile-up, when the pile-up rejection is enabled.
- Baseline Inhibit Time (ns): Inhibition time for the baseline calculation after a trigger signal. It avoids that the tail of an event is measured as part of the baseline (should be greater than 2*Decay Constant).
- Baseline Length (samples): Length in samples of the window for baseline calculation.

Note: The sum of the peaking time and of the flat top could not exceed 4096 ns. In this case the fields related to these parameters become red in the table and the 'Apply Settings' button is disabled.

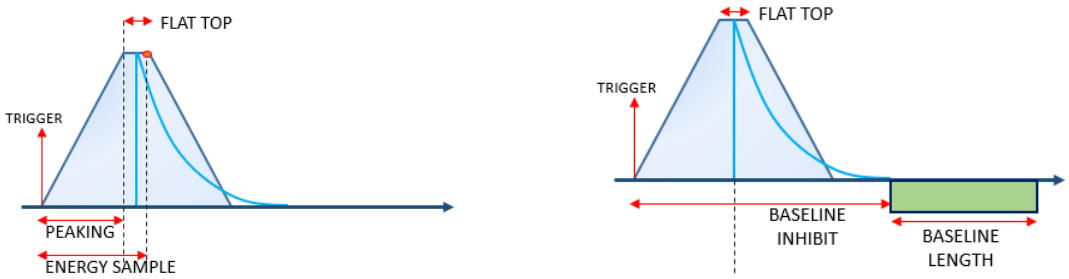


Fig. 15.12: Scheme of the trapezoidal energy filter parameters to be set in Open Hardware Readout Software.

15.3.2.3 AFE settings (DT5560SE/R5560SE only)

These settings (see **Fig. 15.13**) are used to configure the analog frontend parameters for each channel of the board. According to the description of the analog stage given in Sec. 10.2.2, it is possible to configure the following parameters:

- Shaper: Select the shaper type and its timing among DC, AC 1 μs, AC 10 μs, AC 30 μs. This parameter is common for all channels.
- Termination: it is possible to choose between 50 Ω and 1 kΩ input termination.
- Division by 5: enable/disable the analog input signal voltage divider ($\div$ 5).
- Offset Even (mV): set the offset of the even channel of the corresponding couple indicated in the Channel column.
- Offset Odd (mV): set the offset of the odd channel of the corresponding couple indicated in the Channel column.

- Gain: set the analog coarse gain for the relevant couple of channels among the listed possible values.

Note: Gain, Impedance and Division by 5 can be configured for each couple of channels, the Offset independently for each single channel, while the shaper is common for all channels.

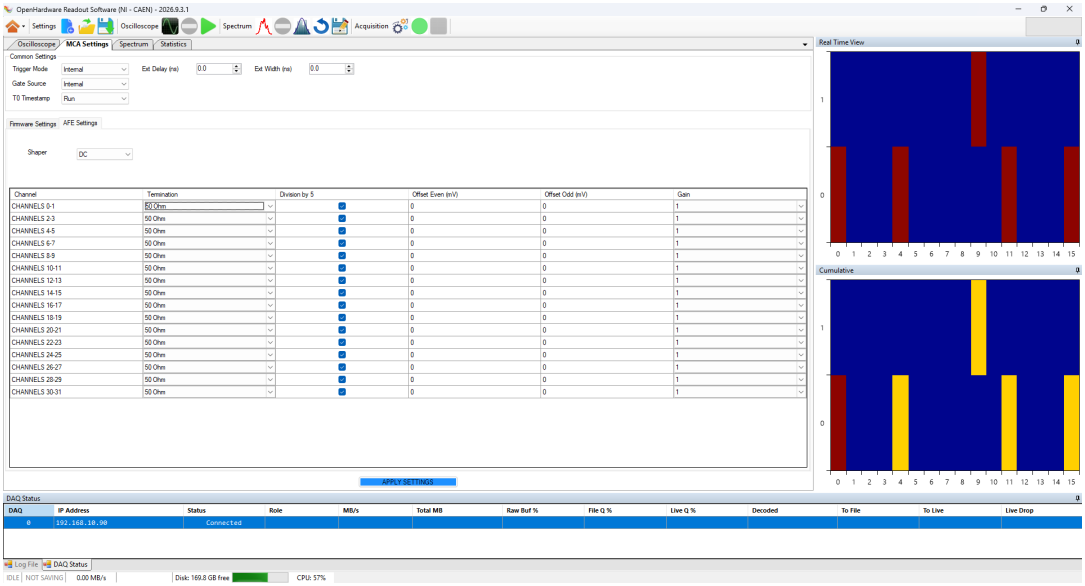


Fig. 15.13: The AFE Settings tab in Open Hardware Readout Software.

15.3.3 Oscilloscope tab

The Oscilloscope Tab allows the user to monitor on plot all analog input signals and digital probes:

- The ANALOG trace displays the waveform sampled by the ADC
- The TRAPEZOIDAL trace displays the trace selected in the **View** menu of the Oscilloscope Settings: the analog signal after the trapezoidal energy filter and the baseline subtraction (**Energy filter**), the trapezoidal trigger filter (**Trigger filter**) or the derivative trigger (**Trigger d**)
- The ENERGY SAMPLE digital signal indicates the position at which the value of the trapezoidal filter is sampled for the energy calculation
- The TRIGGER digital signal toggles when the trapezoidal trigger identifies a signal
- The BASELINE HOLD digital signal displays the state of the baseline restorer. When low, the baseline restorer is running, when high, it is holding

The table on the left of the Oscilloscope Tab has a row for each channel, with the following columns:

- **VIEW**: check it to display the channel; multiple input channels can be captured and displayed on the same plot
- **Thrs**: the threshold of the analog trigger of the oscilloscope (the same parameter as **Oscilloscope Trigger Level (lsb)** in the Firmware Settings of the MCA Settings tab)
- **POL**: the edge of the analog trigger of the oscilloscope, **R** (rising) or **F** (falling)

All oscilloscope configuration parameters can be set in the Oscilloscope Settings, at the top of the tab (see Sec. 15.3.3.1).

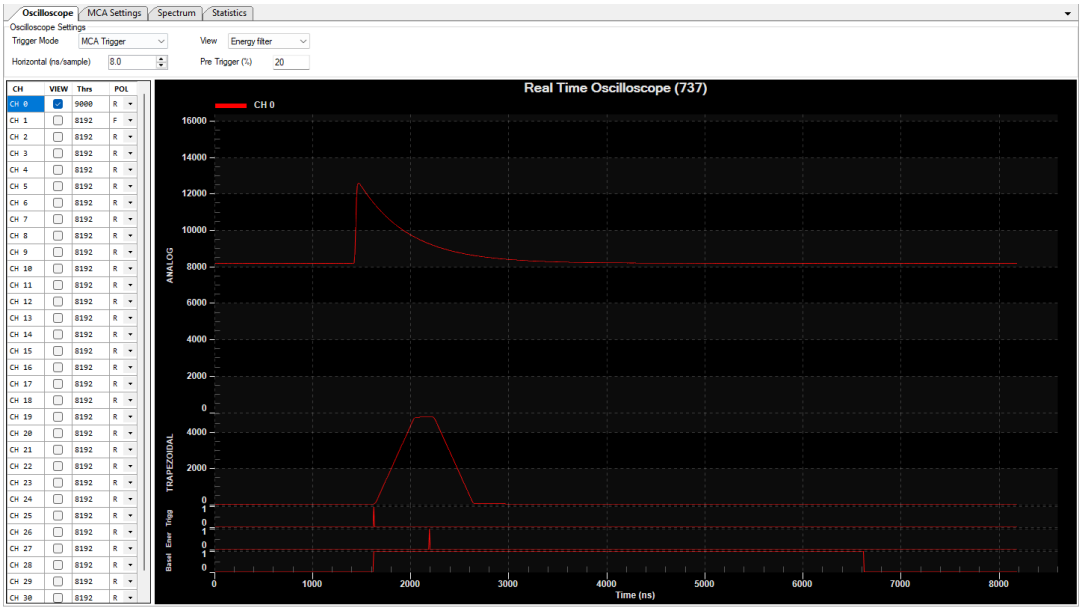


Fig. 15.14: The Oscilloscope Tab of Open Hardware Readout Software. Here only signals for channel 0 are displayed.

15.3.3.1 Oscilloscope settings

These settings are used to configure the same parameters of all the oscilloscopes (i.e. of the waveform recording).

- Trigger Mode: Select the trigger mode of the oscilloscopes. Options are:
  - **Analog Signal**: a leading-edge trigger on the analog input of each channel is implemented to trigger the corresponding oscilloscope waveform acquisition. The threshold (**Thrs**) and the edge (**POL**) must be properly set in the channel table of the Oscilloscope Tab.
  - **MCA Trigger**: the derivative trigger of each channel is used as a trigger for the corresponding oscilloscope waveform acquisition. The threshold to be set is **Trigger level (lsb)** in the Firmware Settings tab.
  - **Free running**: each oscilloscope operates without trigger and acquires a waveform as soon as the previous transfer is completed.
- View: select the trace displayed in the TRAPEZOIDAL plot among **Energy filter** (trapezoidal energy filter), **Trigger filter** (trapezoidal trigger filter) and **Trigger d** (derivative trigger)
- Horizontal (ns/sample): Set the time base for each oscilloscope in ns per sample
- Pre-Trigger (%): Specify the portion of the acquisition window dedicated to the pre-trigger (signal sampled before the trigger)

The value indicated in parentheses in the Oscilloscope plot title represents the current total number of waveform acquisitions.

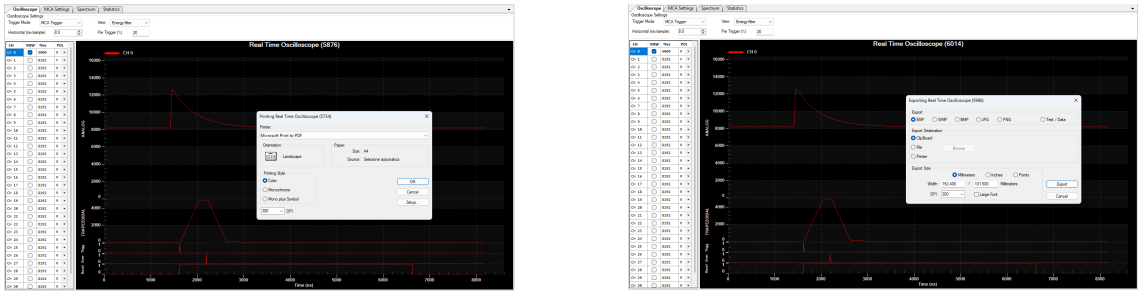
In order to zoom the waveforms, three zoom modes are available. The zoom mode can be selected pressing a key on your keyboard. The same operation can be done from Menu → View. Options are:

- Area (press Z): enables the zoom on both axes. Drag the mouse to zoom a rectangular area.
- Horizontal Zoom (Press H): enables horizontal zoom. Drag the mouse to zoom on the X-axis.
- Vertical Zoom (Press V): enables vertical zoom. Drag the mouse to zoom on the Y-axis.

- Unzoom (press U): restores the full view of the plot.

The user can also:

- print the current view of the plot: Menu → File → Print → Oscilloscope (or press P).
- export the current view of the plot: Menu → File → Export → Oscilloscope (alternatively press X).



(a) Printing.

(b) Saving to file.

Fig. 15.15: Printing (left) and saving to file (right) the current view of the Oscilloscope Tab.

15.3.4 Spectrum tab

The Spectrum Tab displays the cumulative spectrum of each channel calculated in real-time by the board.

The trapezoidal energy filter of the default firmware generates a 64k bin spectrum for each channel. The rebinning feature can be selected from the Menu → Spectrum → Rebin. The rebinning is applied to all channels.

The spectrum of multiple channels can be displayed on the same plot by checking the relative checkboxes in the channels list on the left of the Spectrum Tab. Each time a channel spectrum is added to the display area, a new label will appear in the legend and a color will be automatically selected for the new spectrum. When the spectrum is zoomed, the plot window is divided into two areas: the main area displays the current zoom of the spectrum, while an overview area at the bottom shows the full view of the spectrum.

In order to zoom the spectrum, three zoom modes are available. The zoom mode can be selected pressing a key on your keyboard. The same operation can be done from Menu → View.

- Area (press Z): enables the zoom on both axes. Drag the mouse to zoom a rectangular area.
- Horizontal Zoom (Press H): enables horizontal zoom. Drag the mouse to zoom on the X-axis.
- Vertical Zoom (Press V): enables vertical zoom. Drag the mouse to zoom on the Y-axis.
- Unzoom (press U): restores the full view of the plot.

The spectrum can be displayed in both linear and semi-logarithmic mode. It is possible to switch between Lin/Log mode: Menu → Spectrum → Lin/Log (alternatively Press L).

It is also possible to change the plot type: Menu → Spectrum → Plot Mode (alternatively press O to cycle between plot modes). Available plot modes are:

- Step
- Line
- Line with interpolation

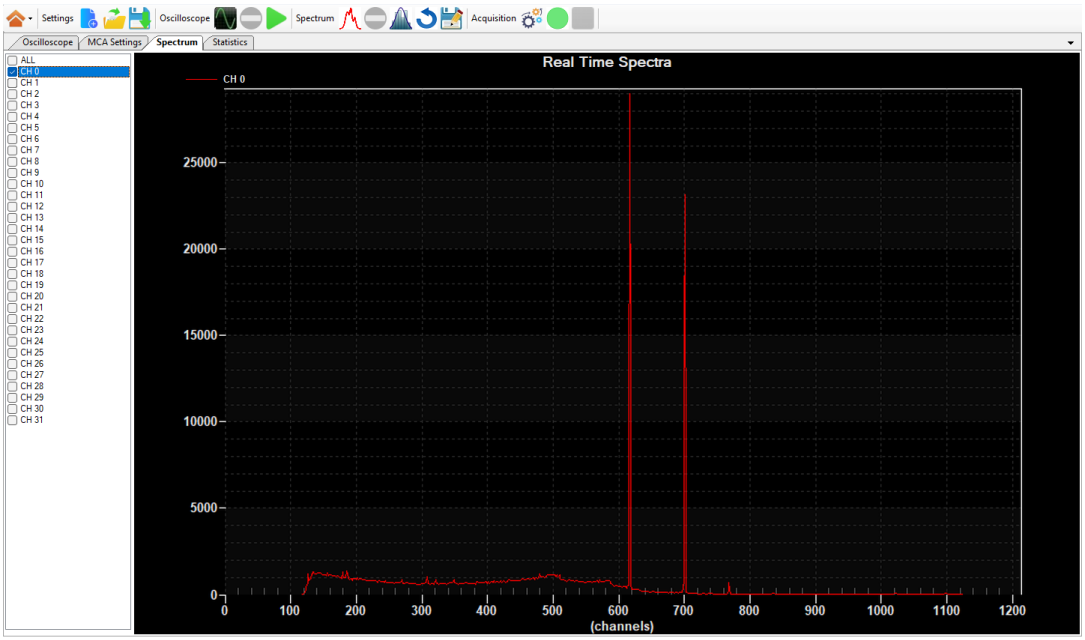
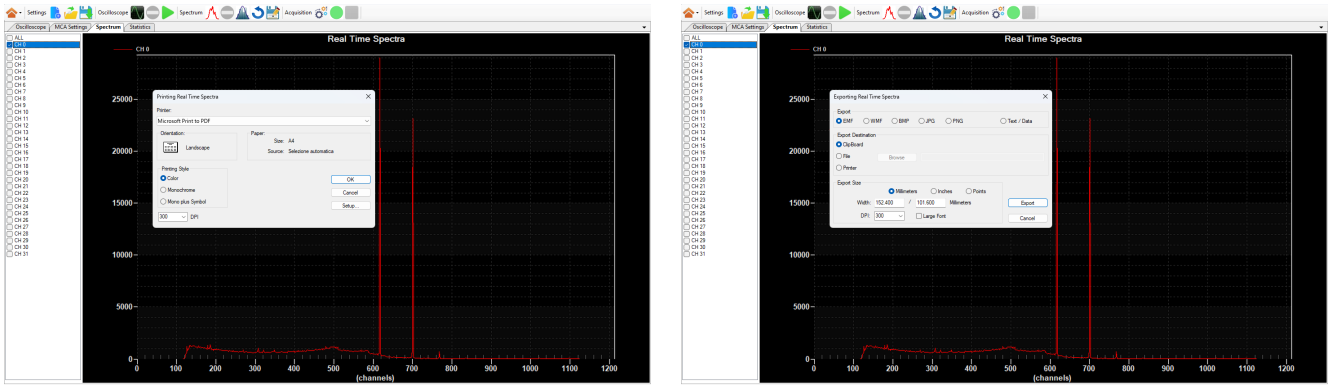


Fig. 15.16: The Spectrum Tab of Open Hardware Readout Software. Here only the spectrum for channel 0 is displayed.

- Bar
- Area
- Area with interpolation
- Dot
- Dot with Line
- Dot with interpolation

The user can:

- print the current view of the plot: Menu → File → Print → Spectrum (alternatively press P).
- save the current view of the plot: Menu → File → Export → Spectrum (alternatively press X).



(a) Printing.

(b) Saving to file.

Fig. 15.17: Printing (left) and saving to file (right) the current view of the Spectrum Tab.

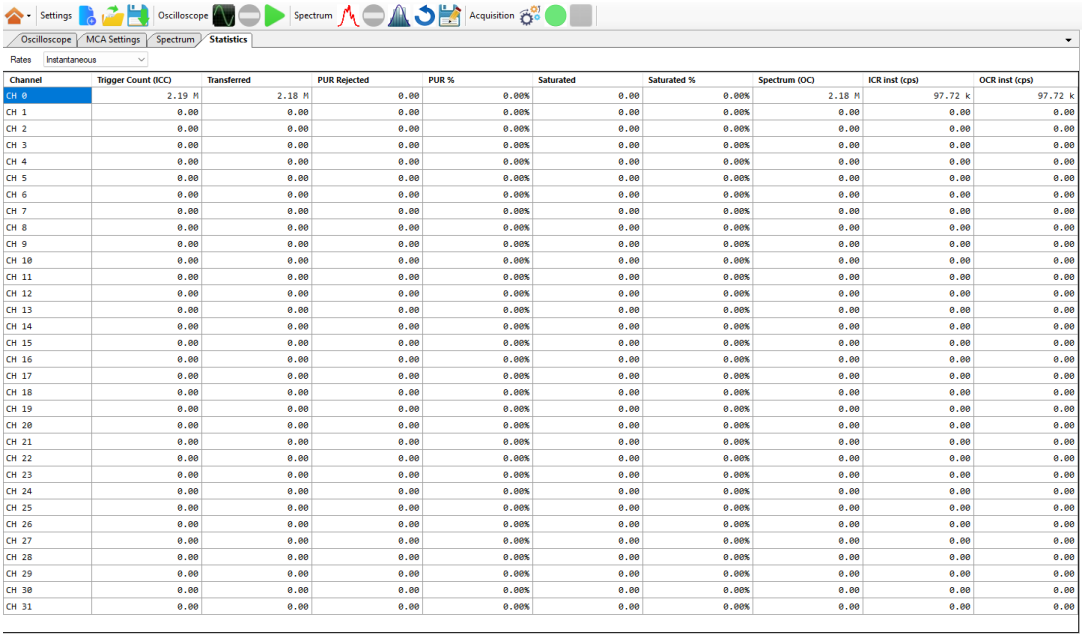
15.3.5 Statistics tab

The Statistics Tab shows, for each channel, the counters and the rates of the acquisition (see **Fig. 15.18**). With **Rates** it is possible to choose whether the rates are displayed as **Instantaneous** or **Average** values.

The table has a row for each channel, with the following columns:

- **Trigger Count (ICC)**: number of triggers (Input Count)
- **Transferred**: number of events transferred
- **PUR Rejected**: number of events rejected as pile-up
- **PUR %**: percentage of the events rejected as pile-up over the total
- **Saturated**: number of saturated events in the spectrum
- **Saturated %**: percentage of the saturated events in the spectrum over the total acquired
- **Spectrum (OC)**: number of good events, which populate the spectrum (Output Count)
- **ICR (cps)**: Input Count Rate, in counts per second
- **OCR (cps)**: Output Count Rate, in counts per second

The counters are displayed with the k (thousands) and M (millions) multipliers.



Channel	Trigger Count (ICC)	Transferred	PUR Rejected	PUR %	Saturated	Saturated %	Spectrum (OC)	ICR inst (cps)	OCR inst (cps)
CH 0	2.19 M	2.18 M	0.00	0.00%	0.00	0.00%	2.18 M	97.72 k	97.72 k
CH 1	0.00	0.00	0.00	0.00%	0.00	0.00%	0.00	0.00	0.00
CH 2	0.00	0.00	0.00	0.00%	0.00	0.00%	0.00	0.00	0.00
CH 3	0.00	0.00	0.00	0.00%	0.00	0.00%	0.00	0.00	0.00
CH 4	0.00	0.00	0.00	0.00%	0.00	0.00%	0.00	0.00	0.00
CH 5	0.00	0.00	0.00	0.00%	0.00	0.00%	0.00	0.00	0.00
CH 6	0.00	0.00	0.00	0.00%	0.00	0.00%	0.00	0.00	0.00
CH 7	0.00	0.00	0.00	0.00%	0.00	0.00%	0.00	0.00	0.00
CH 8	0.00	0.00	0.00	0.00%	0.00	0.00%	0.00	0.00	0.00
CH 9	0.00	0.00	0.00	0.00%	0.00	0.00%	0.00	0.00	0.00
CH 10	0.00	0.00	0.00	0.00%	0.00	0.00%	0.00	0.00	0.00
CH 11	0.00	0.00	0.00	0.00%	0.00	0.00%	0.00	0.00	0.00
CH 12	0.00	0.00	0.00	0.00%	0.00	0.00%	0.00	0.00	0.00
CH 13	0.00	0.00	0.00	0.00%	0.00	0.00%	0.00	0.00	0.00
CH 14	0.00	0.00	0.00	0.00%	0.00	0.00%	0.00	0.00	0.00
CH 15	0.00	0.00	0.00	0.00%	0.00	0.00%	0.00	0.00	0.00
CH 16	0.00	0.00	0.00	0.00%	0.00	0.00%	0.00	0.00	0.00
CH 17	0.00	0.00	0.00	0.00%	0.00	0.00%	0.00	0.00	0.00
CH 18	0.00	0.00	0.00	0.00%	0.00	0.00%	0.00	0.00	0.00
CH 19	0.00	0.00	0.00	0.00%	0.00	0.00%	0.00	0.00	0.00
CH 20	0.00	0.00	0.00	0.00%	0.00	0.00%	0.00	0.00	0.00
CH 21	0.00	0.00	0.00	0.00%	0.00	0.00%	0.00	0.00	0.00
CH 22	0.00	0.00	0.00	0.00%	0.00	0.00%	0.00	0.00	0.00
CH 23	0.00	0.00	0.00	0.00%	0.00	0.00%	0.00	0.00	0.00
CH 24	0.00	0.00	0.00	0.00%	0.00	0.00%	0.00	0.00	0.00
CH 25	0.00	0.00	0.00	0.00%	0.00	0.00%	0.00	0.00	0.00
CH 26	0.00	0.00	0.00	0.00%	0.00	0.00%	0.00	0.00	0.00
CH 27	0.00	0.00	0.00	0.00%	0.00	0.00%	0.00	0.00	0.00
CH 28	0.00	0.00	0.00	0.00%	0.00	0.00%	0.00	0.00	0.00
CH 29	0.00	0.00	0.00	0.00%	0.00	0.00%	0.00	0.00	0.00
CH 30	0.00	0.00	0.00	0.00%	0.00	0.00%	0.00	0.00	0.00
CH 31	0.00	0.00	0.00	0.00%	0.00	0.00%	0.00	0.00	0.00

Fig. 15.18: The Statistics Tab of Open Hardware Readout Software, with the instantaneous rates.

15.3.6 Imaging module

The Imaging Module displays the real-time and the cumulative 2D heatmaps acquired by the board, based on the measured energies for each channel. Each pixel of the image is associated with a channel. The user can rearrange the pixels to create a different shape.

The last captured event is displayed on the **real-time view** (top part of the display area). As the Custom Packet component transfers only one triggered event at a time, only one channel energy value will be displayed.

The bottom part of the display area shows the **cumulative image**.

The acquisition is controlled by the “Start Spectrum” and “Stop Spectrum” buttons in the Control Bar. The “Reset Acquisition” button for spectra in the Control Bar reinitializes also the acquisition of the heatmaps.

The “Save Data” button in the Control Bar saves the cumulative energies of all channels in a .csv file whose saving path can be chosen by the user.

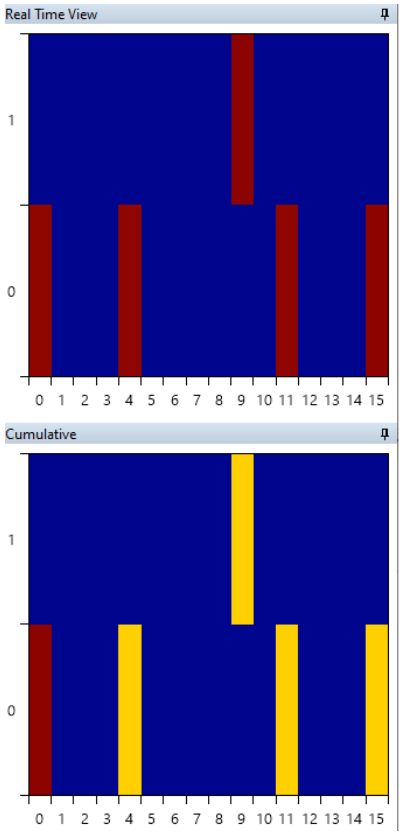


Fig. 15.19: The Imaging Module of Open Hardware Readout Software, where it is possible to see 5 channels acquiring energy data.

15.4 How to perform a fit

The fitting tool of Open Hardware Readout Software allows the user to perform Gaussian fits of the peaks in the spectrum.

In order to create a new fit, follow the instructions below:

1. a channel should be checked in the list of channels on the left of the Spectrum Tab.
2. press the "**Fit Tools**" button in the Control Bar to open the fitting tool: the **Fit** window opens at the bottom of the main window.
3. right-click in the Fit window and select "**Add ROI**": a new row is added to the table. Its **Name** column gives the ROI a name (ROI 1, ROI 2, ...), and the channel of the ROI is selected in the **Channel** column.
4. insert a value in the Cursor 1 and in the Cursor 2 columns of the table to specify respectively the left and the right boundaries of the fit. On the spectrum, the fit boundaries are displayed with white vertical lines and identified with a number corresponding to the table row number. The fit area is shown in green.
5. The fitting tool uses the spectrum data between the two cursors to calculate the Mean, the standard deviation (STD) and the Area of the selected region. For each fitted area the results extracted from the Gaussian fit are also reported: the mean (Mean Fit), the standard deviation (STD Fit), the full width at half maximum (FWHM), the Resolution (R) and the fitting area (Area Fit).
6. A fit can be deleted by selecting the corresponding row in the table and pressing 'Del' on your keyboard.

The ROIs can be hidden from the spectrum, or shown again, by right-clicking in the Fit window and selecting "**Hide/Show ROI**".

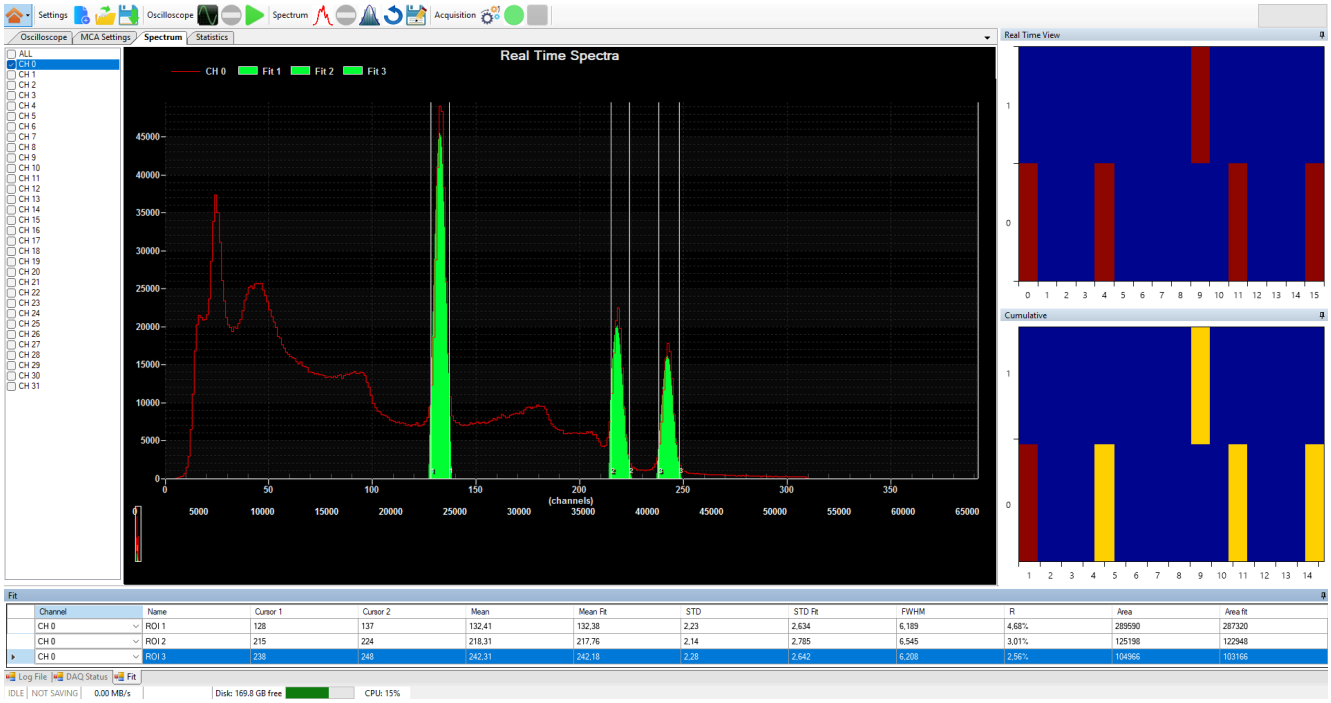


Fig. 15.20: The fitting tool of Open Hardware Readout Software.

15.5 How to save data

The "**Save Data**" button in the Control Bar saves both the spectrum and the cumulative image for the enabled channels on the Spectrum Tab. When the button is pressed a first dialog window is visualized to choose the file path for the spectrum data saving. A second dialog window is shown to set the file path for the cumulative image saving.

- The .csv spectrum file is made of several rows equal to the number of bins in the spectrum. Each row contains the number of events of the corresponding bin for all the channels visualized in the plot.
- The .csv cumulative image file contains the total energy information for all the channels.

The "**Configure Data Recording**" button in the Control Bar opens the "**Data Recording Configuration**" window (see **Fig. 15.21**), where the data recording is set up.

The "**Start/Stop Data Record**" buttons in the Control Bar are used to store on file data from the board. The Data Recording Configuration window also appears after pressing the "Start Data Record" button.

The Data Recording Configuration window is divided into the following areas:

- **Output Location**:
  - **Output Folder**: the folder where the files are saved, selected with the ... button
  - **Base File Name**: the name of the files
  - **File Number**: a number appended to the Base File Name
  - **Auto-increment file number**: if checked, the File Number increases by itself at each new acquisition that is started
- **Output Format**:
  - **Format**: the format of the files, among **CSV (Comma/Semicolon Separated)**, **TXT (Fixed Width)**, **Binary Energy (.sde)**, **NumPy (.npy)**, **MATLAB (.mat)** and **HDF5 (.h5)**

  - **Separator**: the separator of the CSV format, among **Semicolon (;)**, **Comma (,)**, **Tab** and **Pipe (|)**
  - **Write header**: if checked, a header is written at the beginning of the file
- **File Split**: if **Split file at** is checked, the data are split into files of the size set in MB
- **Data Type & Target**: the type of data to be stored and when to stop the recording (see below)
- **Channels**: the channels whose data are saved, selected with the corresponding checkboxes; the **ALL** and **NONE** buttons select or deselect all the channels

The “Data Type” option can be used to choose the type of data to be stored:

- Oscilloscope: Every time the oscilloscope triggers, the downloaded analog waveforms from selected channels are dumped on disk (the digital traces are not saved).
- List Mode: The buffered events consisting of energy and time information from the single channel of the board that event by event fires the trigger are stored on the disk.

The “Target Mode” option sets how to stop data recording:

- Free Running: record data until “Stop Data Record” button is pressed.
- Number of events: record data until a specified number of events is reached.
- Time: record data for a specified amount of time (not available for the Oscilloscope Data Type).

The number of events or the amount of time in the respective recording mode can be set in the “Target Value” field; for the Time mode, the unit can be chosen among s, min and h.

Note: When a target is set, both in Oscilloscope and in List Mode, the data saving stops when the target is reached, even if the acquisition keeps running in the GUI.

The name of the output file is shown at the bottom of the window (for example `acquisition_0001.csv`). To start the Oscilloscope or the List Mode dumping press the **Save** button.

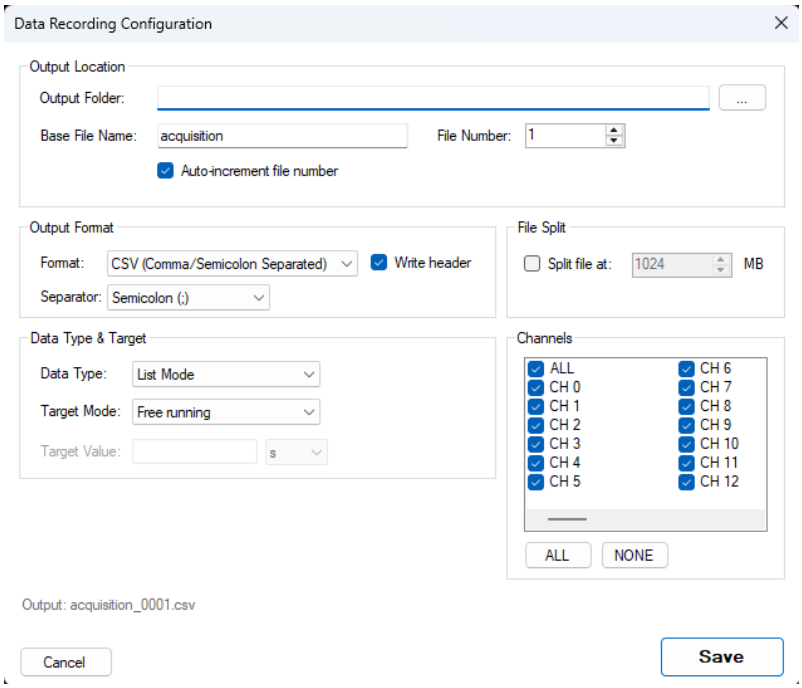


Fig. 15.21: The Data Recording Configuration window.

In the CSV format, each line of the file contains one event; if **Write header** is checked, the first line contains the names of the columns.

Each line of the .csv file for the Oscilloscope Mode contains the following information:

- Channel: channel number;
- Timestamp: time tag of the waveform;
- NSamples: number of waveform samples;
- Samples: the values of the waveform samples, one per column.

Each line of the .csv file for the List Mode contains the following information:

- Channel: event channel;
- Timestamp: event time tag;
- Energy: event energy;
- PileUp: pile-up flag of the event;
- Saturated: saturation flag of the event.

Note: The installation folder of Open Hardware Readout Software contains some Python (.py) examples, among which scripts to read back the saved files in the different formats.

16 Appendix

In this Appendix we give more details about the Ethernet and Optical communication protocols available for the DT5560SE/R5560/R5560SE. The protocols are the same across the family: the examples below are taken from an R5560SE, and the differences of the DT5560SE, which hosts a single DAQ section instead of four, are stated where they apply.

16.1 Ethernet protocol

The standard application implements three different servers:

- a TCP web server listening on the port 8899 that supports binary communication
- a TCP web server listening on ports 8900, 8901, 8902 and 8903 that are directly connected to the DMA managed FIFO. This socket is a monodirectional link that streams any data that the FPGA pushes in the DMA-managed FIFO to the end user client
- a web socket application based on textual JSON messages

The Ethernet protocol is the same for the Gigabit LAN connections and for the USB 2.0 connection. USB 2.0 emulates an Ethernet card on the host client.

16.1.1 TCP web server

The TCP web server listens on the port 8899 for messages. These messages are then converted by the software application in read/write to the FPGA. This server runs on all DAQ boards and, on the R5560/R5560SE only, on the SCSP board (on slow control Ethernet port).

A packet with the following format is sent from the client (PC) to the server (DAQ/SCSP) application:

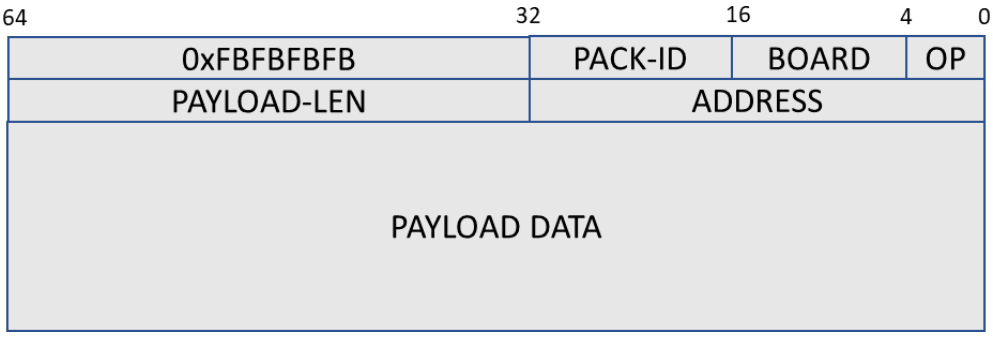


Fig. 16.1: Format of the packet sent from the client to the server.

- `PACK-ID`: incremental number generated by the CB that identifies a particular packet.
- `BOARD`: always 0xFFF if directly connected to the DAQ, which is always the case on the DT5560SE. On the R5560/R5560SE: 0x000 (SCSP), 0x001 (DAQ 1), 0x002 (DAQ 2), 0x003 (DAQ 3), 0x004 (DAQ 4) to identify the sub DAQ board if the client is connected to the Slow Control Board.
- `OP`: identifies the operation

  - 0000: enumerate board. Returns JSON data that enumerates the board and all specs
  - 0001: write with address incrementation
  - 0010: write without address incrementation
  - 0011: read with address incrementation blocking
  - 0100: read without address incrementation blocking
  - 0101: read with address incrementation non-blocking
  - 0110: read without address incrementation non-blocking
- `PAYLOAD-LEN`: number of D-WORD of data to be transferred.
- `ADDRESS`: destination address base address.
- `PAYLOAD`: data to be transferred to the destination.

The DAQ (or the SCSP) answers to the operation with a packet like the following:

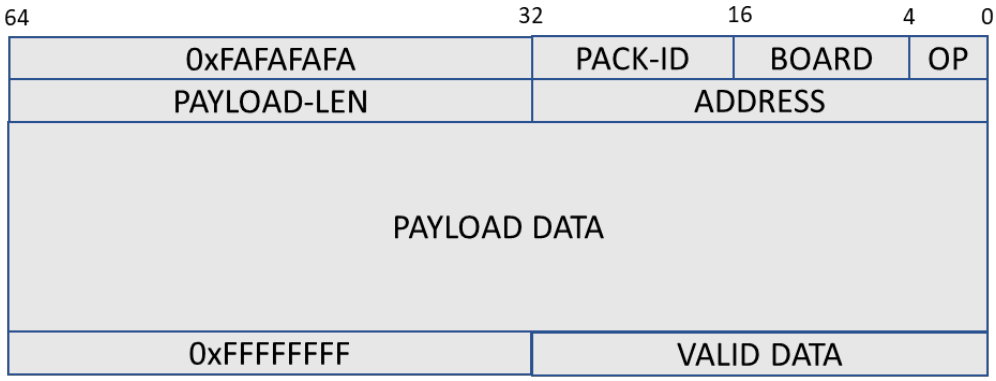


Fig. 16.2: Format of the packet sent from the server to the client.

- `PACK-ID`: copy the PACK-ID of the requesting packet from the CB.
- `BOARD`: always 0xFFF if directly connected to the DAQ. 0x000 (SCSP), 0x001 (DAQ 1), 0x002 (DAQ 2), 0x003 (DAQ 3), 0x004 (DAQ 4) to identify the sub DAQ board if the client is connected to the Slow Control Board.
- `OP`: copy the CB request OP.
- `PAYLOAD-LEN`: copy the CB request PAYLOAD.
- `ADDRESS`: copy the CB request ADDRESS.
- `PAYLOAD`: data transferred from the board to the CB.
- `VALID-DATA`: number of D-WORD effectively transferred in the payload. If the payload is longer than valid data, data after the first VALID-DATA will be discarded.
- `ERROR CODE`: 0xFFFFFFFF means no error, otherwise an error code is returned.

16.1.2 FIFO server

If DMA-based FIFO is instantiated in the firmware, for each FIFO a new TCP server is created starting from the port 8900. These web servers run only on DAQ board and are not accessible from the SCSP of the R5560/R5560SE. There is no protocol on this socket. What is pushed in the DMA FIFO on FPGA side, will appear as is on the end user PC. FIFOs are 32-bit wide and data should be cast to uint32.

16.1.3 JSON server

The JSON server listens on port 8000 for WebSocket connections. WebSocket is a textual protocol supported by all modern browsers, JavaScript and all development tools. The performance on WebSocket is lower than using raw sockets, but it is human readable and produces structured results. All slow control and DAQ configuration are managed through this web socket and all operations listed below can be performed:

```
Set Register    {"command":"set_register", "name":"<register_name>",
                 "value":int_value}
Get Register    {"command":"get_register", "name":"<register_name>",
                 blocking:true/false}
Write Memory    {"command":"write_mem", "name":"<memory_block_name>",
                 "data":[,,,,,,]}
Read Memory     {"command":"read_mem", "name":"<memory_block_name>",
                 blocking:true/false, increment:true/false}
Set timeout     {"command":"set_timeout", int: milliseconds}
```

Blocking indicates if the command stalls until all data are available or timeout.

Increment indicates if address should be incremented during the read.

Other configuration commands can be issued:

```
Change Eth CFG    {"command":"set_eth0", "dhcp":true/false, "ip":"x.x.x.x",
                   "netmask":"x.x.x.x", "gateway":"x.x.x.x"}
Change board id   {"command":"set_id", "value":xxx}
Set ADC clock source
                  {"command":"set_adc_clock",
                   "value":"internal/system/optolink/hlink"}
Set H-LINK clock source (R5560/R5560SE only)
                  {"command":"set_adc_clock",
                   "value":"internal/system/optolink"}
Get Board temperature
                  {"command":"get_temp"}
Get monitor voltages
                  {"command":"get_volts"}
Reboot            {"command":"reboot"}
Save Settings     {"command":"save"}
```

The answer to the command will be:

```
{"command":"<command>", "result":true/false, "reason":"...", "data":[,,,,,]}
```

Data field will contain the read data as uint32 in 10-base digits.

16.2 Optical link protocol

The DT5560SE has **2 SFP+ slots** that can host optical transceivers operating at a bitrate **up to 10 Gbps**, while the R5560/R5560SE has **8 SFP+ connectors**, two per DAQ section, each section managing two independent links, operating at a bitrate **up to 12.5 Gbps**. Each link is bidirectional and can be used for point to point connection or in daisy chain mode.

The Optical Link protocol is not implemented by default and its operation requires firmware/software development on the user's side. The board can be easily interfaced with large readout infrastructure using the SDK and developing in VHDL a custom protocol with daisy chain capabilities.

The connection between the optical link and the FPGA depends on the model of the SoC installed, not on the board model: the Zynq 7030 does not allow the propagation of the clock on the optical link in a daisy-chained system, while the Zynq 7035 allows for it. The DT5560SE and the R5560A/R5560SEA carry a Zynq 7030; the R5560B/R5560SEB carry a Zynq 7035.

In order to propagate the clock in a daisy chain of boards, it is mandatory to recover the clock from the gear box of the GTx (RX), clean it with an external jitter cleaner and provide this clock as a clock reference for the TX GTx.

The RX channel needs a reference clock for correct operation. This should be a clean, low jitter clock generated locally on the RX channel. This clock cannot be recovered from the link. In order to propagate the clock, it is necessary to recover the clock and feed it as a reference clock to the TX.

Xilinx FPGAs have just one QPLL per Quad-block (block of four transceivers). The QPLL is the component that generates all high-speed frequencies that are necessary to serialize the data in the transceiver. If the QPLL is already used for the RX, it cannot also handle the clock for the TX. The TX clock (recovered from RX circuit) must be propagated providing it as a reference clock of the second Quad-Block of transceiver.

Different models of the Zynq SoC are based on different connection schemes for the optical links operation:

16.2.1 Zynq 7030 (DT5560SE, R5560A, R5560SEA)

The Zynq 7030 has just 4 GTx transceivers (in just one quad-block). It is indeed impossible (due to Xilinx FPGA limitation) to propagate the clock in a daisy chain.

On the R5560/R5560SE all four GTx are used, two managing the SFP+ connector while the other two are used for the internal H-LINK. Internal H-LINK and external SFP+ will operate with the same reference clock (it does not mean that they will have the same bit rate; each transceiver has a CPLL that allows the link speed to be changed starting from a common external reference clock).

The recovered clock is cleaned with the jitter cleaner and forwarded back to the FPGA logic. It is used for event timestamping and ADC clocking, and on the R5560/R5560SE it is also propagated on the H-LINK to nearby DAQ sections.

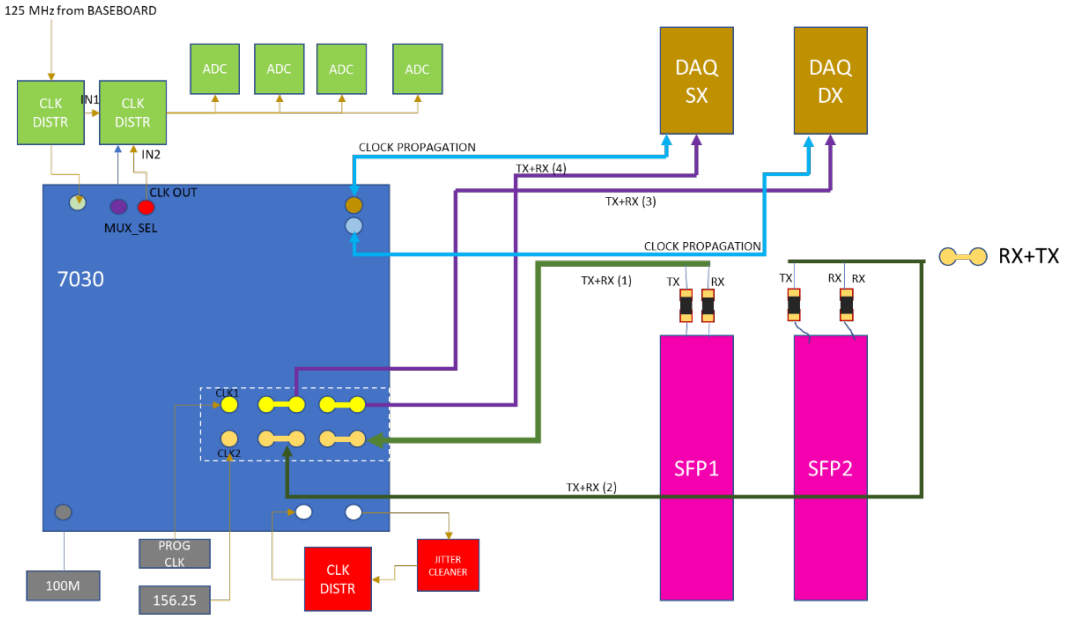


Fig. 16.3: Scheme for optical links operation with the Zynq 7030 SoC.

16.2.2 Zynq 7035 (R5560B, R5560SEB)

The Zynq 7035 has 8 GTx transceivers (in two quad-blocks). So, it is possible to propagate the clock in a daisy chain. The SFP+ connector can be connected to GTX2 of the first quad-block (clock propagation is not possible) or to the GTX1 of the second quad-block (clock propagation is possible). The clock recovered from the RX channel in the first quad-block is cleaned and provided back to the FPGA and to the TX GTX on the second quad-block. The TX will be clocked with the cleaned version of the RX clock, allowing the clock to be propagated down the chain. The recovered clock provided to the FPGA logic can be used for clock timestamping logic, ADCs and it is propagated to nearby DAQ sections.

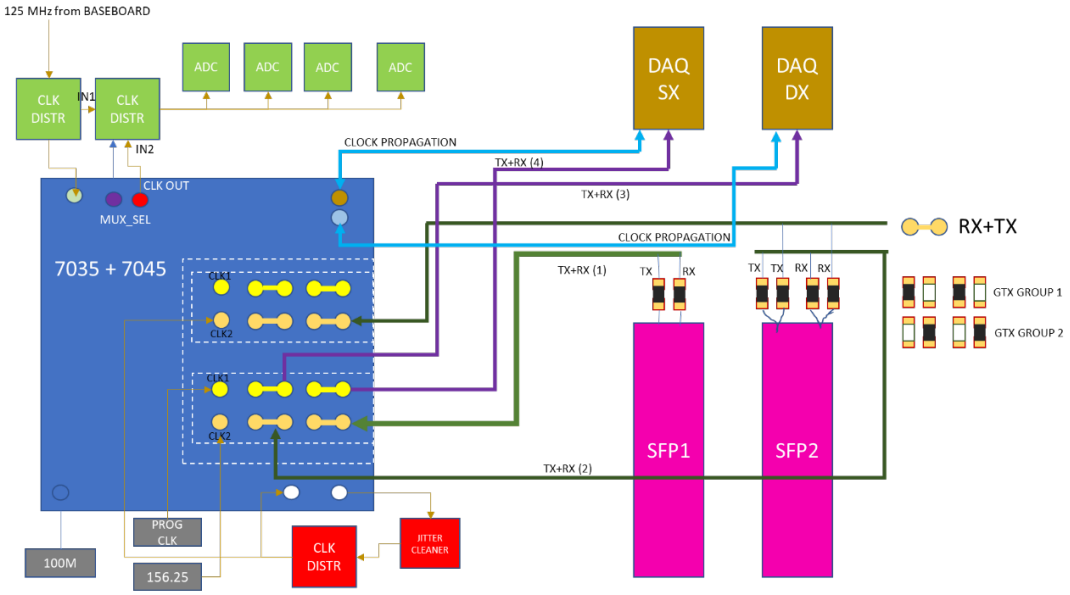


Fig. 16.4: Scheme for optical links operation with the Zynq 7035 SoC.

16.3 Libraries usage examples (Legacy)

Note: The examples of this section use the Legacy library of Sec. 14.3: the supported library for the development of custom software is the SciSDK (see Sec. 14.2).

Note: The examples below use the R5560SE symbol names. On the DT5560SE the connection handle and the connect function carry the board prefix of that model (`tDT5560SE_HANDLE`, `DT5560SE_ConnectTCP`); the register and data access functions (`NI_WriteReg`, `NI_ReadReg`, `NI_ReadData`, `NI_ReadFifo`) are the same on all models.

The following example in C shows how to connect to the board through the IP address, to set the parameters of the Oscilloscope component, to start a data acquisition, to read the Oscilloscope buffer status and download the data array.

```
//define and initialize all the variables
tR5560SE_HANDLE handle;          //board connection handle
uint32_t decimator = 0;          //no decimation of oscilloscope data
uint32_t pre_trigger = 100;      //number of samples between the first data
                                  //and the trigger
uint32_t trigger_level = 10000; //threshold in lsb to be exceeded to fire
                                  //the trigger
uint32_t trigger_mode = 1;       //trigger on the rising edge of the analog
```

```
                                //signal
uint32_t status_osc = 0;       //status of the oscilloscope data buffer
uint32_t position = 0;         //number of samples indicating the trigger
                                //position
uint32_t data_osc[1024];       //oscilloscope data array
uint32_t size_osc = 1024;      //size of the oscilloscope data array
uint32_t read_data_osc;        //number of oscilloscope data read
int Osc_Events = 10;           //number of oscilloscope data array to be
                                //downloaded
int e = 0;                     //iteration variable

#define REG_CONFIG_DECIMATOR     0x00000001
#define REG_CONFIG_PRETRIGGER    0x00000002
#define REG_CONFIG_TRIGGER_LEVEL 0x00000003
#define REG_CONFIG_TRIGGER_MODE  0x00000004
#define REG_CONFIG_ARM           0x00000005
#define REG_READ_STATUS          0x00000006
#define REG_READ_POSITION        0x00000007
#define REG_DATA_ADDRESS         0x00000008

//connect to the R5560SE board
if (R5560SE_ConnectTCP("192.168.50.150", 8888, &handle) != 0)
{ printf("Unable to connect to the board!\n"); return (-1); };

while (e < Osc_Events)
{
    //set oscilloscope parameters
    if (NI_WriteReg(decimator, REG_CONFIG_DECIMATOR, &handle) != 0)
        printf("Set Decimator Error");
    if (NI_WriteReg(pre_trigger, REG_CONFIG_PRETRIGGER, &handle) != 0)
        printf("Set Pre Trigger Error");
    if (NI_WriteReg(trigger_level, REG_CONFIG_TRIGGER_LEVEL, &handle) != 0)
        printf("Set Trigger Level Error");
    if (NI_WriteReg(trigger_mode, REG_CONFIG_TRIGGER_MODE, &handle) != 0)
        printf("Set Trigger Mode Error");

    //start oscilloscope acquisition
    if (NI_WriteReg(0, REG_CONFIG_ARM, &handle) != 0)
        printf("Set Config Error");
    if (NI_WriteReg(1, REG_CONFIG_ARM, &handle) != 0)
        printf("Set Config Error");

    //wait until the oscilloscope status indicates that data are available
    while (status_osc != 1)
        if (NI_ReadReg(&status_osc, REG_READ_STATUS, &handle) != 0)
            printf("Status Error");

    //read the position of the trigger
    if (NI_ReadReg(&position, REG_READ_POSITION, &handle) != 0)
        printf("Position Error");

    //read the oscilloscope data
    if (NI_ReadData(&data_osc, size_osc, REG_DATA_ADDRESS, &handle,
                    &read_data_osc) != 0)
        printf("Get Data Error");

    e++;
```

```
} printf("Download Finished");
```

The following example in C shows how to connect to the board through the IP address, to start a data acquisition from a FIFO buffer provided by the Custom Packet component, to read its buffer status and download the data array. Data are organized inside the Custom Packet with a layout defined by the user during the component creation.

```
//define and initialize all the variables
tR5560SE_HANDLE handle;         //board connection handle
uint32_t status = 0;            //status of the custom packet data buffer
uint32_t data[100000];          //custom packet downloaded data array
uint32_t N_Packet = 100;        //number of data array packets to be
                                 //downloaded
uint32_t N_CP = 4;              //number of custom packet word of the data
                                 //layout
uint32_t bus_mode = 1;          //blocking streaming access to the fifo
                                 //buffer
uint32_t timeout = 1000;        //read data timeout expressed in
                                 //milliseconds
uint32_t read_data;             //number of custom packet data read
uint32_t N_Total_Events = 1000; //total number of data to be downloaded
uint32_t ReadDataNumber = 0;    //iteration variable

#define REG_CONFIG       0x00000001
#define REG_READ_STATUS  0x00000002
#define REG_DATA_ADDRESS 0x00000003

//connect to the R5560SE board
if (R5560SE_ConnectTCP("192.168.50.150", 8888, &handle) != 0)
{ printf("Unable to connect to the board!\n"); return (-1); };

//start custom packet data acquisition
if (NI_WriteReg(0, REG_CONFIG, &handle) != 0)
    printf("Reset Error");
if (NI_WriteReg(1, REG_CONFIG, &handle) != 0)
    printf("Start Error");

//check if the custom packet status indicates that data are available
if (NI_ReadReg(&status, REG_READ_STATUS, &handle) != 0)
    printf("Status Error");

if (status > 0)
{
    while (ReadDataNumber < N_Total_Events)
    {
        //read the data from the custom packet fifo buffer
        if (NI_ReadFifo(&data, N_Packet * (N_CP + 3), REG_DATA_ADDRESS,
                        REG_READ_STATUS, bus_mode, timeout, &handle,
                        &read_data) != 0)
            printf("Data Download Error");
        ReadDataNumber = ReadDataNumber + N_Packet;
    }
    printf("Download completed");
}
else printf("Status Error");
```

Safety Notices

N.B. Read carefully the "Precautions for Handling, Storage and Installation" document provided with the product before starting any operation.

The following HAZARD SYMBOLS may be reported on the unit:

	Caution, refer to product manual
	Caution, risk of electrical shock
	Protective conductor terminal
	Earth (Ground) Terminal
	Alternating Current
	Three-Phase Alternating Current

The following symbol may be reported in the present manual:

	General warning statement

The symbol could be accompanied by the following terms:

- **DANGER**: indicates a hazardous situation which, if not avoided, will result in serious injury or death.
- **WARNING**: indicates a hazardous situation which, if not avoided, could result in death or serious injury.
- **CAUTION**: indicates a situation or condition which, if not avoided, could cause physical injury or damage the product and / or the surrounding environment.



THIS DEVICE MUST NOT BE USED BY UNTRAINED PERSONNEL

CAUTION: Do Not Operate without Covers



TO AVOID ELECTRIC SHOCK OR FIRE HAZARD, DO NOT OPERATE THIS PRODUCT WITH COVERS OR PANELS REMOVED

CAUTION: Do Not Operate in Wet/Damp Conditions



TO AVOID ELECTRIC SHOCK, DO NOT OPERATE THIS PRODUCT IN WET OR DAMP CONDITIONS

CAUTION: Do Not Operate in an Explosive Atmosphere



TO AVOID INJURY OR FIRE HAZARD, DO NOT OPERATE THIS PRODUCT IN AN EXPLOSIVE ATMOSPHERE

CAUTION: Avoid potential hazards



USE THE PRODUCT ONLY AS SPECIFIED.
SERVICE PROCEDURES CAN BE PERFORMED BY QUALIFIED PERSONNEL ONLY

Please, contact the Techincal Support in case Service Procedures are required.



DO NOT OPERATE WITH SUSPECTED FAILURES.
IF YOU SUSPECT THIS PRODUCT TO BE DAMAGED, PLEASE CONTACT THE TECHNICAL SUPPORT

See Chap. **20** for the Technical Support contacts.



IT IS UNDER THE RESPONSIBILITY OF THE CUSTOMER AN IMPROPER USE OF THE PRODUCT

17 Instructions for Cleaning

The equipment may be cleaned with isopropyl alcohol or deionized water and air dried. Clean the exterior of the product only.
Do not apply cleaner directly to the items or allow liquids to enter or spill on the product.

17.1 Cleaning the Air Vents

It is recommended to occasionally clean the air vents (if present) on all vented sides of the board. Lint, dust, and other foreign matter can block the vents and limit the airflow. Be sure to unplug the board before cleaning the air vents and follow the general cleaning safety precautions.

17.2 General Cleaning Safety Precautions

CAEN recommends cleaning the device using the following precautions:

- Never use solvents or flammable solutions to clean the board.
- Never immerse any parts in water or cleaning solutions; apply any liquids to a clean cloth and then use the cloth on the component.
- Always unplug the board when cleaning with liquids or damp cloths.
- Always unplug the board before cleaning the air vents.
- Wear safety glasses equipped with side shields when cleaning the board.

18 Device Decommissioning

After its intended service, it is recommended to perform the following actions:

- Detach all the signal/input/output cable
- Wrap the device in its protective packaging
- Insert the device in its packaging (if present)



THE DEVICE SHALL BE STORED ONLY AT THE ENVIRONMENT CONDITIONS SPECIFIED IN THE MANUAL, OTHERWISE PERFORMANCES AND SAFETY WILL NOT BE GUARANTEED

19 Disposal

The disposal of the equipment must be managed in accordance with Directive 2012/19 / EU on waste electrical and electronic equipment (WEEE).



The crossed bin symbol indicates that the device shall not be disposed with regular residual waste.

20 Technical Support

To contact CAEN specialists for requests on the software, hardware, and board return and repair, it is necessary a MyCAEN+ account on www.caen.it:

https://www.caen.it/support-services/getting-started-with-mycaen-portal/

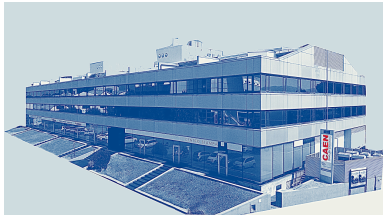
All the instructions for use the Support platform are in the document:



A paper copy of the document is delivered with CAEN boards.
The document is downloadable for free in PDF digital format at:

https:/www.caen.it/safety-information-product-support

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